

Endpoint crossings for random walks with memory

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Abstract

Drop a ball through a Galton board on which every bounce after the first repeats the previous direction with probability p . Kagey asked when the middle bin is as likely as each end bin. For this Markov walk with N steps the crossing comes when the expected number of direction changes is about $\log N$. We ask what changes when the ball remembers more than its last step. In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε . We prove that the crossing is unique and that $N\varepsilon$ at the crossing is $2\log N - \log \log N - \log 16 + o(1)$, about twice the Markov value, because a walk that ends in the middle needs a reversal among its first few steps. At the crossing the law has a hump near each end, and the middle is the lowest bin outside a window of width $O(N^{1/2} \log N)$. Next we take alternating runs whose lengths have tail $k^{-\alpha}$ with $1 < \alpha < 2$, started in equilibrium. Wang, Li and Hänggi observed, as a heuristic for heat conduction, that the ends and the middle decay at the same rate when α is the golden ratio. We prove the lattice form of their observation, with its constant. Namely, the ends beat the middle for large N exactly when α is below the golden ratio. Finally, an aging walk switches at step k with probability c/k . With a fixed first step every bin ties with the end exactly at $c = 1$, for every N . We prove the same for an aging walk with d states, whose vector of counts is exactly uniform at the tie.

Keywords: persistent random walk; elephant random walk; random recursive tree; Pólya urn; alternating renewal process; heavy tails; golden ratio; aging random walk; Galton board.

MSC2020: 60G50; 60K05; 60C05; 60F05; 60J10.

1 Introduction

Suppose you drop a ball through a Galton board with N rows of pegs. At the first peg the ball goes left or right with probability $\frac{1}{2}$ each. At every later peg it repeats its previous direction with probability p and switches with probability $q = 1 - p$. This is the board of Problem 131 in Kagey's Open Problem Collection [1], and one of his questions is when the middle bin is exactly as likely as each of the two end bins. If p is close to 1 the ball rarely switches, so the ends win. If $p = \frac{1}{2}$ the ball does a simple random walk, so the middle wins. Paper A [2] finds the crossing in between. The middle catches up with the ends when the expected number of switches $T = Nq$ is about $\log N$.

Kagey's ball remembers one step, since its next direction depends only on its last one. In this paper we ask what happens to the middle-versus-ends question if the ball remembers more. We study three kinds of memory.

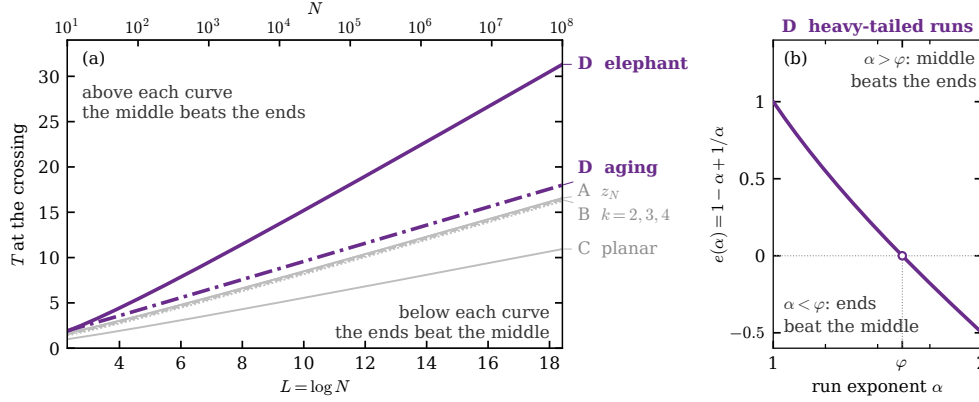


Figure 1: The crossing on each model's switching scale against $L = \log N$ (a), and the heavy-tailed exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of this paper for a stationary start (b). The scale is T for Papers A to C, $x = N\varepsilon$ for the elephant walk (drawn through the root g_N of Theorem 2.7), and the expected number $H_N - 1$ of switches at the fixed-start crossing $c = 1$ for the aging walk. The curves of this paper are in bold, and a version of this figure also appears in Paper C.

- In the elephant random walk of Schütz and Trimper [5] the ball remembers its whole past. At each peg it picks one of its earlier directions uniformly at random and repeats it, except that with probability ε it reverses it. We write $x = N\varepsilon$, which is the expected number of reversals up to ε .
- In a walk with heavy-tailed runs the ball remembers how long it has gone in its current direction. Its path is a sequence of runs of alternating direction, and the run lengths ξ are independent with $\mathbb{P}(\xi \geq k) = k^{-\alpha}$ for $k \geq 1$.
- In an aging walk the ball remembers how old it is. It switches at peg $k \geq 2$ with probability c/k . We also give the ball d directions. Then at peg $k \geq 2$ it turns to each of the other $d - 1$ directions with probability $\theta/(k + d - 2)$, which is c/k with $c = \theta$ when $d = 2$.

The answer depends on the kind of memory. For the elephant walk the middle needs about twice as many reversals as Kagey's board needs switches. Namely, the crossing value of x is about $2 \log N$, not $\log N$. For heavy-tailed runs started in equilibrium the ends win for large N if α is below the golden ratio $\varphi = (1 + \sqrt{5})/2$, and the middle wins if α is above φ . For the aging walk with a fixed first step the crossing is exactly $c = 1$, for every N and every bin. With d directions the same happens at $\theta = 1$, where the vector of counts is exactly uniform (Section 5).

Why the elephant needs twice as many reversals. The ends are easy. The ball lands in the right end bin exactly when its first step goes right and it never changes direction. This has probability $\frac{1}{2}(1 - \varepsilon)^{N-1} \approx \frac{1}{2}e^{-x}$ for the elephant walk, and the same with q in place of ε for Kagey's board. So only the middle differs. On Kagey's board the T switches are spread over the whole walk, and each moves the final position by about a run length N/T . So the number of right steps spreads over about $N/\sqrt{T}$ bins, the middle bin has probability of order $\sqrt{T}/N$, and the ends lose when e^{-T} falls to about $1/N$, that is, when $T \approx \log N$.

The elephant walk spreads a reversal in a different way. A reversal at step $t + 1$ is copied by later steps, those copies are copied, and so on. The steps that descend from it form a family, whose share of the walk behaves like the share of one ball in Pólya's urn started with one ball against t . So a late reversal has a small family and hardly moves the ball. To land exactly in the middle,

the walk needs a family that fills half the walk, so it needs a reversal among its first few steps. This has probability of order ε , and hitting the exact middle brings another factor of order $1/N$. So the middle bin has probability of order $\varepsilon/N = x/N^2$ (in fact about $4x/N^2$), and the ends lose only when e^{-x} falls to about $1/N^2$, that is, when $x \approx 2 \log N$. More precisely, at Kagey's crossing $\frac{1}{2}e^{-T} \approx \sqrt{2T/\pi}/N$, and squaring this relation turns it into the elephant relation $\frac{1}{2}e^{-x} \approx 4x/N^2$ with $x \approx 2T - \log(2\pi)$.

The small families also give the shape of the law. About x/m^2 reversals have a family of m steps, so the bin at distance m from an end has probability about $x/(2m^2)$ over a wide range of m . So the law has a hump near each end, and the middle is where the two tails meet. Indeed, twice $x/(2m^2)$ at $m = N/2$ is the $4x/N^2$ above.

Why the golden ratio. For heavy-tailed runs take $1 < \alpha < 2$, so a run has a finite mean μ but an infinite variance. Start the walk in equilibrium, at a typical time of a long sequence of runs. Then the first run is already in progress, and long runs are more likely to be in progress. The chance that this run covers all N steps is of order $N^{1-\alpha}$, and this is the end. For the middle, the runs to the right and the runs to the left form two independent renewal processes, each making about $N/(2\mu)$ runs, and the ball ends in the middle when they meet at level $N/2$. Each process fluctuates on the scale $N^{1/\alpha}$ of a stable law, so they meet exactly with probability of order $N^{-1/\alpha}$. Hence the ratio of end to middle is of order $N^{1-\alpha}/N^{-1/\alpha} = N^{1-\alpha+1/\alpha}$. The exponent vanishes when $\alpha^2 - \alpha - 1 = 0$, that is, at $\alpha = \varphi$. Below φ the ends win, and above φ the middle wins. If the first run is a fresh run instead, the end is of order $N^{-\alpha}$, a factor N smaller, and the middle wins for every $1 < \alpha < 2$.

1.1 Relation to Papers A, B and C

This is the fourth of four papers on Problem 131, which we call Papers A, B, C and D. Paper A [2] solves Kagey's board. Paper B [3] treats d directions, where the ball turns to each other direction with the same probability. Paper C [4] gives the general theory for slowly switching chains. These three papers study Markov walks, whose next step depends only on the current one, and their answer is that the middle (or a face) ties with the vertices when T is of order $\log N$. This paper, Paper D, keeps Kagey's question and changes the walk. It can be read on its own. From the earlier papers we use only the crossing equation of Paper A, namely (1) below, and we use it only to compare the elephant crossing with the Markov crossing in Section 2. The four papers share the following notation.

- Kagey's board has N steps (Kagey's row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . We put $q = 1 - p$ and $u = q/p$ (the odds of a switch).
- $P_N(k)$ is the probability of k right steps, that is, the probability that the ball lands in bin k . The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.
- $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step).
- The crossing is the value of p (or of T) at which a bin is exactly as likely as an endpoint. For the middle bin it is p_N , and $z_N = N(1 - p_N)$.
- Papers B and C have d states. A face F is a nonempty set of states, and the vertices are the faces of size 1. For a chain with transition matrix $I + (T/N)Q$, λ_F is the principal Dirichlet eigenvalue of $-Q$ on F , and K_F is a second-order constant.

For the elephant walk $x = N\varepsilon$ corresponds to T , and its crossing x_N corresponds to z_N . We compare the two crossings through the crossing equation of Paper A [2, Proposition 3.8], which

says that for every $N \geq 2$

$$z_N + \frac{1}{2} \log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2}), \quad c_0 = \frac{1}{2} \log \frac{\pi}{8}, \quad (1)$$

where the error term is at most $\frac{1}{4L^2} + \frac{L^2+L}{N}$ in absolute value, and that $L/2 \leq z_N \leq L$. As in Paper B the start law matters. For heavy-tailed runs it decides whether the threshold is φ or 1, and for the aging walk it decides whether the crossing is exactly $c = 1$. Figure 1 shows the crossings of all four papers.

1.2 Setting and notation

Every walk in this paper takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \leq N : X_i = +1\}$ be the number of right steps, and keep writing $P_N(k) = \mathbb{P}(A_N = k)$ for the probability that the ball lands in bin k , now for whichever walk we study. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N - k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \leq k} 1/j$ is the k -th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \rightarrow \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0, 1)$ and $x = N\varepsilon$. Its crossing is ε_N , and $x_N = N\varepsilon_N$. Heavy-tailed runs (Section 4) have the exponent α , the mean run length $\mu = \mathbb{E}\xi = \zeta(\alpha)$ for $\alpha > 1$, and the length D of the first run. A fresh start has D distributed as ξ , and the stationary start has $\mathbb{P}(D = d) = \mathbb{P}(\xi \geq d)/\mu$. We put $R_N = E_N/C_N$, so the ends win exactly when $R_N > 1$. The aging walk (Section 5) has the constant $c \in (0, 2)$, and its version with d states has the constant θ and the count vector $(n_1, \dots, n_d)$, where n_i is the number of steps in state i . The symbols that only one section uses are defined at the start of that section. The letters p , q and u have their meaning for Kagey's board only in this introduction, and later sections reuse them locally.

1.3 Results

The elephant walk. For every even $N \geq 2$ the crossing ε_N exists, is unique and is below $\frac{1}{2}$ (Proposition 2.3). The center has derivative exactly $4/(N+2)$ at $\varepsilon = 0$ (Proposition 2.5), and $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ for even $N \geq 64$ and $\varepsilon(1 + \log N) \leq \frac{1}{100}$, with explicit constants (Theorem 2.6). Let g_N be the root of $g + \log g = 2L - \log 8$. Then $x_N = g_N + O(L^2/N)$, and so $x_N = 2L - \log L - \log 16 + O(\log L/L)$ (Theorem 2.7). Compared with Kagey's board, $x_N = 2z_N - \log(2\pi) + c_*/L + O(\log L/L^2)$ with $c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4}$ (Theorem 2.8).

We also go one order further. In the same range of ε the center is $\frac{4\varepsilon}{N+2}N^{2\varepsilon}(1 + O(\varepsilon))$, with explicit constants (Proposition 2.9). This gives $x_N = g_N - (2g_N L + g_N^2/2)/(N(1 + 1/g_N)) + O(L/N)$, so the root g_N is above the crossing for large N (Proposition 2.10). At fixed N the center is a polynomial $c_1\varepsilon + c_2\varepsilon^2 + \dots$ in ε , and $c_2/c_1 - 2L \rightarrow 2\gamma - 2 - \log 2$ (Proposition 2.11).

The remaining results describe the whole law near the crossing. At the crossing the end is below its neighbor for every even $N \geq 8$ (Corollary 3.6 and Proposition 3.8). For $N < 1688$ this rests on a floating-point computation whose rounding error we bound (Lemma 3.7). Let M_N be the number of steps that disagree with the first step. For $N \geq 2$ and $\varepsilon \leq \frac{1}{4}$ the law of M_N is within an explicit

δ_B of a compound Poisson law, bin by bin, and $\delta_B = O(x^2 \log^2 N/N)$ when $x^2 \leq N$ (Theorem 3.9). If $x \rightarrow \infty$ and $x^3 \log^2 N/N \rightarrow 0$, then $(M_N - x \log x)/x$ converges in law to the Landau law, with a bound for each bin (Theorem 3.10). So at the crossing every mode lies at distance $x_N \log x_N + \lambda_0 x_N + O(\sqrt{x_N \log x_N})$ from the nearer end, where $\lambda_0 = -0.2228\dots$ is the mode of the Landau density (Corollary 3.12, and Corollary 3.17 gives the weaker bound $O(x_N \log^2 x_N)$). Farther from the ends, bin $N - m$ has probability $\frac{x}{2m^2}(1 + o(1))$ for $x \log^2(x + 1) \ll m \ll N$, when $x \geq 1$ and $x \log^2 N/N \rightarrow 0$ (Theorem 3.14). Finally, for large even N the center at the crossing is below every interior bin outside a window of width of order $N^{1/2} \log N$ around it (Theorem 3.15).

Heavy-tailed runs. For $1 < \alpha < 2$ and every first run that is finite almost surely, C_N is asymptotic to an explicit constant times $N^{-1/\alpha}$ (Theorem 4.4). With the stationary start $R_N \sim C(\alpha)N^{1-\alpha+1/\alpha}$, and $C(\varphi) = 0.7761\dots$, so at $\alpha = \varphi$ the center still wins. With a fresh start $R_N \rightarrow 0$ (Corollary 4.5). For $\alpha \geq 2$ the center is of order $N^{-1/2}$, with an extra factor $(\log N)^{-1/2}$ at $\alpha = 2$, and $R_N \rightarrow 0$ with either start (Theorem 4.7). For $0 < \alpha < 1$ the runs have infinite mean and only the fresh start makes sense. Then $NC_N \rightarrow \frac{2}{\pi} \tan \frac{\pi\alpha}{2}$ and $R_N \rightarrow \infty$ (Theorem 4.8). So with a fresh start the ends win for $\alpha < 1$ and the center wins for $\alpha > 1$. We do not prove what happens at $\alpha = 1$ (Remark 4.9).

Aging. With a fixed first step and $c = 1$ the number of right steps is exactly uniform on $\{1, \dots, N\}$. The ratio of each interior bin to the end increases strictly in c , so every interior bin ties with the end exactly at $c = 1$, for every $N \geq 2$ (Proposition 5.1). With d states, a start in state 1 and $\theta = 1$, the count vector is exactly uniform over all compositions $(n_1, \dots, n_d)$ of N with $n_1 \geq 1$, and every other such bin ties with the corner $(N, 0, \dots, 0)$ exactly at $\theta = 1$, for every $N \geq 2$ (Theorem 5.2). A small change of the aging rate gives a chain whose counts have exactly the law of Pólya's urn for every weight (Proposition 5.4). With a fair first step each interior bin is twice as likely as an end at $c = 1$, and its crossing is unique and below 1 (Proposition 5.6). For the center the crossing is $c_N = 1 - \log 2/(\log N + \gamma + 2 - 2 \log 2) + O((\log N)^{-3})$ (Proposition 5.7).

1.4 Prior work

The steps of the elephant walk split into clusters, since a step either copies an earlier step exactly, with probability $1 - 2\varepsilon$, or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [6], and Baur and Bertoin [7] relate the walk to Pólya-type urns. Our regime $N\varepsilon \asymp \log N$ lies in the strongly supercritical regime of Baur [8], who extends the results of Bertoin [9] for retention probability $1 - t/\log n$. Baur's Corollary 1 gives the sizes of the largest clusters other than the root's (a Poisson limit with intensity $a^{-2} da$), which is the picture of about x/m^2 families of size m behind Theorem 3.14. These results describe cluster sizes, not the center probability, which is a large deviation of the root cluster. The law that appears in the bulk (Theorems 3.9 and 3.10) is also known. Without its truncation, our compound Poisson law is the Lea–Coulson form of the Luria–Delbrück distribution, and Kessler and Levine [15, Eqs. (14), (18) and (19)] describe its profile $x/(m(m+1))$ for small x and its Landau limit for large x (see also [16]). The same stable law governs the fluctuations of the root cluster when the retention probability is $1 - c/\log n$ [14]. What we add is the comparison of the elephant walk with this law in our regime, with explicit errors, and the location of the mode. Guérin, Laulin, Raschel and Simon [10, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal, and their other results fix the memory parameter as N grows. So do the local limit theorem of Trinas and Valle [11, Theorem 3.1] for the superdiffusive walk and their bounds

on where log-concavity fails. We did not find an asymptotic formula for the center probability when $\varepsilon \rightarrow 0$ as N grows.

For heavy-tailed runs, Godrèche and Luck [12] give the transform of the occupation time of an alternating renewal process and the scalings of the center and of both edges, and [18, 19] describe the profile of a Lévy walk. Wang, Li and Hänggi [20] already observe, as a heuristic for heat conduction, that equal decay of the central peak and the side peaks gives the golden ratio. The two decay rates of the end, $N^{1-\alpha}$ and $N^{-\alpha}$, are the scalings of the ballistic peaks of a Lévy walk with and without an equilibrated start [17, Table I]. We prove the lattice statement with a local limit at the center, treat both starts, and compute $C(\alpha)$. For $\alpha < 1$ the limit law of A_N/N is Lamperti's [13]. For aging walks, Dietz and Sethuraman [30] proved the limit law of A_N/N , Engländer and Volkov [31] gave a second proof, and Engländer, Volkov and Wang [34] found the scaling limit of the walk. For d states the limit is a Dirichlet law [30, Theorem 1.4]. Our statements are exact at every N .

1.5 Organization

Section 2 reduces the elephant walk to a counting chain, proves that its crossing is unique, finds the crossing, and then gives the center and the crossing to second order. Section 3 describes the whole law near the crossing, namely the dip at each end, the compound Poisson and Landau approximations of the bulk with the place of the modes, the tail $x/(2m^2)$, and the window around the center. Section 4 treats heavy-tailed runs, first the golden ratio for $1 < \alpha < 2$ and then the exponents $\alpha \geq 2$ and $\alpha < 1$. Section 5 treats the aging walk with a fixed first step, with d states, and with a fair first step. Section 6 describes the checks. The longer proofs are in the appendices. Appendix A holds the proofs for Section 2. Appendix B holds the proofs for the tail and the modes in Section 3, and Appendix C holds the proofs for the bulk. Appendix D holds the proofs for Section 4, and Appendix E proves Proposition 5.7.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain, prove that the crossing is unique, compute the center to first and second order in ε , and locate the crossing. The last subsection gives one more term of the center and of the crossing.

Definition 2.1 (elephant walk). Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε . All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.2, $A_N = \#\{i \leq N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$, and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. In the notation of [5] the memory parameter is $1 - \varepsilon$, and [10] write $a = 1 - 2\varepsilon$. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \leq t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1 - 2\varepsilon)\frac{b}{t}, \quad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t \geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b + 1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1 . For $0 \leq k \leq N$, $P_N(k) = \frac{1}{2}[f_N(N - k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1 - \varepsilon)^{N-1}$, and $P_N(k) = P_N(N - k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t . So $X_{t+1} \neq X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1-M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1 , which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N - k$. Finally $M_N \leq N - 1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1} (1 - r_t(0)) = (1 - \varepsilon)^{N-1}$. $\square$

So the end is Kagey's end $p^{N-1}/2$ with $p = 1 - \varepsilon$. More flips always help the middle against the end.

Proposition 2.3. *Let $N \geq 2$ and $1 \leq k \leq N - 1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0, 1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $C_N \geq 2E_N$ for $\varepsilon \geq \frac{1}{2}$.*

Proof. The probability of a path of M is a product of $N - 1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s . We divide it by the probability $f_N(0) = (1 - \varepsilon)^{N-1}$ of the path that never flips. With $y = b_s/s \in [0, 1)$ the factors become $\frac{\varepsilon + (1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \geq 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε , and strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the claim for $f_N(k)/f_N(0)$, and the claim for $P_N(k)/E_N$ follows by Lemma 2.2. As $\varepsilon \rightarrow 0$, $C_N \rightarrow 0$ and $E_N \rightarrow \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1 - \varepsilon, \varepsilon]$, so each path of M that ends at h has probability at least $(1 - \varepsilon)^{N-1} = 2E_N$. Hence $C_N \geq 2E_N$. The ratio C_N/E_N is continuous and strictly increasing, so it equals 1 at exactly one ε , and this ε lies in $(0, \frac{1}{2})$. $\square$

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε , it is Pólya's urn.

Lemma 2.4 (Pólya's urn). *Let $2 \leq v \leq N$. At time v an urn holds one ball of one color and $v - 1$ of the other. At each time $k = v, \dots, N - 1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2} / \binom{N-1}{v-1}$ for $1 \leq d \leq N - v + 1$, and $\beta_v(d) = 0$ for other d . Also β_v is nonincreasing in d and $\sum_d d \beta_v(d) = N/v$. For $1 \leq d \leq N - 1$,*

$$\sum_{v=2}^N \beta_v(d) = \frac{N}{d(d+1)}, \quad \sum_{v=2}^N (v-2) \beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. If the urn holds s balls of the first color at time k , the next draw is of that color with probability s/k . So a given order of i draws of the first color among the $N - v$ draws has probability $\frac{i!(v-1)v \dots (N-i-2)}{v(v+1) \dots (N-1)} = \frac{(v-1)i!(N-i-2)!}{(N-1)!}$. Multiplying by $\binom{N-v}{i}$ and putting $d = i + 1$ gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, which equals $\binom{N-d-1}{v-2} / \binom{N-1}{v-1}$. Hence $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d}$ for $2 \leq d \leq N - v + 1$, and β_v is nonincreasing. The fraction of balls of the first color is a martingale that starts at $1/v$, which gives the mean. For the two sums put $i = N - v$, so that $v - 1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. By the Chu–Vandermonde identity, $\sum_i \binom{i}{d-1} \binom{N-1-i}{1} = \binom{N}{d+1}$ and $\sum_i \binom{i}{d-1} \binom{N-1-i}{2} = \binom{N}{d+2}$. Multiplying by a_d and $2a_d$ gives the two sums. $\square$

We need some more notation. Let $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ be the chance of ending at the center from state b at time s . For $1 \leq t \leq h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{h-i}{N-i}, \quad c_1 = \sum_{t=1}^h \pi_t, \quad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \leq t \leq h}$ is a probability vector.

Proposition 2.5. *Let $N \geq 2$ be even.*

(a) $C_N = \sum_{t=1}^h \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1).$

(b) C_N is a polynomial in ε with $C_N(0) = 0$. At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C'_N(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) $\sum_t w_t t = 3 - \frac{6}{h+2}$, and $w_t \leq 2t2^{-t}$ for $1 \leq t \leq h$.

Proof. (a) Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \leq t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. At time $t+1$ there are t steps equal to X_1 , and this number never decreases. So $V_{t+1}(1) = 0$ for $t > h$. Paths with no flip end at $0 \neq h$.

(b) Every transition probability is affine in ε , so C_N and $V_{t+1}(1)$ are polynomials in ε . By (a), $C_N(0) = 0$ and $C'_N(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^N \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4 gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$, so $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2}$. Next $\frac{h-i}{N-i} \leq \frac{1}{2}$, so $w_t = \frac{N+2}{4} \pi_t \leq \frac{N+2}{2(N-t)} t2^{-t} \leq 2t2^{-t}$ for $t \leq h$. $\square$

In words, to first order the walk flips once, at some step $t+1$, and then runs as Pólya's urn from t balls of one color and one of the other. The minority fraction is then close to a Beta(1, t) variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N+2)$. Since $w_t \leq 2t2^{-t}$, only an early flip counts.

2.2 Second order

Theorem 2.6. *Let $N \geq 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon).$

(b) $C_N \leq \frac{4\varepsilon}{N+2} N^{2\varepsilon} e^{5200\varepsilon}$ whenever $N \geq 64$ and $0 < \varepsilon(1 + \log N) \leq \frac{1}{100}$.

In particular $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon \log N)$ in the range of (b), with an absolute implied constant.

In words, the first-order term is correct up to a factor $1 + O(\varepsilon \log N)$, which is $1 + O(L^2/N)$ at the crossing.

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with βs extra balls of each color, where $\beta = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s+10)$ extra balls, where $\kappa = 2\varepsilon/(1-4\varepsilon)$. The step at time s then loses a factor of about $e^{\kappa/s}$, and the product of these factors is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

N	x_N	g_N	$x_N - x_{2\text{nd}}$	$x_N / \log N$	$D(N) \log N$
50	3.940487	4.288636	+0.3452	1.007	-0.083
400	7.629217	7.843768	+0.0621	1.273	-0.199
1600	10.240565	10.340051	+0.0179	1.388	+0.215
3200	11.546834	11.610464	+0.0097	1.431	+0.386
12800	14.158845	14.182921	+0.0028	1.497	+0.607
51200	16.778468	16.786946	+0.0008	1.547	+0.707

Table 1: Exact elephant crossings, the root g_N of $G(g) = \Lambda_N$, and the difference $D(N)$ of Theorem 2.8 times $\log N$. Here $x_{2\text{nd}}$ is the formula of Proposition 2.10 without its error term.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2 \log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \quad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0, \infty)$.

Theorem 2.7. *As $N \rightarrow \infty$ over even N , the following hold.*

- (a) $x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.
- (b) $x_N = 2 \log N - \log \log N - \log 16 + O(\log \log N / \log N)$.

In words, the root g_N locates the crossing to within $O(L^2/N)$, and the expected number of flips at the crossing is about $2 \log N$.

Idea of the proof. Let $F(x)$ be the logarithm of C_N/E_N at $\varepsilon = x/N$. It increases in x , and its only zero is x_N . Theorem 2.6 and the formula for E_N show that $F(x)$ is within $O(L^2/N)$ of $G(x) - \Lambda_N$ when x is of order L . Since $G' \geq 1$, the zero of F is within $O(L^2/N)$ of the zero g_N of $G - \Lambda_N$.

Proof. (a) *Step 1: both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε . By Proposition 2.3, F is strictly increasing and x_N is its only zero. Let $1 \leq x \leq 2\Lambda_N$. Then $x/N \leq 4L/N$, so for large N Theorem 2.6 applies with $\varepsilon = x/N$. As $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$ and $x \geq 1$, it gives $\log C_N = \log x + \log 4 - 2L + O(xL/N)$. Also $\log(1 - y) = -y + O(y^2)$ for $0 \leq y \leq \frac{1}{2}$, so $\log E_N = -\log 2 - (N-1)x/N + O(x^2/N) = -\log 2 - x + O(x^2/N)$. Since $\log 4 + \log 2 = \log 8$ and $x \leq 2\Lambda_N \leq 4L$, we get $|F(x) - G(x) + \Lambda_N| \leq \rho_N$ on $[1, 2\Lambda_N]$, with $\rho_N = O(L^2/N)$.

Step 2: trap the zero. Now $G' \geq 1$ and $G(g_N) = \Lambda_N$. So $F(x) < 0$ for $1 \leq x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \leq 2\Lambda_N$. As $\Lambda_N - \log \Lambda_N \leq g_N \leq \Lambda_N$ (because $G(\Lambda_N - \log \Lambda_N) \leq \Lambda_N \leq G(\Lambda_N)$), both ranges are nonempty for large N . Since F is increasing, its zero x_N lies between them, so $|x_N - g_N| \leq \rho_N$. Then $F(x_N) = 0$ gives $|G(x_N) - \Lambda_N| \leq \rho_N$.

(b) Since $g_N = \Lambda_N - \log g_N$ and $\Lambda_N - \log \Lambda_N \leq g_N \leq \Lambda_N$, we have $\log g_N = \log \Lambda_N + O(\log \Lambda_N / \Lambda_N)$, and $\log \Lambda_N = \log 2 + \log \log N + O(1/\log N)$. So $g_N = 2 \log N - \log 8 - \log 2 - \log \log N + O(\log \log N / \log N)$, which is (b) for g_N , and (a) gives it for x_N . $\square$

Table 1 gives exact crossings on the grid $N = 50 \cdot 2^k$, computed from the recursion for f_N in Lemma 2.2. The limit 2 of $x_N / \log N$ is approached slowly, because of the term $-\log \log N$.

Now we compare with Kagey's board. Doubling (1) without its last two terms gives $2z_N + \log(2z_N) = \Lambda_N + \log(2\pi)$, so $2z_N - \log(2\pi)$ nearly solves $G(x) = \Lambda_N$. What is left is of order $1/L$.

Theorem 2.8. *Let z_N be the crossing value of T for Kagey's board with N steps, as in Section 1.1. As $N \rightarrow \infty$ over even N ,*

$$D(N) := x_N - (2z_N - \log(2\pi)) = \frac{c_*}{\log N} + O\left(\frac{\log \log N}{(\log N)^2}\right), \quad c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4} = 0.668939 \dots$$

In words, the elephant needs twice as many flips as Kagey's board needs switches, less $\log(2\pi) = 1.837 \dots$, and the difference from this tends to 0 like $c_*/\log N$.

Idea of the proof. Both crossings solve an equation of the form $x + \log x = 2L + \text{constant}$ up to small errors. We put the Markov crossing into the equation $G(x) = \Lambda_N$ of the elephant, find that $2z_N - \log(2\pi)$ misses it by c_*/L , and divide by $G' \approx 1$.

Proof. Write $\ell = \log(2\pi)$. By Section 1.1, the error term in (1) is $O(L^{-2})$ and $L/2 \leq z_N \leq L$. So (1) gives $z_N + \frac{1}{2} \log z_N = L + O(1)$ with $\log z_N = \log L + O(1)$, hence $z_N = L - \frac{1}{2} \log L + O(1)$. Since $z_N \geq L/2$, this gives $\frac{1}{8z_N} - \frac{1}{8L} = \frac{L - z_N}{8Lz_N} = O(\log L/L^2)$. Put $Y = 2z_N$. Twice (1) plus $\log 2$, together with $2c_0 + \log 2 = \log \frac{\pi}{4}$, gives $Y + \log Y = 2L + \log \frac{\pi}{4} + \frac{1}{4L} + O(\log L/L^2)$. Also $Y = 2L - \log L + O(1)$, so $1/Y = 1/(2L) + O(\log L/L^2)$ and $\log(1 - \ell/Y) = -\ell/(2L) + O(\log L/L^2)$. Using $\log \frac{\pi}{4} - \ell = -\log 8$, we get

$$G(Y - \ell) = Y + \log Y - \ell + \log\left(1 - \frac{\ell}{Y}\right) = \Lambda_N + \frac{1}{4L} - \frac{\ell}{2L} + O\left(\frac{\log L}{L^2}\right) = \Lambda_N - \frac{c_*}{L} + O\left(\frac{\log L}{L^2}\right).$$

By Theorem 2.7(a), $G(x_N) = \Lambda_N + O(L^2/N)$. Both x_N and $Y - \ell = 2z_N - \log(2\pi)$ are $2L + O(\log L)$, so $G' = 1 + 1/x = 1 + O(1/L)$ between them, and the mean value theorem gives $x_N - (Y - \ell) = c_*/L + O(\log L/L^2)$. $\square$

We computed z_N from the exact law of Kagey's board [2]. The difference $D(N)$ changes sign between $N = 400$ and $N = 1600$, and $D(N) \log N$ is 0.707 at $N = 51200$ against the limit $c_* = 0.669$ (Table 1). So the asymptotic regime is reached slowly.

2.4 The crossing to second order

Theorem 2.6 gives $\log C_N$ up to an error $O(\varepsilon \log N)$. A lower bound that matches part (b) of that theorem brings the error down to $O(\varepsilon)$.

Proposition 2.9 (the center to second order). *Let $N \geq 64$ be even and $0 < \varepsilon(1 + \log N) \leq \frac{1}{100}$. Then*

$$-11\varepsilon - \varepsilon^2(84 \log N + 1.05 \cdot 10^5) \leq \log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N \leq 5200\varepsilon.$$

In particular $|\log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N| \leq 5200\varepsilon$ there.

In words, C_N is $\frac{4\varepsilon}{N+2} N^{2\varepsilon}$ times a factor $1 + O(\varepsilon)$.

Idea of the proof. The proof is in Appendix A. The upper bound is Theorem 2.6(b). For the lower bound we compare M after the first flip with Pólya's urn conditioned to end at h , which we call the bridge. Then $V_{t+1}(1)/\pi_t$ is the mean over the bridge of a product of factors $1 + O(\varepsilon)$, one for each step (Proposition A.5). By Jensen's inequality the mean of the product is at least the exponential of the mean of the sum of the logarithms. The step at time u adds about $2\varepsilon/u$ to this mean, and the sum over $u \leq N$ is about $2\varepsilon \log N$.

With this bound the proof of Theorem 2.7 gives one more term of the crossing.

Proposition 2.10 (the crossing to second order). *As $N \rightarrow \infty$ over even N ,*

$$x_N = g_N - \frac{2g_N \log N + g_N^2/2}{N(1 + 1/g_N)} + O\left(\frac{\log N}{N}\right).$$

So $x_N < g_N$ for large N .

In words, the root g_N overshoots the crossing by about $6 \log^2 N/N$.

Proof. Let F be as in the proof of Theorem 2.7, and let $1 \leq x \leq 2\Lambda_N$ and $\varepsilon = x/N$. For large N Proposition 2.9 applies and gives $\log C_N = \log x + \log 4 - 2L + \frac{2xL}{N} + O(\frac{x}{N})$. For $0 \leq y \leq \frac{1}{2}$ we have $\log(1-y) = -y - \frac{y^2}{2} + O(y^3)$, so $\log E_N = -\log 2 + (N-1) \log(1 - \frac{x}{N}) = -\log 2 - x - \frac{x^2}{2N} + O(\frac{x}{N})$, because x^2/N^2 and x^3/N^2 are $O(x/N)$ for $x \leq 4L$. Hence

$$F(x) = G(x) - \Lambda_N + \frac{2xL + x^2/2}{N} + O\left(\frac{x}{N}\right) \quad (1 \leq x \leq 2\Lambda_N).$$

By Theorem 2.7, x_N lies in this range for large N . So $F(x_N) = 0$ gives $G(x_N) = \Lambda_N - \eta_N$ with $\eta_N = (2x_N L + x_N^2/2)/N + O(L/N)$. By the mean value theorem, $x_N - g_N = -\eta_N/G'(\xi)$ for some ξ between x_N and g_N . Here $G'(\xi) = 1 + 1/\xi = (1 + 1/g_N)(1 + O(1/N))$, since $|\xi - g_N| = O(L^2/N)$ and $g_N \geq L$ for large N . As $\eta_N = O(L^2/N)$, we get $x_N - g_N = -\eta_N/(1 + 1/g_N) + O(L^2/N^2)$. Replacing x_N by g_N in the main term of η_N changes it by $O(L^3/N^2)$. This gives the formula. Its main term is of order L^2/N and is larger than the error for large N , so $x_N < g_N$. $\square$

Proposition 2.9 does not identify the term of order ε . For fixed N and small ε we can find it. By Proposition 2.5(b), C_N is a polynomial in ε , and we write $C_N = c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)$ as $\varepsilon \rightarrow 0$ and $k_2(N) = c_2/c_1$.

Proposition 2.11 (the coefficient of ε^2). *As $N \rightarrow \infty$ over even N ,*

$$k_2(N) - 2 \log N \rightarrow 2\gamma - 2 - \log 2 = -1.5387 \dots$$

In words, for fixed large N and small ε the factor $1 + O(\varepsilon)$ in Proposition 2.9 is about $1 - 1.54\varepsilon$.

Idea of the proof. The proof is in Appendix A. Differentiating the bridge formula at $\varepsilon = 0$ gives $k_2(N) = \sum_t w_t[A_t - (t-1)]$, where A_t is the mean over the bridge of an explicit sum (Lemma A.8). For fixed t and large N the bridge is close to a walk with fair steps, and the sum can be computed in closed form. It is $2 \log N + 2(\gamma - 1 - \log 2) + (t-2)H_{t-1} + o(1)$. The weights w_t tend to $t2^{-t-1}$, and the series over t sums to $2\gamma - 2 - \log 2$.

Remark 2.12 (numerical values and open points). From the Taylor coefficients of C_N , computed in double precision (`verify_elephant_crossing.py`), we get $k_2(N) - 2 \log N = -1.585878, -1.546572$ and -1.541005 at $N = 1600, 12800$ and 51200 . The formula of Proposition 2.10 without its error term is within 0.07 of x_N already at $N = 400$ (Table 1). We do not know an upper bound $\log C_N \leq \log(c_1\varepsilon) + \varepsilon k_2(N) + O(\varepsilon^2(\log N)^k)$, or an explicit N beyond which $g_N > x_N$.

3 The elephant walk near its crossing

We now look at the whole law P_N near the crossing (Figure 2). The picture is that of the introduction. Each flip turns a whole family of steps, most families are small, and the law falls like $x/(2m^2)$ from each end. Lemmas 3.1 to 3.4, the results of Section 3.2 and Theorem 3.9 hold as stated. Elsewhere in this section and in Appendices B and C we assume, unless we say otherwise, that

$$N \geq 64, \quad 0 < \varepsilon \leq \frac{1}{4}, \quad x = N\varepsilon \geq 1.$$

Recall that $P_N(N - m) = \frac{1}{2}[f_N(m) + f_N(N - m)]$. Some numerical statements also use odd N , and there x_N is the value of x at which the two central bins are as likely as the ends (it is unique by Proposition 2.3). For a mode $a^* \geq N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N .

3.1 Families and the first flip

For $k_0 \geq 1$ let $(M_k^{[k_0]})_{k \geq k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M . It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M . We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \geq 2$ and $t \geq j - 1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k + 1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, N - 1$. We write $\mathcal{T}_v$ for the subtree of v , that is, v and its descendants, and we call it the family of v .

Lemma 3.1 (tree and marks). *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k + 1$ is U_k . For $2 \leq v \leq N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \geq k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k + 1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k + 1$ to the root is $k + 1$ followed by the path from U_k , so $\sigma(k + 1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k , $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \dots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \geq k_0$, then $k + 1$ is odd with conditional probability $(1 - \varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. $\square$

So a single flip at v turns every step of its family. Kürsten's coupling [6] uses bond percolation with retention probability $1 - 2\varepsilon$ instead, which gives the same law of the walk. For $k \geq v$, if $\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k + 1$ joins $\mathcal{T}_v$ with probability s/k . This is the urn of Lemma 2.4, so $|\mathcal{T}_v|$ has the law β_v and mean N/v , and on average there are $N/(d(d + 1))$ families of size d . If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N , and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m + 1))$, for $1 \leq m \leq N - 1$ (at $m = h$ this is $c_1\varepsilon$ of Proposition 2.5). This is also proved in Lean (Section 6).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting). *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \leq v \leq N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S , the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v , which is uniform on $[v - 1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \geq 2$ has its parent outside S . So, given this event, each element of S other than v picks its parent freely among the smaller elements of S , v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. $\square$

Lemma 3.3 (the first flip). *Let $V = \min\{v \geq 2 : \mu_v = 1\}$, with $V = \infty$ if there is no mark, and let $D = |\mathcal{T}_V|$. Then $\mathbb{P}(V = j, D = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \geq 1$*

$$f_N(m) = \sum_{j=2}^N \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d) \mathbb{P}(d - Y_d + Z_{N-d}^{(j)} = m),$$

where Y_d and $Z_{N-d}^{(j)}$ are independent.

Proof. Marks and tree are independent, which gives the law of (V, D) . Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is odd exactly when the path from u up to j , with j excluded, has an even number of marks. So by Lemma 3.1 for the tree on S rooted at j , the number of odd vertices in S has the law of $d - Y_d$. Paths from outside S stay outside S , and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 3.1 its odd count has the law of $Z_{N-d}^{(j)}$. The two counts are independent, and their law depends on S only through d . Sum over j and d , and note that $M_N = 0$ if there is no mark. $\square$

In words, the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1 . Only the law of each pair $(Y_d, Z_{N-d}^{(j)})$ enters the lemma. So from now on $(Y_d)_{d \geq 1}$ is one path of $M^{[1]}$ and $(Z_t^{(j)})_{t \geq j-1}$ one path of $M^{[j-1]}$, independent of each other and of (V, D) .

Lemma 3.4 (unimodality given the first step). *The function f_N is unimodal.*

This is shown in the proof of Theorem 1.1 of Guérin, Laulin, Raschel and Simon [10], for every N and every memory parameter $1 - \varepsilon$, and it is also proved in Lean (Section 6). So f_N is nondecreasing on $\{0, \dots, m_h\}$ and nonincreasing on $\{m_h, \dots, N\}$. With a fair first step, P_N mixes f_N and its mirror image. When the mode of f_N is far from the center, the mixture can have two humps, one near each end.

3.2 The end and its neighbor

We first say why the end should fall below its neighbor once $x > 2$. Given $X_1 = +1$, the end needs no flip at all. Its neighbor, bin $N-1$, needs one flip that is never copied. A flip at step j is never copied with probability about j/N . Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

Theorem 3.5 (endpoint dip). *Let $N \geq 3$ and $0 < \varepsilon \leq \frac{1}{2}$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\frac{x}{2(1-\varepsilon)} < \rho_1 \leq \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \geq 2(1-\varepsilon)$, and strictly above it if the right side is below 1.

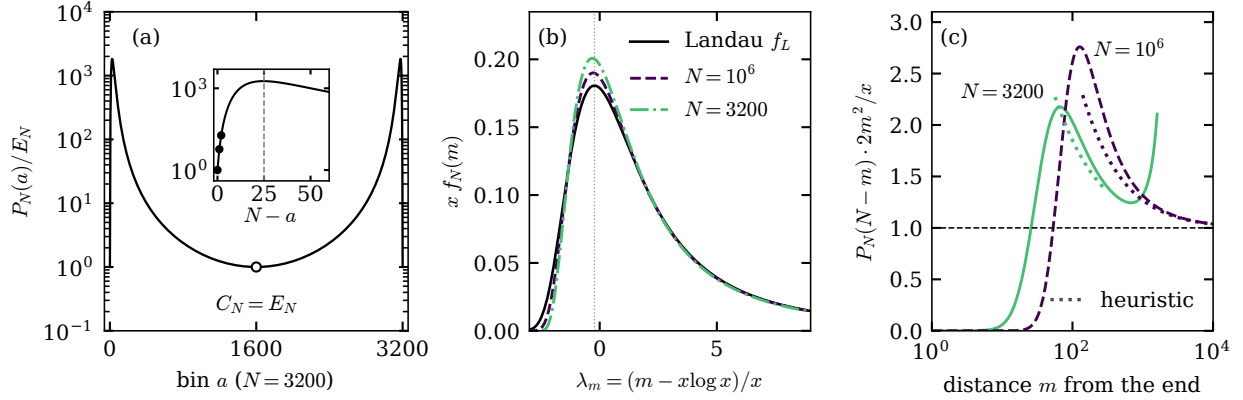


Figure 2: The elephant walk at its crossing. (a) The law P_N at $N = 3200$ and its crossing $x = 11.5468$, divided by the end E_N . The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins N , $N - 1$ and $N - 2$. The end is below its neighbor, and the modes are 25 steps from each end (dashed). (b) The law of M_N , scaled as $x f_N(m)$, against $\lambda_m = (m - x \log x)/x$, with the Landau density f_L . The dotted line is its mode λ_0 . (c) The tail $P_N(N - m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem 3.14, and the dotted curves are the heuristic correction $1 + x(2 \log m + 2\gamma - 3)/m$ for $2x \log x \leq m \leq N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N - m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

In words, when ε is small the neighbor of the end is about $x/2$ times as likely as the end.

Proof. By Lemma 2.2, $\rho_1 = (f_N(1) + f_N(N - 1))/f_N(0)$ with $f_N(0) = (1 - \varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up once, from time $j - 1$ to time j for some $2 \leq j \leq N$, and then stays at 1. Since $r_t(0) = \varepsilon$ and $1 - r_k(1) = (1 - \varepsilon)(1 - c/k)$ with $c = (1 - 2\varepsilon)/(1 - \varepsilon) \in [0, 1]$,

$$f_N(1) = \varepsilon(1 - \varepsilon)^{N-2} \sum_{j=2}^N P_j, \quad P_j = \prod_{k=j}^{N-1} \left(1 - \frac{c}{k}\right).$$

Since $c \leq 1$, $P_j \geq \prod_{k=j}^{N-1} (1 - 1/k) = (j - 1)/(N - 1)$, and these lower bounds sum to $N/2$. So $f_N(1)/f_N(0) \geq x/(2(1 - \varepsilon))$. Next, $1 - c/k \leq e^{-c/k}$ and $\sum_{k=j}^{N-1} 1/k \geq \log(N/j)$ give $P_j \leq (j/N)^c$. As $(t/N)^c$ is nondecreasing in t ,

$$\sum_{j=2}^N P_j \leq \int_2^{N+1} \left(\frac{t}{N}\right)^c dt \leq \frac{N(1 + 1/N)^{c+1}}{c + 1} \leq \frac{N(1 + 1/N)^2(1 - \varepsilon)}{2 - 3\varepsilon},$$

using $c \leq 1$ and $1/(c + 1) = (1 - \varepsilon)/(2 - 3\varepsilon)$. So $f_N(1)/f_N(0) \leq x(1 + 1/N)^2/(2 - 3\varepsilon)$.

The term $f_N(N - 1)$. The event $M_N = N - 1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \geq 2$ has probability $r_k(k - 1) = 1 - \varepsilon - (1 - 2\varepsilon)/k \in [\frac{1}{2}, 1 - \varepsilon]$. So $0 < f_N(N - 1) \leq \varepsilon(1 - \varepsilon)^{N-2}$. This makes the lower bound strict and gives the term $\varepsilon/(1 - \varepsilon)$ in the upper bound. $\square$

Corollary 3.6 (the dip at the crossing). *Let $N \geq 4$ be even and suppose that $x_N \geq 2(1 - \varepsilon_N)$. Then at the crossing $E_N = C_N < P_N(N - 1)$, and P_N is not unimodal.*

Proof. By Proposition 2.3, $\varepsilon_N < \frac{1}{2}$, so Theorem 3.5 gives $P_N(N-1) > E_N = C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h , so we may replace a_0 by $N - a_0$ and assume $a_0 \leq h$. Then $P_N(a) \leq C_N$ for every $a \geq h$, including $a = N - 1$, which is a contradiction. $\square$

The hypothesis $x_N \geq 2(1 - \varepsilon_N)$ says that $\varepsilon_N \geq 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \leq E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \geq 8$. For $N \geq 1688$ we use Theorem 2.6(b). For smaller N that theorem does not apply, and we compute C_N/E_N in floating-point arithmetic and bound the rounding error. At $\varepsilon = 2/(N+2)$ the transition probabilities of M are ratios of integers, namely

$$1 - r_t(b) = \frac{N(t-b) + 2b}{(N+2)t}, \quad r_t(b) = \frac{2(t-b) + Nb}{(N+2)t}. \quad (2)$$

By Lemma 2.2, $C_N/E_N = 2f_N(h)/f_N(0)$. Write $f_t(b) = \mathbb{P}_+(M_t = b)$ for the law of M_t at this ε , for $1 \leq t \leq N$. The law of M_{t+1} follows from the law of M_t by

$$f_{t+1}(b) = f_t(b)(1 - r_t(b)) + f_t(b-1)r_t(b-1), \quad f_1(0) = 1. \quad (3)$$

Here $f_t(b) > 0$ exactly for $0 \leq b \leq t-1$. So the first term is absent for $b = t$ and the second for $b = 0$. The values $f_{t+1}(b)$ with $b \leq h$ only need the values $f_t(b)$ with $b \leq h$. Every number in (3) is positive, and nothing is subtracted. This is what makes the rounding error easy to bound.

Lemma 3.7 (rounding error). *Let $N \geq 2$ be even with $N(N+2) < 2^{53}$, let $\varepsilon = 2/(N+2)$, and put $u = 2^{-53}$. Compute numbers $\hat{f}_t(b)$ for $1 \leq t \leq N$ and $0 \leq b \leq \min(t-1, h)$ in IEEE 754 double precision with rounding to nearest, as follows. Put $\hat{f}_1(0) = 1$. In (3) each weight is one division of the two integers in (2), each term on the right is one multiplication of a computed value $\hat{f}_t$ by a computed weight, and if both terms are present they are added by one addition. Finally let $\hat{R}$ be twice the computed quotient of $\hat{f}_N(h)$ by $\hat{f}_N(0)$. Suppose that no computed weight, product or sum in the recursion is below 2^{-1000} . Then*

$$(1-u)^{6N-5} \leq \frac{C_N/E_N}{\hat{R}} \leq (1-u)^{-(6N-5)}.$$

For $N < 1688$ the right side is at most $1 + 3 \cdot 10^{-12}$.

In words, for $N < 1688$ the computed ratio is correct to about 11 digits, because each of the $N-1$ steps of the recursion rounds each term three times and all terms are positive.

Proof. We use the standard model of floating-point arithmetic [27, Section 2.2]. If a and b are floating-point numbers and $\circ$ is addition, multiplication or division, then the computed value of $a \circ b$ is $(a \circ b)(1 + \delta)$ with $|\delta| \leq u$, as long as the result does not overflow or underflow. Underflow means that the exact result is nonzero and below the smallest normal number 2^{-1022} , and then the rounded result is at most 2^{-1022} . So a computed result that is at least 2^{-1000} did not underflow, and by hypothesis this covers every operation in the recursion. The integers in (2) are positive and at most $N(N+2) < 2^{53}$, so they are floating-point numbers, and each computed weight is the true weight times some $1 + \delta$.

We claim that $\hat{f}_t(b) = f_t(b)(1 + \vartheta)$ with $(1-u)^{3(t-1)} \leq 1 + \vartheta \leq (1+u)^{3(t-1)}$. We use induction on t , and the case $t = 1$ is clear. In the step from t to $t+1$, each term on the right of (3) is computed as $\hat{f}_t(b')$ times the true weight times $(1 + \delta_1)(1 + \delta_2)$, and the sum of two terms gets one more factor $1 + \delta_3$. All numbers are positive. So $\hat{f}_{t+1}(b)$ lies between $(1-u)^3$ and $(1+u)^3$ times

the right side of (3) with $\hat{f}_t$ in place of f_t , and the claim follows. Nothing overflows on the way, because the true weights and the $f_t(b)$ are at most 1 and $(1+u)^{3N} < 2$.

In the last division both numbers lie between 2^{-1000} and 2. So the quotient lies between 2^{-1001} and 2^{1001} , it neither underflows nor overflows, and the division adds one more factor $1 + \delta$. Doubling is exact. Hence $(C_N/E_N)/\hat{R}$ lies between $(1-u)^{3(N-1)}(1+u)^{-3(N-1)-1}$ and $(1+u)^{3(N-1)}(1-u)^{-3(N-1)-1}$. Since $1+u \leq (1-u)^{-1}$, this gives both bounds with the exponent $6(N-1)+1 = 6N-5$. For $N \leq 1686$ put $k = 6N-5 \leq 10111$. Then $(1-u)^{-k} \leq (1-ku)^{-1} \leq 1+2ku \leq 1+3 \cdot 10^{-12}$. $\square$

Proposition 3.8 (the hypothesis holds from $N = 8$ on). *For every even $N \geq 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \geq 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2, 4$ and 6 the hypothesis of Corollary 3.6 fails.*

In words, from $N = 8$ on the end is below its neighbor at the crossing, so the law has a hump on each side of the center.

Proof. The range $N \geq 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \leq \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \leq \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \geq -y - y^2$ for $0 \leq y \leq \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1 - \frac{2}{N})^{N-1} \geq \frac{1}{2}e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \leq N < 1688$. The script `certify_elephant.py` runs the computation of Lemma 3.7 for each of the 840 even N in this range, with exactly the operations listed there (elementwise operations on arrays of doubles, with no fused operation, and for a sample of N the same bits come out of plain scalar arithmetic). The smallest weight, product or sum that occurs is above 10^{-9} , so the lemma applies. The largest computed ratio is $\hat{R} = 0.947307\dots$, at $N = 8$, and the next largest is $0.729947\dots$, at $N = 10$. By Lemma 3.7, $C_N/E_N \leq 0.9474$ for every even $8 \leq N < 1688$.

The three smallest N . At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2, 4$ and 6 , so $\varepsilon_N < 2/(N+2)$ there. The last claim of the proposition about $N \geq 8$ is Corollary 3.6. $\square$

As an independent check, the same script runs (3) in exact integer arithmetic for 20 values of N between 8 and 1686. In each case $C_N < E_N$ exactly, and the double-precision ratio differs from the exact one by a relative error below 10^{-13} . At $N = 8$ the exact ratio is $\frac{67903}{71680}$. By Lemma 3.4, P_N is always a mixture of two unimodal laws (namely f_N and its mirror image), as is the limit law in [10, eq. (1)].

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Let CP_N be the law of $\sum_{d=1}^{N-1} d\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. On average $x/(d(d+1))$ flipped vertices have a family of size d . If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The law CP_N is a known one. Without the truncation at $d = N-1$ its generating function is $(1-s)^{x(1-s)/s}$, which is the Lea–Coulson form of the Luria–Delbrück distribution with parameter x [15, Eq. (14)]. Its limit after centering at $x \log x$ and dividing by x is known as well (see [15, Eq. (19)] and [16]), and the same stable law appears for the root cluster of a random recursive

tree [14]. So the limit law in the two theorems below is not new. What they add is the comparison with the elephant walk in sup norm, the explicit errors, and (in Lemma 3.11) the location of the mode.

Theorem 3.9 (compound Poisson bulk). *For $N \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$,*

$$\sup_{m \geq 0} |f_N(m) - \text{CP}_N(m)| \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}.$$

In words, bin by bin the law of the number of disagreeing steps is within δ_B of the compound Poisson law, and $\delta_B = O(x^2 \log^2 N/N)$ when $x^2 \leq N$.

Idea of the proof. The proof is in Appendix C. Let Z_+ be the total size of the flipped families. Then $M_N = Z_+$ unless some flipped step has a flipped descendant, and this has probability at most $x^2 \log N/N$. Given the tree, Z_+ is a sum of independent terms, so its characteristic function is a product, and the product is close to $e^{\varepsilon F}$ with $F = \sum_v (e^{i\theta|\mathcal{T}_v|} - 1)$. The mean of εF is exactly the logarithm of the characteristic function of CP_N . So it remains to show that F is close to its mean. For this we use the Efron–Stein inequality [25]. Changing the parent of one vertex changes the family sizes only along two paths to the root, and these paths have length of order $\log N$.

The families of size up to about x add up to nearly $x \log x$, and the few families of size of order x give fluctuations of order x . So we center M_N at $x \log x$ and divide by x . Put $\lambda_m = (m - x \log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)} dt, \quad B(t) = -it \log|t| - \frac{\pi}{2}|t|.$$

This is the Landau density, a stable density with index one. We show in Appendix C that e^B is the characteristic function of a probability law of class L , so that f_L is a unimodal probability density by [24].

Theorem 3.10 (Landau limit). *There is an absolute constant C such that the following hold under the standing assumptions of this section.*

- (a) *For every integer $m \geq 0$, $|x \text{CP}_N(m) - f_L(\lambda_m)| \leq C(\frac{1+\log x}{x} + \frac{x}{N})$.*
- (b) *For every integer $m \geq 0$, $|x f_N(m) - f_L(\lambda_m)| \leq C(\frac{1+\log x}{x} + \frac{x}{N}) + x\delta_B$.*
- (c) *If $x \rightarrow \infty$ and $x^3 \log^2 N/N \rightarrow 0$, then $(M_N - x \log x)/x$ converges in law to the law with density f_L . Both conditions hold at the crossing.*

In words, near the crossing the number of disagreeing steps is about $x \log x$, with fluctuations of order x that follow the Landau law, bin by bin.

Idea of the proof. The proof is in Appendix C. If we allow every jump size $d \geq 1$, the characteristic function of the compound Poisson law is $e^{x\chi(e^{i\theta})}$ with $\chi(s) = (1-s) \log(1-s)/s$, and the jumps $d \geq N$ change it little. For small θ we have $\chi(e^{i\theta}) = -i\theta \log(-i\theta) + O(\theta^2 |\log|\theta||)$. With $\theta = t/x$ the exponent becomes $B(t) + it \log x$ plus an error of order $t^2 \log x/x$. The term $it \log x$ is the centering at $x \log x$. Both characteristic functions decay like $e^{-c|t|}$, so Fourier inversion turns this into a bound for each bin. Part (b) then follows from Theorem 3.9.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.11 (the mode of the Landau density). (a) The density f_L has exactly one mode λ_0 , and $-0.22278299 < \lambda_0 < -0.22278297$.

(b) $f'_L(\lambda_0) = 0$, and $-0.0792 \leq f''_L(\lambda) \leq -0.0786$ for $-0.2229 \leq \lambda \leq -0.2227$.

(c) There is a constant $\kappa_L > 0$ with $f_L(\lambda_0) - f_L(\lambda) \geq \kappa_L \min\{1, (\lambda - \lambda_0)^2\}$ for every real λ .

The proof is in Appendix C. It is computer-assisted. We turn the Fourier integral into the real integral $f_L(\lambda) = \frac{1}{\pi} \int_0^\infty e^{-s \log s - \lambda s} \sin(\pi s) ds$. Then we enclose f'_L at two points and f''_L on the interval between them in ball arithmetic, with the rigorous integration algorithm of [28] on $[2^{-20}, 30]$ and explicit bounds for the rest (`certify_elephant.py`). Part (c) follows from (b) and the unimodality of f_L .

Corollary 3.12 (where the modes are). Let $N \rightarrow \infty$ over even N . At the crossing, with $x = x_N$, every mode $a^* \geq N/2$ of P_N has

$$m^* = x \log x + \lambda_0 x + O(\sqrt{x \log x}),$$

and so has the largest mode m_h of f_N .

In words, each hump of the law at the crossing sits at distance about $x \log x - 0.2228x$ from its end.

Remark 3.13 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\text{CP}_N(m) - 1| \leq 3.5 \cdot 10^{-4}$ for $1 \leq m \leq 10^4$. At every tested crossing m^* equals the mode of CP_N . The error term in Corollary 3.12 is not effective. Its proof needs the bound of Theorem 3.10(b) to be below $\kappa_L/2$, where κ_L is the constant of Lemma 3.11(c), and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

Theorem 3.14 (the tail). Let N be even, and let $x = N\varepsilon \geq 1$ depend on N with $x \log^2 N/N \rightarrow 0$. Let $\omega_N \rightarrow \infty$ with $\omega_N x \log^2(x+1) = o(N)$, and let $\delta'_N \rightarrow 0$. Then

$$f_N(m) = \frac{x}{m(m+1)}(1 + o(1)) \quad \text{and} \quad P_N(N-m) = \frac{x}{2m^2}(1 + o(1)),$$

uniformly for $\omega_N x \log^2(x+1) \leq m \leq \delta'_N N$.

In words, the chance of landing at distance m from an end is the chance that a flipped family has m steps, halved because the first step must point to that end.

Idea of the proof. The proof is in Appendix B. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.4), while the other flips change the count only by a relative amount of order $x \log m/m$. For the upper bound we show $\mathbb{P}(M_N \geq m) \leq \frac{x}{m}(1 + O(x \log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N-m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the exact ratio $P_N(N-m)/(x/m^2)$ is 0.51889. At moderate m , CP_N suggests $f_N(m)m(m+1)/x \approx 1 + x(2 \log m + 2\gamma - 3)/m$, and this factor is still about 2 near $m = 70$ when $N = 1600$ (Figure 2(c)).

3.5 The center

Theorem 3.15 (shape at the crossing). *There are absolute constants C_w and N_0 with the following property. Let $N \geq N_0$ be even with $x_N \in [2, N^{1/4}]$. Then at the crossing $P_N(a) > C_N$ for every $1 \leq a \leq N-1$ with $|a-h| \geq C_w(Nx_N \log N)^{1/2}$.*

By Theorem 2.7, $x_N \in [2, N^{1/4}]$ for large N , so at the crossing the window has width of order $N^{1/2} \log N$. In words, the middle bin is lower than every interior bin except possibly those within about $N^{1/2} \log N$ of it.

Idea of the proof. Up to the mode of f_N the law is above $f_N(1)/2$, which beats $C_N = E_N$ by Theorem 3.5. Farther in, it is close to $x/(2m^2)$, far above the center $4x/N^2$. Near the middle, the main term $x\Pi(m)$ below exceeds its value at h by a factor of about $1 + 12t^2/N^2$ at distance t from h , while we know C_N and the law only up to relative errors of order $x \log N/N$. The two meet at t of order $(Nx \log N)^{1/2}$.

Proof. Write $x = x_N$ and $\varepsilon = \varepsilon_N$, so that $C_N = E_N$. By symmetry we take $a = N - m$ with $1 \leq m \leq h$, and put $t = h - m$. Since $2 \leq x \leq N^{1/4}$, every error term below is bounded by a function of N alone, and we take N large. Put $\Pi(m) = \frac{1}{2} \left[\frac{1}{m(m+1)} + \frac{1}{(N-m)(N-m+1)} \right]$, so that $x\Pi(h) = \frac{4\varepsilon}{N+2}$. As $\varepsilon(1 + \log N) \rightarrow 0$, Theorem 2.6(b) gives $C_N \leq x\Pi(h)(1 + \eta)$ with $\eta = N^{2\varepsilon} e^{5200\varepsilon} - 1$. Put $y = \varepsilon(2 \log N + 5200)$. Then $y \leq \frac{1}{2}$ for large N , and $e^y - 1 \leq 2y$ for $0 \leq y \leq \frac{1}{2}$, so $\eta \leq 2y \leq 10^4 \varepsilon \log N$.

Step 1: $m \leq m_h$. Then $f_N(m) \geq f_N(1)$ by Lemma 3.4. The proof of Theorem 3.5 gives $f_N(1) \geq \frac{x}{2(1-\varepsilon)} f_N(0)$, and $f_N(0) = 2E_N = 2C_N$. So $P_N(N-m) \geq \frac{1}{2} f_N(m) \geq \frac{x}{2(1-\varepsilon)} C_N > C_N$, since $x \geq 2$.

Step 2: $m_h < m \leq N/8$. Then $f_N(m) \geq f_N(k)$ with $k = \lfloor N/8 \rfloor$, by Lemma 3.4. Proposition B.5(a) at k gives $f_N(k) \geq \frac{x}{k(k+1)} (1 - O(\frac{x \log N}{N}))$. So $P_N(N-m) \geq \frac{32x}{N(N+8)} (1 - o(1))$, which exceeds $\frac{4x}{N(N+2)} (1 + \eta) \geq C_N$.

Step 3: $N/8 < m \leq h$. Then m and $N-m$ are at least $N/8$, and Proposition B.5 at m and at $N-m$ gives $P_N(N-m) \geq x\Pi(m)(1 - \bar{e})$ with $\bar{e} = O(x \log N/N)$. Put $\bar{N} = N+1$ and $s = 2t/\bar{N}$, so that $m + \frac{1}{2} = \frac{\bar{N}}{2}(1-s)$ and $N-m + \frac{1}{2} = \frac{\bar{N}}{2}(1+s)$. Since $\frac{1}{k(k+1)} \geq (k + \frac{1}{2})^{-2}$ and $\Pi(h) = \frac{1}{h(h+1)} = (\frac{\bar{N}}{2})^{-2} (1 - \bar{N}^{-2})^{-1}$, we get

$$\frac{\Pi(m)}{\Pi(h)} \geq \frac{1}{2} [(1-s)^{-2} + (1+s)^{-2}] (1 - \bar{N}^{-2}) \geq (1 + 3s^2)(1 - \bar{N}^{-2}),$$

because $\frac{1}{2} [(1-s)^{-2} + (1+s)^{-2}] = \sum_{k \geq 0} (2k+1)s^{2k}$. With $b = \bar{e} + \bar{N}^{-2} \leq \frac{1}{2}$ this gives $P_N(N-m) \geq x\Pi(h)(1 + 3s^2)(1 - b) \geq x\Pi(h)(1 + \frac{3}{2}s^2 - b)$.

Step 4: the window. The bound of Step 3 exceeds $x\Pi(h)(1 + \eta) \geq C_N$ as soon as $\frac{3}{2}s^2 > \eta + b$. Now $\eta + b \leq A_2 x \log N/N$ for an absolute constant A_2 , since $x \geq 2$, and $s \geq t/N$. So the condition holds when $t \geq C_w(Nx \log N)^{1/2}$ with $C_w^2 > 2A_2/3$. $\square$

We did not compute C_w or N_0 . Inside the window the proof gives nothing, and closing it needs second differences of the law.

Conjecture 3.16 (convexity between the humps). *For every $N \geq 10$ and $x = x_N$, the second difference $P_N(a-1) - 2P_N(a) + P_N(a+1)$ is positive for $2m^* + 1 \leq a \leq N - 2m^* - 1$.*

With the symmetry of the law, the conjecture makes P_N strictly convex on $[2m^*, N - 2m^*]$, so h is its unique minimizer there (the two central bins for odd N). The next corollary shows that $m^* = o(N)$ at the crossing, so together with Theorem 3.15 the conjecture would make the center the unique interior minimum for large even N .

Corollary 3.17 (the modes). *For large even N the modes a^* of P_N at the crossing satisfy $\min(a^*, N - a^*) = O(x_N \log^2 x_N)$.*

Proof. Write $x = x_N$ and $\varepsilon = \varepsilon_N$. By Theorem 2.7, $x \sim 2 \log N$, so for large N the hypotheses of Theorem 2.6(b) and Lemma B.7 hold. The law is symmetric about h , so $m^* = \min(a^*, N - a^*)$ maximizes $f_N(m) + f_N(N - m)$ over $0 \leq m \leq h$. Lemma B.7 gives $m_h \leq h$ for large N . So Lemma 3.4 and Theorem 2.6(b) give $f_N(N - m) \leq f_N(h) = C_N = O(x/N^2)$ for $m \leq h$, and hence $f_N(m^*) \geq f_N(m_h) - C_N$. Let ℓ be as in the proof of Lemma B.7. Then $f_N(m_h) \geq f_N(\ell) \geq \frac{x}{2\ell(\ell+1)}$, so $f_N(m^*) \geq \frac{x}{4\ell(\ell+1)}$ for large N . If $m^* > m' = \max(m_h, \ell)$, then Lemmas B.1 and B.2 give $(m^* - m' + 1)f_N(m^*) \leq \mathbb{P}(M_N \geq \ell) \leq xH_\ell/\ell$. Hence $m^* \leq \max(m_h, \ell) + 4(\ell + 1)H_\ell$ in every case. Finally Lemma B.7 gives $m_h = O(x \log^2 x)$, and $\ell = O(x \log x)$ and $H_\ell = O(\log x)$. $\square$

The conjecture holds for every $10 \leq N \leq 600$ and for $N = 800, 1600, \dots, 25600$. It leaves out the two endpoints $a = 2m^*$ and $a = N - 2m^*$, because there the second difference is not positive for some $N \leq 51$. Heuristically, a concave stretch of f_N far from its mode is added to the tail $x/(2r^2)$ of the other branch, with $r = N - a$, and the second difference $3x/r^4$ of this tail beats the concave curvature, which is of order $x^2/(N^2 r^3)$. We leave open the window around the center, the tail between $x \log x$ and $x \log^2 x$, and effective constants in Theorems 3.14 and 3.15.

4 Heavy-tailed runs

In this section the walk remembers the time since its last switch. The steps form maximal runs whose directions alternate. The first run has length D , and the later runs have i.i.d. lengths $\xi_1, \xi_2, \dots$ distributed as ξ , where

$$\bar{F}(k) = \mathbb{P}(\xi \geq k) = k^{-\alpha} \quad (k \geq 1)$$

for some $\alpha > 0$. So $\mathbb{P}(\xi = k) = k^{-\alpha} - (k+1)^{-\alpha} > 0$ for every $k \geq 1$. The first direction is fair, and D , (ξ_i) and the first direction are independent. With a fresh start D has the law of ξ . For $\alpha > 1$ the stationary start is $\mathbb{P}(D = d) = \bar{F}(d)/\mu$, where $\mu = \mathbb{E}\xi = \zeta(\alpha)$. This law is stationary in the following sense. The number of steps left in the current run, counting the current one, is a Markov chain that moves from $k \geq 2$ to $k - 1$ and from 1 to a fresh copy of ξ . Since $\bar{F}(k) = \bar{F}(k+1) + \mathbb{P}(\xi = k)\bar{F}(1)$, the law $\bar{F}(k)/\mu$ is invariant for this chain. So the stationary start sees the runs at a typical time. We define A_N , P_N , C_N and E_N as in Section 1.2, put $R_N = E_N/C_N$, and write $\mathbb{P}_+$ for the law given that the first run is $+$. In this section the constants implied by $O(\cdot)$ may depend on α .

Proposition 4.1 (the edge). *For every law of D , $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$. With a fresh start $E_N = \frac{1}{2}N^{-\alpha}$. With the stationary start*

$$E_N = \frac{\zeta(\alpha, N)}{2\zeta(\alpha)} = \frac{N^{1-\alpha}}{2(\alpha-1)\zeta(\alpha)} \left(1 + \frac{\theta_N(\alpha-1)}{N}\right)$$

for some $\theta_N \in [0, 1]$, where $\zeta(\alpha, N) = \sum_{k \geq N} k^{-\alpha}$ is the Hurwitz zeta function.

Proof. All N steps are $+1$ exactly when the first direction is $+$ and $D \geq N$. Hence $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$, and $\mathbb{P}(\xi \geq N) = N^{-\alpha}$. For the stationary start $\mathbb{P}(D \geq N) = \zeta(\alpha, N)/\zeta(\alpha)$. Since $t^{-\alpha}$ is decreasing, $\int_N^\infty t^{-\alpha} dt \leq \zeta(\alpha, N) \leq N^{-\alpha} + \int_N^\infty t^{-\alpha} dt$, and the integral equals $N^{1-\alpha}/(\alpha-1)$. $\square$

So the edge only sees the first run, while the center depends on all the runs. For the center we need the renewal quantities

$$\begin{aligned} \tau_m &= \xi_1 + \dots + \xi_m, \quad \tau_0 = 0, \quad u_m(k) = \mathbb{P}(\tau_m = k), \\ q_m(k) &= \mathbb{P}(\tau_{m-1} < k \leq \tau_m) = \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j) \end{aligned}$$

for $m, k \geq 1$. So $u_m(k)$ is the chance that the m -th run ends at level k , and $q_m(k)$ the chance that the m -th run covers level k . For the delayed versions let D be independent of (ξ_i) , and put $u_m^D(k) = \mathbb{P}(D + \tau_{m-1} = k)$, $q_1^D(k) = \mathbb{P}(D \geq k)$ and $q_m^D(k) = \mathbb{P}(D + \tau_{m-2} < k \leq D + \tau_{m-1})$ for $m \geq 2$. For the stationary start $u_m^D(k) = \sum_d \frac{\bar{F}(d)}{\mu} u_{m-1}(k-d) = q_m(k)/\mu$.

The $+$ runs and the $-$ runs form two independent renewal processes. The walk is in bin a at time N exactly when one of them sits at its level and the other is inside a run that covers its level.

Proposition 4.2 (two renewal processes). *Let D have any law on $\{1, 2, \dots\}$. For $a + b = N$ with $a, b \geq 1$,*

$$\mathbb{P}_+(A_N = a) = \sum_{m \geq 1} \left[u_m(b) q_{m+1}^D(a) + u_m^D(a) q_m(b) \right],$$

and $\mathbb{P}_+(A_N = N) = \mathbb{P}(D \geq N)$. So for even N and $h = N/2$, $C_N = \sum_{m \geq 1} [u_m(h) q_{m+1}^D(h) + u_m^D(h) q_m(h)]$. For the fresh start this is $C_N = \sum_m u_m(h) [q_m(h) + q_{m+1}(h)]$.

Proof. Let the $+$ runs have lengths $D, \xi_2^+, \xi_3^+, \dots$ and the $-$ runs $\xi_1^-, \xi_2^-, \dots$, and let $s_j^\pm$ be the sum of the first j lengths of each sign, with $s_0^\pm = 0$. So s_j^+ has law u_j^D and s_j^- has law u_j , and the two sequences are independent. The j -th $+$ run covers the steps in $(s_{j-1}^+ + s_{j-1}^-, s_j^+ + s_{j-1}^-]$, and the j -th $-$ run covers $(s_j^+ + s_{j-1}^-, s_j^+ + s_j^-]$. We split according to the run that contains step $N = a + b$. If it is the first run, then $A_N = N$, and this happens exactly when $D \geq N$. If it is the j -th $+$ run with $j \geq 2$, then $A_N = N - s_{j-1}^-$, so $A_N = a$ means $s_{j-1}^- = b$ and $s_{j-1}^+ < a \leq s_j^+$. This has probability $u_{j-1}(b) q_j^D(a)$. If it is the j -th $-$ run, then $A_N = s_j^+$, so $A_N = a$ means $s_j^+ = a$ and $s_{j-1}^- < b \leq s_j^-$. This has probability $u_j^D(a) q_j(b)$. The cases are disjoint. Summing them, with $m = j-1$ in the second and $m = j$ in the third, gives the formula. For even N changing the sign of every step gives $C_N = \frac{1}{2}[\mathbb{P}_+(A_N = h) + \mathbb{P}_+(A_N = N - h)] = \mathbb{P}_+(A_N = h)$. $\square$

This identity is the coefficient form of the double transform of Godrèche and Luck [12, Section 7] for the occupation time of an alternating renewal process. To evaluate it we need the stable limit of τ_m and its local form.

Lemma 4.3 (stable and local limits). *Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. Put $K_\alpha = \Gamma(2-\alpha)/(\alpha-1) = -\Gamma(1-\alpha) > 0$ and $\sigma_\alpha = 2K_\alpha |\cos \frac{\pi\alpha}{2}|$.*

(i) $\mathbb{E} e^{i\theta\xi} = 1 + i\mu\theta + K_\alpha(-i\theta)^\alpha + O(\theta^2)$ as $\theta \rightarrow 0$, with the principal branch.

(ii) $(\tau_m - m\mu)/a_m$ converges in law to a variable G with $\mathbb{E} e^{-sG} = e^{K_\alpha s^\alpha}$ for $s \geq 0$ and $\mathbb{E} e^{i\theta G} = \exp(K_\alpha(-i\theta)^\alpha)$. The law of G is totally skewed to the right and has a bounded, uniformly continuous density g with $g(z) \rightarrow 0$ as $|z| \rightarrow \infty$.

α	$e(\alpha)$	$C(\alpha)$	$C_{\text{fr}}(\alpha)$	R_{1600}	R_{10^4}	R_{10^5}
1.3	+0.469231	0.422717	0.498631	17.7593 [1.3180]	38.7537 [1.2171]	106.149 [1.1317]
1.5	+0.166667	0.647971	0.846371	2.14409 [0.9675]	2.9528 [0.9818]	4.37642 [0.9914]
φ	0	0.776132	1.073675	0.661763 [0.8526]	0.697331 [0.8985]	0.727689 [0.9376]
5/3	-1/15	0.835101	1.182237	0.411815 [0.8064]	0.389307 [0.8614]	0.352822 [0.9102]
1.7	-0.111765	0.880046	1.265508	0.298369 [0.7733]	0.262005 [0.8334]	0.215839 [0.8881]
1.9	-0.373684	1.438986	2.266075	0.0449706 [0.4923]	0.0256505 [0.5569]	0.0121612 [0.6242]

Table 2: Constants of Corollary 4.5 and exact values of $R_N = E_N/C_N$ for the stationary start, with $R_N/(C(\alpha)N^{e(\alpha)})$ in brackets.

(iii) (Gnedenko) The constant $\bar{u} := \sup_{m \geq 1} \sup_k a_m u_m(k)$ is finite, and $a_m u_m(k) - g((k - m\mu)/a_m) \rightarrow 0$ uniformly in k as $m \rightarrow \infty$.

(iv) If G' is an independent copy of G , then $G - G'$ is symmetric α -stable with $\mathbb{E}e^{i\theta(G-G')} = e^{-\sigma_\alpha|\theta|^\alpha}$, and $I_\alpha := \int_{\mathbb{R}} g(z)^2 dz = \Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$.

Proof. (i) Summation by parts gives $1 - \mathbb{E}w^\xi = (1 - w) \sum_{k \geq 1} \bar{F}(k)w^{k-1}$ for $|w| \leq 1$. With $w = e^{-v}$ and $\text{Li}_\alpha(w) = \sum_{k \geq 1} k^{-\alpha}w^k$ this is $1 - \mathbb{E}e^{-v\xi} = (e^v - 1)\text{Li}_\alpha(e^{-v})$. For $\text{Re } v \geq 0$ and $0 < |v| < 2\pi$, [29, 25.12.12] gives $\text{Li}_\alpha(e^{-v}) = \Gamma(1 - \alpha)v^{\alpha-1} + \zeta(\alpha) + O(|v|)$. So $1 - \mathbb{E}e^{-v\xi} = \mu v + \Gamma(1 - \alpha)v^\alpha + O(|v|^2)$, and $v = -i\theta$ gives (i). (ii) By (i), $e^{-i\mu t}\mathbb{E}e^{it\xi} = 1 + K_\alpha(-i\theta)^\alpha/m + O(\theta^2 m^{-2/\alpha})$ for $t = \theta/a_m$. As $2/\alpha > 1$, the characteristic function of $(\tau_m - m\mu)/a_m$ tends to $\exp(K_\alpha(-i\theta)^\alpha)$, whose modulus $\exp(-\frac{1}{2}\sigma_\alpha|\theta|^\alpha)$ is integrable. Lévy's continuity theorem, Fourier inversion and the Riemann-Lebesgue lemma give G and g (the domain of attraction also follows from [22, Theorem XVII.5.2]). For real $s > 0$, the expansion of $1 - \mathbb{E}e^{-v\xi}$ above with $v = s/a_m$ gives $\mathbb{E}e^{-s(\tau_m - m\mu)/a_m} \rightarrow e^{K_\alpha s^\alpha}$ in the same way, and with $2s$ in place of s this gives uniform integrability, so $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$. (iii) Every $\mathbb{P}(\xi = k)$ is positive, so ξ has maximal span 1, and Gnedenko's local limit theorem [21, Chapter 4] gives the limit. Then $a_m u_m(k) \leq \|g\|_\infty + 1$ for large m and $a_m u_m(k) \leq a_m$ for the others, so $\bar{u} < \infty$. (iv) The function $|\mathbb{E}e^{i\theta G}|^2 = e^{-\sigma_\alpha|\theta|^\alpha}$ is the characteristic function of $G - G'$, and Plancherel's theorem gives $I_\alpha = \frac{1}{2\pi} \int e^{-\sigma_\alpha|\theta|^\alpha} d\theta$, which is $\Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$. $\square$

Theorem 4.4 (the center for $1 < \alpha < 2$). *Let $1 < \alpha < 2$ and let D be any a.s. finite first run, in particular either start. As $N \rightarrow \infty$ over even N ,*

$$C_N \sim 2I_\alpha \left(\frac{\mu}{h}\right)^{1/\alpha} = \frac{2\Gamma(1 + 1/\alpha)}{\pi} \left(\frac{\mu}{K_\alpha |\cos(\pi\alpha/2)| N}\right)^{1/\alpha}.$$

In words, the center decays like $N^{-1/\alpha}$, and the first run does not matter.

Idea of the proof. Both kinds of runs must fill about h steps, which takes about h/μ runs of each kind. Each renewal process is then within about $N^{1/\alpha}$ of its mean, so the two meet at a given level with probability of order $N^{-1/\alpha}$. In the proof (Appendix D) we split the sum in Proposition 4.2 by the number m of runs. With $b_h = (h/\mu)^{1/\alpha}$, the terms with $|m\mu - h| \leq Mb_h$ form a Riemann sum for $2b_h^{-1} \int_{-M}^M g^2$ by Lemma 4.3(iii). The terms with $m \leq h/(4\mu)$ add up to $O(h^{-\alpha}) = o(b_h^{-1})$, since a large sum of few runs is most likely made by one long run. The others add up to $O(b_h^{-1})\mathbb{P}(|G| \geq M - 1) + o(b_h^{-1})$, and we let $M \rightarrow \infty$.

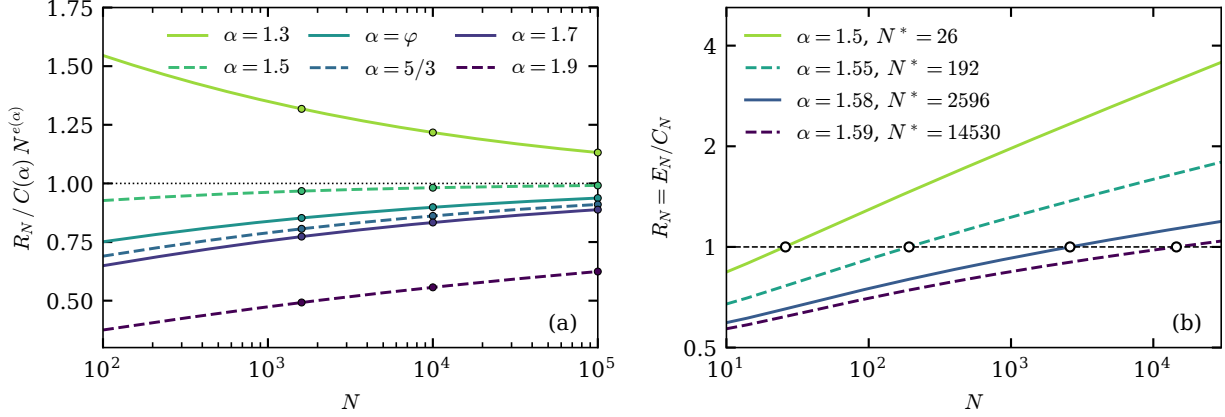


Figure 3: Alternating runs with a stationary start. (a) $R_N / (C(\alpha) N^{e(\alpha)})$ for even N from 100 to 10^5 , with the entries of Table 2 as dots. (b) R_N for $\alpha = 1.5, 1.55, 1.58, 1.59$ below φ . Each curve crosses 1 once on $10 \leq N \leq 3 \cdot 10^4$, at $N^*(\alpha)$ (circles).

Corollary 4.5 (the golden-ratio exponent). *Let $1 < \alpha < 2$ and $e(\alpha) = 1 - \alpha + 1/\alpha$. If $\mathbb{P}(D \geq N) \sim c_D N^{-\beta_D}$ with $c_D, \beta_D > 0$, then $R_N \sim c_D N^{1/\alpha - \beta_D} / (4I_\alpha(2\mu)^{1/\alpha})$. In particular $R_N \sim C(\alpha) N^{e(\alpha)}$ for the stationary start and $R_N \sim C_{\text{fr}}(\alpha) N^{e(\alpha)-1}$ for the fresh start, where*

$$C(\alpha) = \frac{\pi(K_\alpha |\cos \frac{\pi\alpha}{2}|)^{1/\alpha}}{4(\alpha-1)\Gamma(1+\frac{1}{\alpha})\zeta(\alpha)^{1+1/\alpha}}, \quad C_{\text{fr}}(\alpha) = (\alpha-1)\zeta(\alpha)C(\alpha).$$

Since $e(\alpha) > 0$ exactly when $\alpha < \varphi$, with a stationary start the ends win for all large N if $\alpha < \varphi$, and the center wins if $\alpha > \varphi$. At $\alpha = \varphi$,

$$R_N \rightarrow C(\varphi) = \frac{\pi\varphi(\varphi\Gamma(\varphi^{-2})|\cos \frac{\pi\varphi}{2}|)^{1/\varphi}}{4\Gamma(\varphi)\zeta(\varphi)^\varphi} = 0.7761317206\dots,$$

so the center still wins by the factor $1/C(\varphi) = 1.2884$. With a fresh start $e(\alpha) - 1 < 0$, and the center wins for every $1 < \alpha < 2$.

Proof. By Proposition 4.1 and Theorem 4.4, with $h = N/2$, we have $R_N = \mathbb{P}(D \geq N)/(2C_N)$, and this is asymptotic to $c_D N^{-\beta_D} / (4I_\alpha(2\mu/N)^{1/\alpha})$, which is the general formula. By Lemma 4.3(iv), $4I_\alpha(2\mu)^{1/\alpha} = 4\Gamma(1+\frac{1}{\alpha})\mu^{1/\alpha} / (\pi(K_\alpha |\cos \frac{\pi\alpha}{2}|)^{1/\alpha})$. The fresh start has $c_D = 1$ and $\beta_D = \alpha$, which gives $C_{\text{fr}}(\alpha)$. By Proposition 4.1 the stationary start has $c_D = 1/((\alpha-1)\mu)$ and $\beta_D = \alpha-1$, which gives $C(\alpha)$. Next, $e(\alpha) = -(\alpha^2 - \alpha - 1)/\alpha$, and φ is the positive root of $\alpha^2 - \alpha - 1$. At $\alpha = \varphi$ we have $\alpha - 1 = 1/\varphi$, $1 + 1/\alpha = \varphi$ and $2 - \alpha = \varphi^{-2}$. So $K_\varphi = \varphi\Gamma(\varphi^{-2})$, and substituting these into $C(\alpha)$ gives the closed form. $\square$

Corollary 4.5 is the lattice form, with its constant, of the heuristic of [20, Section 2.1]. The approach to $C(\varphi)$ is slow. At $\alpha = \varphi$ we have $R_{10^5}/C(\varphi) = 0.938$, and R_N increases on the computed grid (Table 2 and Figure 3).

The golden ratio also appears for the elephant walk of Section 2 with first step $+1$. There the law of A_N is log-concave for every N exactly when $1 - 2\varepsilon \leq 1/\varphi$ [10, Proposition 2.2]. We know of no connection. Log-concavity should be kept apart from unimodality, which is weaker. With first step $+1$, the limit law of the superdiffusive elephant walk (the case $\varepsilon < \frac{1}{4}$) is unimodal for every

α	start	center C_N	ratio R_N	status
$\alpha > 2$	stationary	$\sim \sqrt{2\mu/(\pi v N)}$	$\sim C_2(\alpha)N^{3/2-\alpha} \rightarrow 0$	Theorem 4.7(a)
$\alpha = 2$	stationary	$\sim \sqrt{2\mu/(\pi N \log N)}$	$\sim \sqrt{\pi \log N/(8\zeta(2)^3 N)} \rightarrow 0$	Theorem 4.7(b)
$1 < \alpha < 2$	stationary	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C(\alpha)N^{e(\alpha)}$	Corollary 4.5
$1 < \alpha < 2$	fresh	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C_{\text{fr}}(\alpha)N^{e(\alpha)-1} \rightarrow 0$	Corollary 4.5
$\alpha = 1$	fresh	$\approx 2/(\pi^2 \tilde{m}_h)$	$\approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$	heuristic, not proved (Remark 4.9)
$0 < \alpha < 1$	fresh	$\sim \frac{2}{\pi} \tan \frac{\pi\alpha}{2} \cdot \frac{1}{N}$	$\sim \frac{\pi}{4 \tan(\pi\alpha/2)} N^{1-\alpha} \rightarrow \infty$	Theorem 4.8

Table 3: The center and the ratio $R_N = E_N/C_N$ for each exponent. The center wins when $R_N \rightarrow 0$ and the ends win when $R_N \rightarrow \infty$. For $\alpha \geq 2$ the center formula holds for every a.s. finite first run, and a fresh start also gives $R_N \rightarrow 0$.

such ε [10, Theorem 1.1]. For log-concavity, the authors of [10] conjectured in their Remark 2.3 that for every $\varepsilon < \frac{1}{4}$ the law of A_N is log-concave for all large N . Trinas and Valle disproved this [11, Theorem 4.4]. They show that for $1 - 2\varepsilon > 0.80399$ the law of A_N is not log-concave for arbitrarily large N , and they prove a uniform local limit theorem for the superdiffusive walk [11, Theorem 3.1].

Remark 4.6 (crossover sizes and heat conduction). For $\alpha < \varphi$ and a stationary start the crossover is late when α is close to φ . For $\alpha = 1.5, 1.55, 1.58, 1.59$ and every even N in $[10, 3 \cdot 10^4]$, $R_N - 1$ changes sign once, at $N^*(\alpha) = 26, 192, 2596, 14530$. Heuristically, $C(\alpha)N^{e(\alpha)} = 1$ gives $\log N^*(\alpha) = 0.1834/(\varphi - \alpha) + O(1)$ as $\alpha \uparrow \varphi$. At the value $\alpha = 5/3$ of Lévy-walk models of heat conduction [19, 18], $R_N \sim C(5/3)N^{-1/15}$ with $C(5/3) = 0.8351$.

We turn to the other exponents. Table 3 collects them. For $\alpha \geq 2$ the sums τ_m have a Gaussian limit, and the center wins with either start.

Theorem 4.7 ($\alpha \geq 2$). *Let D be any a.s. finite first run, and let $N \rightarrow \infty$ over even N .*

- (a) *Let $\alpha > 2$ and $v = \text{Var } \xi = 2\zeta(\alpha - 1) - \zeta(\alpha) - \zeta(\alpha)^2$. Then $C_N \sim \sqrt{2\mu/(\pi v N)}$. With the stationary start $R_N \sim C_2(\alpha)N^{3/2-\alpha}$, where $C_2(\alpha) = \sqrt{\pi v/(2\mu)}/(2(\alpha - 1)\mu)$.*
- (b) *Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}\xi^2 = \infty$. Then $(\tau_m - m\mu)/\sqrt{m \log m}$ converges in law to the standard normal law, and $C_N \sim \sqrt{2\mu/(\pi N \log N)}$. With the stationary start $R_N \sim \sqrt{\pi \log N/(8\zeta(2)^3 N)}$.*

In both cases $R_N \rightarrow 0$ with the fresh start too.

In words, for $\alpha \geq 2$ the center is of the Gaussian order $N^{-1/2}$, with an extra factor $(\log N)^{-1/2}$ at $\alpha = 2$, where the variance is infinite. With either start the end is at most of order $N^{1-\alpha}$, so the center wins. The exponent of R_N is continuous at $\alpha = 2$, since $e(2) = -\frac{1}{2} = \frac{3}{2} - 2$.

Idea of the proof. The proof of Theorem 4.4 uses the exponent α in three places only. These are the limit law of τ_m with its local form, the bound $u_m(k) \leq C_B m k^{-\alpha-1}$ for a sum far above its mean, and the fact that the norming sequence varies regularly with index less than 1. All three hold for $\alpha \geq 2$ with the normal law in place of the stable law. For $\alpha > 2$ the norming is $a_m = \sqrt{vm}$. For $\alpha = 2$ the truncated second moment $\mathbb{E}[\xi^2; \xi \leq x]$ grows like $2 \log x$, which gives the norming $a_m = \sqrt{m \log m}$. Since $\int g^2 = 1/(2\sqrt{\pi})$ for the standard normal density g , the center is $1/(\sqrt{\pi} b_h)$ to leading order, where b_h is the norming at $m = h/\mu$. We give the proof in Appendix D, where the proof of Theorem 4.4 is written so that it covers both theorems.

The convergence is slow at $\alpha = 2$. There R_N divided by the leading term in (b) is 0.885 at $N = 1600$ and 0.943 at $N = 10^5$. At $\alpha = 3$ the ratio of R_N to the leading term in (a) is 0.981 at $N = 16000$.

For $0 < \alpha < 1$ the runs have infinite mean, and only the fresh start makes sense. Then A_N/N has a limit law that is not concentrated at a point, namely Lamperti's occupation-time law [13, 12], with density

$$\ell_\alpha(y) = \frac{\sin \pi \alpha}{\pi} \frac{y^{\alpha-1}(1-y)^{\alpha-1}}{y^{2\alpha} + 2y^\alpha(1-y)^\alpha \cos \pi \alpha + (1-y)^{2\alpha}} \quad (0 < y < 1).$$

So the center should be of order $1/N$. The next theorem is the local form of this at the center.

Theorem 4.8 ($0 < \alpha < 1$). *Let $0 < \alpha < 1$, with a fresh start. As $N \rightarrow \infty$ over even N ,*

$$NC_N \rightarrow \ell_\alpha(\tfrac{1}{2}) = \frac{2}{\pi} \tan \frac{\pi \alpha}{2}.$$

Hence $R_N \sim \frac{\pi}{4 \tan(\pi \alpha/2)} N^{1-\alpha} \rightarrow \infty$.

In words, for $\alpha < 1$ the center is an ordinary bin of size about $1/N$, while the end has the much larger probability $\frac{1}{2}N^{-\alpha}$ that the first run never stops. So the ends win for every $\alpha < 1$.

Idea of the proof. Now τ_m is of order $a_m = m^{1/\alpha}$ with no centering, and τ_m/a_m converges to a positive stable variable G with density g . The number m of runs of each sign is no longer concentrated. It ranges over all m with $m^{1/\alpha}$ of order h , and on this range the terms of Proposition 4.2 are $\frac{2}{\alpha h m} t^2 g(t)^2$ with $t = h/a_m$. Summing over m is integrating in $\log m$, which gives $NC_N \rightarrow 4 \int_0^\infty t g(t)^2 dt$. This integral is the density at $\frac{1}{2}$ of $G_1/(G_1 + G_2)$ for two independent copies of G , and a Mellin transform identifies the law of G_1/G_2 and hence this density. The proof is in Appendix D. It does not use Lamperti's theorem.

Here too the convergence is slow near $\alpha = 1$. At $N = 16000$ the ratio of NC_N to its limit is 1.042, 1.001, 0.928 and 0.554 for $\alpha = 0.3, 0.5, 0.7$ and 0.9 . We do not prove the local limit at bins other than the center.

Remark 4.9 (the exponent $\alpha = 1$ and the fresh start). With a fresh start the threshold is $\alpha = 1$. The ends win for $\alpha < 1$ by Theorem 4.8, and the center wins for $\alpha > 1$ by Corollary 4.5 and Theorem 4.7. At $\alpha = 1$ we only have a heuristic. Suppose we repeat the proof of Theorem 4.4 with a skewed Cauchy limit, the centering $m(H_{m+1} - 1)$, and the density $1/\pi^2$ at 0 of the difference of two copies of the limit. This suggests the row of Table 3, namely $R_N \approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$, where $\tilde{m}_h$ solves $m(H_{m+1} - 1) = h$. We have not controlled the error terms, so the claim that the center wins by a logarithm at $\alpha = 1$ is not proved. The exact R_N divided by $\pi^2 \tilde{m}_h/(8h)$ is 0.796, 0.850 and 0.886 at $N = 1600, 10^4$ and 10^5 .

We do not know a rate in Theorem 4.4, whether R_N is monotone with a single sign change of $R_N - 1$, how to make the case $\alpha = 1$ rigorous, or the full shape of P_N for heavy-tailed runs.

5 Aging walks

In this section the walk remembers its age. After its first step it switches direction at step $k \geq 2$ with probability c/k , independently of the past, where $0 < c < 2$. So the walk switches less and less often, and the expected number of switches in N steps is $c(H_N - 1)$, which is about $c \log N$. With

a fair first step, A_N/N converges in law to $\text{Beta}(c, c)$ [30, Example 1.1 and Theorem 1.4] (their chain is this walk for $c \leq 1$), [31, Theorem 1], and the limit is uniform exactly when $c = 1$ [31, Remark 2]. At finite N the first step matters. We first fix it, then pass to d states, and then return to the fair first step.

We write $\mathbb{P}_+$ for the law given $X_1 = +1$, as in Section 4, and X_N for the last step. A path is a sequence $x \in \{\pm 1\}^N$, and it switches at time $t \geq 2$ if $x_t \neq x_{t-1}$. Step $k \geq 2$ contributes the factor c/k to the probability of a path if the path switches there, and $1 - c/k$ otherwise. So relative to no switch, a switch at time t multiplies the probability of a path by the odds

$$\omega_t = \frac{c/t}{1 - c/t} = \frac{c}{t - c} \quad (t \geq 2),$$

and

$$\mathbb{P}_+(X = x) = \prod_{k=2}^N \left(1 - \frac{c}{k}\right) \prod_{t: x_t \neq x_{t-1}} \omega_t \quad (x_1 = +1). \quad (4)$$

Each ω_t is positive and continuous on $(0, 2)$, increases strictly in c , and tends to 0 as $c \downarrow 0$.

5.1 A fixed first step

Proposition 5.1 (two states, fixed first step). *Let $N \geq 2$ and let the first step be $+1$.*

(a) *At $c = 1$, for $1 \leq s \leq N$,*

$$\mathbb{P}_+(A_N = s, X_N = +1) = \frac{s-1}{N(N-1)}, \quad \mathbb{P}_+(A_N = s, X_N = -1) = \frac{N-s}{N(N-1)}.$$

So A_N is uniform on $\{1, \dots, N\}$.

(b) *For $1 \leq s \leq N-1$ the ratio $\mathbb{P}_+(A_N = s)/\mathbb{P}_+(A_N = N)$ increases strictly in $c \in (0, 2)$, and it equals 1 exactly at $c = 1$.*

In words, with a fixed first step every bin from 1 to $N-1$ ties with the end bin N at $c = 1$, for every N . For $c < 1$ the end bin is the most likely bin, and for $c > 1$ it is the least likely of the bins $1, \dots, N$ (bin 0 has probability 0). So the fixed-start crossing is exactly $c = 1$, and it does not depend on N or on the bin.

Proof. (a) We use induction on N . Put $u_N^\pm(s) = \mathbb{P}_+(A_N = s, X_N = \pm 1)$, which is 0 unless $1 \leq s \leq N$. For $N = 2$ the second step is $+1$ or -1 with probability $\frac{1}{2}$ each, so $u_2^+(2) = u_2^-(1) = \frac{1}{2}$ and $u_2^+(1) = u_2^-(2) = 0$, as claimed. At $c = 1$ step $N+1$ switches with probability $1/(N+1)$. Conditioning on X_N gives, for every s ,

$$(N+1)u_{N+1}^+(s+1) = Nu_N^+(s) + u_N^-(s), \quad (N+1)u_{N+1}^-(s) = u_N^+(s) + Nu_N^-(s).$$

Suppose the claim holds for N , and let $1 \leq s \leq N$. Then $N(s-1) + (N-s) = s(N-1)$ gives $u_{N+1}^+(s+1) = \frac{s}{(N+1)N}$, and $(s-1) + N(N-s) = (N-1)(N+1-s)$ gives $u_{N+1}^-(s) = \frac{N+1-s}{(N+1)N}$. The two remaining values are $u_{N+1}^+(1) = 0$ and $u_{N+1}^-(N+1) = 0$, since $u_N^\pm(0) = u_N^\pm(N+1) = 0$. This is the claim for $N+1$. Summing over X_N gives $\mathbb{P}_+(A_N = s) = \frac{(s-1)+(N-s)}{N(N-1)} = \frac{1}{N}$.

(b) Only the path with every step $+1$ has $A_N = N$, and it never switches. By (4),

$$\frac{\mathbb{P}_+(A_N = s)}{\mathbb{P}_+(A_N = N)} = \sum_x \prod_{t: x_t \neq x_{t-1}} \omega_t, \quad (5)$$

where the sum is over the paths x with $x_1 = +1$ and s steps equal to $+1$. For $1 \leq s \leq N-1$ there is such a path, and each of them switches at least once. So the right side is a sum of products of the ω_t with at least one factor each, and it increases strictly in c . By (a) it equals 1 at $c = 1$. $\square$

After a shift of one step, part (a) is the exercise in [34, footnote 1 on p. 6]. The argument for (b) is the one in Proposition 2.3. Both parts of Proposition 5.1 are also proved in Lean (Section 6).

5.2 d states

The same thing happens for a chain on d states that leaves its current state for each other state at the same aging rate. For such chains Dietz and Sethuraman [30, Theorem 1.4] proved that the fractions of time converge in law to the Dirichlet law with every parameter θ , which is uniform on the simplex exactly at $\theta = 1$ (see also [33]). Part (a) of the theorem below is an exact form of this at finite N . Fix $d \geq 2$ and θ with $0 < \theta < d/(d-1)$. The chain $X_1, X_2, \dots$ on $\{1, \dots, d\}$ starts at $X_1 = 1$. At step $k \geq 2$ it moves to each of the other $d-1$ states with probability $\theta/(k+d-2)$, and it stays with probability $1 - (d-1)\theta/(k+d-2)$. The condition on θ says that the probability of staying is positive at every step $k \geq 2$. Let $L_N = (L_N(1), \dots, L_N(d))$ be the vector of visit counts, namely $L_N(i) = \#\{k \leq N : X_k = i\}$. Its values are the compositions $n = (n_1, \dots, n_d)$ of N into d nonnegative parts with $n_1 \geq 1$. There are $\binom{N+d-2}{d-1}$ of them (subtract one from n_1). We call them bins. The corner Ne_1 , where e_i is the i -th unit vector, is the bin of the chain that never moves.

Theorem 5.2 (d states). *Let $d \geq 2$, $0 < \theta < d/(d-1)$ and $N \geq 2$.*

(a) *At $\theta = 1$, for every bin n and every state j ,*

$$\mathbb{P}(L_N = n, X_N = j) = \frac{n_j - \mathbf{1}\{j = 1\}}{(N-1)\binom{N+d-2}{d-1}}. \quad (6)$$

So L_N is uniform on the $\binom{N+d-2}{d-1}$ bins, the corner included.

(b) *For every bin $n \neq Ne_1$ the ratio $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ increases strictly in θ , and it equals 1 exactly at $\theta = 1$.*

In words, at $\theta = 1$ all bins are equally likely, and given the counts the last state is distributed as the state at a uniformly chosen one of the steps $2, \dots, N$ (the forced first visit to state 1 is left out). Every other bin ties with the corner exactly at $\theta = 1$, for every $N \geq 2$ and every d . Below $\theta = 1$ the corner is the most likely bin, and above it the corner is the least likely.

Idea of the proof. For (a) we check (6) by induction on N . At $\theta = 1$ step N stays with weight $N-1$ and moves to each other state with weight 1. We read the weight $N-1$ as $(N-2) + 1$. So the chain copies its last state with weight $N-2$, and it picks each of the d states with weight 1. By the induction hypothesis the last state is like the state at a uniformly chosen one of the steps $2, \dots, N-1$. So given the counts m after $N-1$ steps, the new state is j with weight $(m_j - \mathbf{1}\{j = 1\}) + 1$. This is the rule of Pólya's urn, and it keeps the counts uniform. Part (b) is the path-weight argument of Proposition 5.1(b).

Proof. (a) We use induction on N . Write $P_N(n, j) = \mathbb{P}(L_N = n, X_N = j)$ and $\delta_j = \mathbf{1}\{j = 1\}$. Let $N = 2$. At $\theta = 1$ the second step goes to each of the d states with probability $1/d$ (the probability of staying is $1 - (d-1)/d$). So $P_2(2e_1, 1) = \frac{1}{d}$ and $P_2(e_1 + e_j, j) = \frac{1}{d}$ for $j \neq 1$, and P_2 vanishes

otherwise. Since $\binom{d}{d-1} = d$, the right side of (6) is $(n_j - \delta_j)/d$. This is $\frac{1}{d}$ in these d cases. In every other case $n_j = \delta_j$, and it is 0.

Now let $N \geq 3$ and suppose that (6) holds for $N - 1$. Fix a bin n and a state j . If $n_j = \delta_j$, then the right side of (6) is 0. The left side is 0 too. Indeed, if $j \neq 1$ then state j is never visited, and if $j = 1$ then state 1 is visited only at the first step, so $X_N \neq j$ in both cases. Otherwise $m = n - e_j$ is a composition of $N - 1$ with $m_1 \geq 1$. At $\theta = 1$ step N moves to each other state with probability $1/(N + d - 2)$ and stays with probability $(N - 1)/(N + d - 2)$. Conditioning on X_{N-1} gives

$$P_N(n, j) = \frac{(N - 1) P_{N-1}(m, j) + \sum_{i \neq j} P_{N-1}(m, i)}{N + d - 2} = \frac{(N - 2) P_{N-1}(m, j) + \sum_i P_{N-1}(m, i)}{N + d - 2}.$$

By the induction hypothesis and $\sum_i (m_i - \delta_i) = N - 2$, the numerator is

$$\frac{(N - 2)(m_j - \delta_j) + (N - 2)}{(N - 2) \binom{N+d-3}{d-1}} = \frac{n_j - \delta_j}{\binom{N+d-3}{d-1}}.$$

Since $\binom{N+d-2}{d-1} = \binom{N+d-3}{d-1} \frac{N+d-2}{N-1}$, this gives (6) for N . Summing (6) over j and using $\sum_j (n_j - \delta_j) = N - 1$ gives $\mathbb{P}(L_N = n) = 1/\binom{N+d-2}{d-1}$.

(b) A path $x \in \{1, \dots, d\}^N$ with $x_1 = 1$ has probability $\prod_{k=2}^N (1 - \frac{(d-1)\theta}{k+d-2})$ times the product of the odds

$$\omega_t^{(d)} = \frac{\theta/(t + d - 2)}{1 - (d - 1)\theta/(t + d - 2)} = \frac{\theta}{t + d - 2 - (d - 1)\theta}$$

over the times $t \geq 2$ with $x_t \neq x_{t-1}$. The denominator is positive and decreases in θ , so each $\omega_t^{(d)}$ increases strictly in θ . Only the constant path has counts Ne_1 . So $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ is the sum of the products of the odds over the paths with counts n . For a bin $n \neq Ne_1$ there is such a path, and each of them moves at least once. Hence the ratio increases strictly in θ , and by (a) it equals 1 at $\theta = 1$. $\square$

Remark 5.3. For $d = 2$ the chain switches at step k with probability θ/k , so Theorem 5.2 with $\theta = c$ is Proposition 5.1, and (6) reads $(s - 1)/(N(N - 1))$ for $j = 1$ and $n = (s, N - s)$.

Take Pólya's urn with d colors, started with one ball of each color. At each draw we take a ball uniformly at random and return it with one more ball of its color. A sequence of $N - 1$ draws with counts m has probability $(d - 1)! \prod_i m_i! / (N + d - 2)!$. So the counts are uniform on the compositions of $N - 1$ into d parts, and given the counts m the last draw has color j with probability $m_j/(N - 1)$. Hence (6) says that, for each fixed N , the pair $(L_N - e_1, X_N)$ has the law of the counts and the last color after $N - 1$ draws. The two processes are different, though. The chain is a Markov chain, and the draws are exchangeable. For $d = 2$ and $N = 4$ the path 1, 1, 2, 2 has probability $\frac{1}{8}$ for the chain, and the draws 1, 2, 2 have probability $\frac{1}{12}$. Also, at $\theta = 1$ the chain moves at step k with probability $(d - 1)/(k + d - 2)$, independently over k . This is the probability that customer k opens a new table in the Chinese restaurant process with parameter $d - 1$. So the expected number of moves in N steps at the tie is $\sum_{k=2}^N \frac{d-1}{k+d-2} = (d - 1) \log N + O(1)$, which is the first-order scale of the vertex crossing in Paper B [3].

We checked (6) and the strict ordering in (b) in exact rational arithmetic for $d = 2, 3, 4$ and $2 \leq N \leq 8$ (`verify_aging.py`).

The urn is not special to $\theta = 1$. It appears for every weight once the aging rate is adjusted a little. Fix $d \geq 2$ and $\theta > 0$, and let $Y_1, Y_2, \dots$ be the following chain on $\{1, \dots, d\}$. At step $t \geq 1$,

with probability

$$\varphi_t = \frac{d\theta}{d\theta + t - 1}$$

the state Y_t is drawn afresh, uniformly from the d states, and otherwise $Y_t = Y_{t-1}$ (so Y_1 is uniform, since $\varphi_1 = 1$). In other words, at step $t \geq 2$ the chain moves to each other state with probability $\theta/(d\theta + t - 1)$. Let M_T be the vector of visit counts of $Y_1, \dots, Y_T$. At $\theta = 1$ this is the chain of Theorem 5.2 without its first step, namely $Y_t = X_{t+1}$ and $M_T = L_{T+1} - e_1$. We write $x^{(m)} = x(x+1) \cdots (x+m-1)$.

Proposition 5.4 (fresh or repeat). *For every $T \geq 1$, every composition m of T into d nonnegative parts and every state j ,*

$$\mathbb{P}(M_T = m, Y_T = j) = \frac{m_j}{T} D_T(m), \quad D_T(m) = \frac{T!}{(d\theta)^{(T)}} \prod_{i=1}^d \frac{\theta^{(m_i)}}{m_i!}. \quad (7)$$

So M_T has the law D_T of the counts in Pólya's urn with d colors and initial weight θ on each color. For $d = 2$ and even T ,

$$\frac{\mathbb{P}(M_T = (T/2, T/2))}{\mathbb{P}(M_T = (T, 0))} = \binom{T}{T/2} \prod_{i=0}^{T/2-1} \frac{\theta + i}{\theta + T/2 + i}, \quad (8)$$

which increases strictly in θ and equals 1 exactly at $\theta = 1$.

In words, the chain repeats its last state where the urn repeats a uniformly chosen earlier draw, and at each fixed time this makes no difference to the counts or to the last state. For two states the center and the end tie at $\theta = 1$ for every T , with ends ahead below it and the center ahead above it, and now the whole law is explicit on both sides of the tie.

Proof. We use induction on T . For $T = 1$ both sides of (7) are $\frac{1}{d}$ when $m = e_j$ and 0 otherwise. Let $T \geq 2$ and suppose (7) holds for $T - 1$. If $m_j = 0$ both sides vanish. Otherwise put $m' = m - e_j$. Summing (7) over the last state gives $\mathbb{P}(M_{T-1} = m') = D_{T-1}(m')$. A fresh draw gives j with probability $\frac{1}{d}$ whatever the last state was, and a repeat gives j only when the last state was j . So

$$\mathbb{P}(M_T = m, Y_T = j) = \frac{\varphi_T}{d} D_{T-1}(m') + (1 - \varphi_T) \frac{m_j - 1}{T - 1} D_{T-1}(m') = \frac{\theta + m_j - 1}{d\theta + T - 1} D_{T-1}(m').$$

On the other hand $D_T(m) = D_{T-1}(m') \cdot \frac{T}{m_j} \cdot \frac{\theta + m_j - 1}{d\theta + T - 1}$ by the definition of D_T . This gives (7). Summing over j gives $\mathbb{P}(M_T = m) = D_T(m)$, and the ratio (8) is $D_T(T/2, T/2)/D_T(T, 0)$ with $\theta^{(T)} = \theta^{(T/2)}(\theta + T/2)^{(T/2)}$. Each factor $(\theta + i)/(\theta + T/2 + i)$ increases strictly in θ , and at $\theta = 1$ the product is $(T/2)!^2/T!$. $\square$

Remark 5.5. Here is another way to see Proposition 5.4. The fresh draws cut $Y_1, \dots, Y_T$ into runs, and each run gets a uniform state. The urn can be run in the same way, as a Chinese restaurant process in which each table gets a uniform color. There a table is opened with the same probabilities φ_t . The lengths of the runs are the spacings between the successes of independent trials with success probabilities φ_t , and it is a classical fact (the Feller coupling, see [32]) that these spacings have the same law as the table sizes. The footnote of Engländer, Volkov and Wang [34] credits M. Balázs with a relation to Pólya urns for two states. We do not know whether it is this one. We checked (7) in exact arithmetic for $d = 2, 3, 4$, four values of θ and $T \leq 7$ (verify_aging_urn.py).

5.3 A fair first step

We return to two states and let the first step be fair. Then both ends can be reached, and $E_N = \mathbb{P}(A_N = N) = \frac{1}{2} \prod_{k=2}^N (1 - c/k)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. So

$$\mathbb{P}(A_N = s) = \frac{1}{2} (\mathbb{P}_+(A_N = s) + \mathbb{P}_+(A_N = N - s)), \quad E_N = \frac{1}{2} \mathbb{P}_+(A_N = N), \quad (9)$$

since $\mathbb{P}_+(A_N = 0) = 0$.

Proposition 5.6 (fair first step). *Let $N \geq 2$ and $1 \leq s \leq N - 1$, with a fair first step.*

- (a) *At $c = 1$, $\mathbb{P}(A_N = s) = \frac{1}{N}$ and $E_N = \frac{1}{2N}$.*
- (b) *The ratio $\mathbb{P}(A_N = s)/E_N$ is continuous and strictly increasing in $c \in (0, 2)$. It tends to 0 as $c \downarrow 0$ and equals 2 at $c = 1$. So there is exactly one c in $(0, 2)$ with $\mathbb{P}(A_N = s) = E_N$, and it is less than 1.*

For even N we write c_N for the crossing in (b) with $s = h = N/2$, so that $C_N = E_N$ exactly at $c = c_N$. In words, at $c = 1$ every interior bin still has probability $\frac{1}{N}$, and the two ends share the probability $\frac{1}{N}$ that the fixed start gave to one end. So at $c = 1$ the center is twice as likely as an end, and the crossing moves below 1.

Proof. (a) This follows from (9) and Proposition 5.1(a). (b) By (9) the ratio is the sum of the right sides of (5) for s and for $N - s$. Both are finite sums of products of at least one ω_t . So the ratio is continuous and strictly increasing, and it tends to 0 as $c \downarrow 0$. By (a) it equals 2 at $c = 1$. Hence it equals 1 exactly once, at some $c < 1$. $\square$

How far below 1 is c_N ? The end is of order N^{-c} and the center is of order $1/N$, so they meet when N^{1-c} is about 2.

Proposition 5.7 (the fair-start crossing). *For even $N \rightarrow \infty$,*

$$c_N = 1 - \frac{\log 2}{\log N + \gamma + 2 - 2 \log 2} + O((\log N)^{-3}),$$

where γ is Euler's constant.

Idea of the proof. The end is explicit, namely $E_N = \Gamma(N + 1 - c)/(2\Gamma(2 - c)\Gamma(N + 1))$, which is $N^{-c}/(2\Gamma(2 - c))$ with relative error $O(1/N)$. For the center we prove a local limit theorem with a rate, namely $NC_N = d_c + O(N^{-0.4})$ uniformly for c near 1, where $d_c = 4^{1-c}/B(c, c)$ is the density of Beta(c, c) at $\frac{1}{2}$. So $\log(C_N/E_N) = (c - 1) \log N + \Theta(c) + O(N^{-0.4})$ with $\Theta(c) = \log(2\Gamma(2 - c)d_c)$. Here $\Theta(1) = \log 2$ and $\Theta'(1) = \gamma + 2 - 2 \log 2$, and solving for the zero gives the expansion. The proof is in Appendix E. The local limit theorem comes from an averaging identity for C_N and a coupling of the walk with a sign process driven by a Poisson process of intensity $c dt/t$, whose occupation fraction is exactly Beta(c, c).

The convergence is slow. The exact crossing at $N = 200$ is $c_{200} = 0.8925$, against 0.8932 from the expansion without its error term. The constant in the error term is not computed.

6 Verification

All computations are in Python, and the main ones were done by two independent methods (for example, forward and backward recursions for the elephant center agree to about 10^{-14} at $N = 12800$). The proofs use computation in three places only. First, some constants are values of explicit sums and integrals, such as the 60-term sum in the proof of Theorem 2.6(b), which we evaluated in 30-digit arithmetic (`verify_elephant_crossing.py`). Second, Lemma 3.11 rests on enclosures of f'_L at two points and of f''_L on the interval between them, computed in ball arithmetic with rigorous error bounds. Third, Proposition 3.8, which makes Corollary 3.6 hold for every even $N \geq 8$, rests for $N < 1688$ on a double-precision computation of C_N/E_N at $\varepsilon = 2/(N+2)$ whose rounding error is bounded in Lemma 3.7. The last two computations are in `certify_elephant.py`, which also repeats the third one in exact integer arithmetic for 20 values of N . Conjecture 3.16, the tables, the figures and the numbers quoted outside the proofs rest on computation alone.

Some of the exact and finite statements are also proved in Lean 4, among them Proposition 2.3, the value $C'_N(0) = 4/(N+2)$ of Proposition 2.5(b), Lemma 3.4, Theorem 3.5, Proposition 5.1 and the identities behind Corollary 4.5. The Lean files are in the folder `kagey-problems/problem131/lean/PaperD`. The file `MAP.md` in that folder lists each formalized statement with its Lean name. The asymptotic results and Theorem 5.2 were not formalized.

Data and code availability. Everything is in the folder `kagey-problems/problem131` of <https://github.com/apemm/Kagey-Problems>, at the release `paperD-arxiv-v1`. To rerun the checks, run `python -B verify_all.py --paper d` in that folder, which takes several minutes. The scripts are in the subfolder `paperD` and need `numpy`, `scipy`, `mpmath` and `python-flint`. The file `SUPPLEMENT.md` says what each script checks and where to find the predictions that we wrote down before each computation.

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A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

Statements and declarations

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Data availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of <https://github.com/apemm/Kagey-Problems>, at the release `paperD-arxiv-v1`. Section 6 describes the checks and how to rerun them.

A Proofs for Section 2

We first prove Theorem 2.6. Then Section A.1 proves Proposition 2.9, and Section A.2 proves Proposition 2.11. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a)

we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1 , and for $b \geq 1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h, 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b+c > 0$ and $s-b+c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b} B(h+c, h+c) / B(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \leq h-b \leq N-s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b+c > 0$ and $s-b+c > 0$.*

- (a) $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N . So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.
- (b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.
- (c) If $0 < \varepsilon < \frac{1}{2}$ and $\beta = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\beta s}(b)$ for $0 \leq b \leq s$.
- (d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \beta s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s-1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b+c$, $y = s-b+c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N-s$ draws with $h-b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = B(h+c, h+c)/B(x, y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $B(x, y)/B(x+1, y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1 + 2\beta = 1/(1-2\varepsilon)$, we have $\frac{b+\beta s}{s+2\beta s} = (1-2\varepsilon)(\frac{b}{s} + \beta) = r_s(b)$.

(d) Let $F_u = \Phi_u^c$ on $\{1, \dots, u-1\}$ for $s \leq u \leq N$. Then F_u vanishes off $\mathcal{S}_u$ and $F_N(b) = \mathbf{1}\{b = h\}$. By (a) and (c), $\mathcal{L}_u F_{u+1}(b) - F_u(b) = (r_u(b) - r_u^c(b))(F_{u+1}(b+1) - F_{u+1}(b))$. Since $r_u^c(b) = \frac{1}{2} - (\frac{u}{2} - b)/(u+2c)$, this is the product of $\frac{u}{2} - b$, $\frac{1}{u+2c} - \frac{1}{u(1+2\beta)} \geq 0$ (as $c \leq \beta s \leq \beta u$) and $F_{u+1}(b+1) - F_{u+1}(b)$. Let $b \in \mathcal{S}_u$. If $b = h$ then $b \geq u/2$ and $F_{u+1}(b+1) = 0$. If $u-b = h$ then $b \leq u/2$ and $F_{u+1}(b) = 0$. Otherwise $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$, which is ≥ 1 exactly when $b \leq u/2$ (write $b = \frac{u}{2} - d$, and then the difference of the two sides of the inequality is $2d(h+c)$). So in all three cases the first and last factors have the same sign. Hence $\mathcal{L}_u F_{u+1} \geq F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) below gives $V_s \geq F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. $\square$

Lemma A.2. *Let $1 \leq s_0 \leq N$, and for $s_0 \leq s \leq N$ let $F_s \geq 0$ be a function on $\{1, \dots, s-1\}$.*

- (a) If $F_N(h) \geq 1$ and $\mathcal{L}_s F_{s+1} \leq K_s F_s$ on $\mathcal{S}_s$ for $s_0 \leq s < N$, with constants $K_s \geq 0$, then $V_s \leq (\prod_{u=s}^{N-1} K_u) F_s$ on $\mathcal{S}_s$.
- (b) If $F_N(b) \leq \mathbf{1}\{b = h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \geq F_s$ on $\mathcal{S}_s$ for $s_0 \leq s < N$, then $V_s \geq F_s$.

Proof. We use backward induction on s , from $V_N(b) = \mathbf{1}\{b = h\}$. In (a), V_{s+1} vanishes at infeasible states and $F_{s+1} \geq 0$, so on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \leq \prod_{u>s} K_u \mathcal{L}_s F_{s+1} \leq \prod_{u \geq s} K_u F_s$. In (b) we get $V_s \geq \mathcal{L}_s F_{s+1} \geq F_s$ on $\mathcal{S}_s$, and $F_s = 0 \leq V_s$ off $\mathcal{S}_s$. $\square$

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s+10)$ of each color at time s , grows a little faster than the βs of M . Its hitting probability $\Phi_s^{\kappa s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s , and Proposition A.4 shows that $\log \Delta_s \leq \kappa/s$ up to a smaller term. Two effects enter Δ_s . Raising the extra balls from κ_s to κ_{s+1} changes Φ , and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \leq s \leq N$, $b \in \mathcal{S}_s$ and $\tau \geq 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s - 2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1-z) + \frac{1+z}{(1-z)\Sigma} + \frac{2(1+z)}{3\Sigma^2(1-z)^2}.$$

Proof. Since $\log B(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x+y)$, the derivative is $2\psi(\Sigma) - \psi(X) - \psi(Y) - 2\psi(N+2\tau) + 2\psi(h+\tau)$. For $x > 0$, [29, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2 \int_0^\infty \frac{t dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t dt}{e^{2\pi t}-1} = \frac{1}{24}$. We use this upper bound for $\psi(\Sigma)$ and lower bound for $\psi(X)$ and $\psi(Y)$. The duplication formula [29, 5.5.5] gives $2\psi(2y) = \psi(y) + \psi(y + \frac{1}{2}) + 2\log 2$, and ψ is increasing, so $2\psi(N+2\tau) - 2\psi(h+\tau) > 2\log 2$. Hence the derivative is less than $\log \frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $4XY = \Sigma^2(1-z)$ and $X^2 + Y^2 = \Sigma^2(1+z)/2$, so the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. $\square$

Proposition A.4. *Let $0 < \varepsilon < \frac{1}{4}$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s + 2\kappa_s$. For $1 \leq s \leq N-1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s \Phi_{s+1}^{\kappa_{s+1}}(b) / \Phi_s^{\kappa_s}(b)$.*

(a) *Let $c = \kappa_s$, $c' = \kappa_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s+2c'}$, $v = \frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\beta)}$. Then $|u| < \frac{1}{2}$ and*

$$\Delta_s(b) = \mathcal{J} \frac{\Phi_s^{c'}(b)}{\Phi_s^c(b)}, \quad \mathcal{J} = 1 - (\rho - 1) \frac{u(u+v)}{\frac{1}{4} - u^2}.$$

(b) $\log \Delta_s(b) \leq \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) *For $1 \leq t \leq h$, $V_{t+1}(1) \leq \pi_t \exp(\kappa[H_{N-1} - H_t + \frac{2}{3t}] + \kappa_{t+1}[\log \frac{(t+1)^2}{4t} + \frac{2}{3}])$.*

(d) $C_N \leq U_N := \varepsilon c_1 e^{\kappa H_{N-1}} \sum_{t=1}^h w_t (1-\varepsilon)^{t-1} e^{\kappa \mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)(\log \frac{(t+1)^2}{4t} + \frac{2}{3}) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa - \beta)/(1 + 2\beta) = \kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c' .

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar{r} = \bar{r}_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c' , and the recursion in Lemma A.1(a), we have $r' \Phi_{s+1}^{c'}(b+1) = \bar{r} \Phi_s^{c'}(b)$ and $(1-r') \Phi_{s+1}^{c'}(b) = (1-\bar{r}) \Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r')(1-\bar{r})}{1-r'} + \frac{r\bar{r}}{r'}$. Here $r' = \frac{1}{2} - u$, $\bar{r} = \frac{1}{2} + v$ and $r = \frac{1}{2} - \rho u$, since $\frac{1}{2} - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac{1}{4} - u^2$ the numerator is $f(u, v) + f(-u, -v)$ with $f(u, v) = (\frac{1}{2} + \rho u)(\frac{1}{2} - v)(\frac{1}{2} - u)$, and this equals $\frac{1}{4} - \rho u^2 - (\rho - 1)uv$. Also $|d| \leq \frac{s}{2} - 1$, so $|u| < \frac{1}{2}$.

(b) *Step 1: the factor $\mathcal{J}$.* Write $c = \kappa_s$, $c' = c + \kappa$ and $\Sigma = \Sigma_s$. Since $c' - \beta s = (\kappa - \beta)s + 11\kappa$, the choice of κ gives $\rho - 1 = \kappa + \nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\beta)}$. As u and v have the sign of d , $u(u+v) \geq u^2$.

So $\mathcal{J} \leq 1 - (\kappa + \nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J} > 0$ by (a), since $0 < r < 1$ and $\bar{r}$, $1 - \bar{r}$ are not both 0. So $\log \mathcal{J} \leq -(\kappa + \nu_s)W'$.

Step 2: the ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ , while z decreases and Σ increases in τ . So we may take the bound at $\tau = c$, where $X + Y = \Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1-z) \leq W$, so

$$\log \Delta_s(b) \leq \kappa \left[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2} \right] - (\kappa + \nu_s)W'. \quad (10)$$

Step 3: three facts. (i) $X, Y \geq 1 + c$, since $b \geq 1$ and $s - b \geq 1$. So $XY \geq (1+c)\Sigma/2$ and $1 + W = \frac{\Sigma^2}{4XY} \leq \frac{\Sigma}{2(1+c)}$. (ii) $z' = \lambda z$ with $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \geq 1 - \frac{4\kappa}{\Sigma}$. So $W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \leq (1-\lambda)W(1+W) \leq \frac{2\kappa}{1+c}W$ by (i). As $c \geq 11\kappa$, this gives $W' \geq \frac{9}{11}W$. (iii) $\Sigma \geq s$ and $1 + c > \kappa s$. If $W > 0$ then $s \neq 2b$, and with $b \geq 1$, $s - b \geq 1$ this forces $s \geq 3$, so $\Sigma^2 \geq 3s$.

Step 4: collect the terms. By (ii) and (iii), $\kappa(W - W') \leq \frac{2\kappa^2}{1+c}W \leq \frac{2\kappa}{s}W$ and $\frac{2\kappa W}{\Sigma} \leq \frac{2\kappa}{s}W$. Since $(1+2W)(1+W) = 1 + 3W + 2W^2$, κ times the last term in the bracket of (10) is $\frac{2\kappa}{3\Sigma^2} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2}$. Here $\frac{2\kappa W}{\Sigma^2} \leq \frac{2\kappa}{3s}W$ by (iii), and $\frac{4\kappa W^2}{3\Sigma^2} \leq \frac{4\kappa W(1+W)}{3\Sigma^2} \leq \frac{2\kappa W}{3\Sigma(1+c)} \leq \frac{2\kappa}{3s}W$ by (i). With $-\nu_s W' \leq -\frac{9}{11}\nu_s W$ this gives

$$\log \Delta_s(b) \leq \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W \left[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\beta)} \right],$$

and the bracket is negative because $1 + 2\beta < 2$ for $\varepsilon < \frac{1}{4}$, so $18/(1+2\beta) > 9 > 16/3$.

(c) Apply Lemma A.2(a) with $s_0 = t + 1$, $F_s = \Phi_s^{\kappa s}$ and $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h) = 1$, $F_s > 0$ on $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} = \Delta_s F_s \leq K_s F_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s \geq s$. Since $1 \in \mathcal{S}_{t+1}$ and $\sum_{s>t} s^{-2} \leq \frac{1}{t}$, we get $V_{t+1}(1) \leq e^{\kappa[H_{N-1}-H_t+2/(3t)]} \Phi_{t+1}^{\kappa_{t+1}}(1)$. Now $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau \in [0, \kappa_{t+1}]$ of the derivative in Lemma A.3, whose bound is largest at $\tau = 0$. There $X = 1$, $Y = t$ and $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$, and the bound is $\log \frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \leq \log \frac{(t+1)^2}{4t} + \frac{2}{3}$.

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1} = \kappa(t+1)$ and $\pi_t = c_1 w_t$. $\square$

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \geq \varepsilon c_1 \sum_t w_t (1 - \varepsilon)^{t-1}$. Now $(1 - \varepsilon)^{t-1} \geq 1 - (t-1)\varepsilon$ and $\sum_t w_t (t-1) = 2 - \frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \geq \varepsilon c_1 (1 - 2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1: the constants.* Let $N \geq 64$ and $\varepsilon(1 + \log N) \leq \frac{1}{100}$. Then $\varepsilon < 0.002$ and $\kappa \leq 2\varepsilon(1 + 8\varepsilon) \leq 2.1\varepsilon$. Since $\log \frac{(t+1)^2}{4t} \geq 0$ and $\frac{2}{3}(t+1) > 1 + \log t \geq H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \leq \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \geq \frac{2}{3t}$, also $\mathcal{G}(t) \leq (t+1)\log(t+1) \leq 12t\log(t+1)$. So $\kappa\mathcal{G}(t) \leq 25.2\varepsilon t\log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \leq h$, since then $25.2\varepsilon\log(t+1) \leq 25.2\varepsilon\log N \leq 0.252 < \frac{1}{2}\log 2$.

Step 2: the sum over t . Since $e^y - 1 \leq ye^y$ for $y \geq 0$, Step 1 gives $e^{\kappa\mathcal{G}(t)} - 1 \leq \kappa\mathcal{G}(t)2^{t/2}$. By Proposition A.4(d), $(1 - \varepsilon)^{t-1} \leq 1$, $\sum_t w_t = 1$ and $w_t \leq 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \leq \varepsilon c_1 e^{\kappa H_{N-1}} \left(1 + 50.4\varepsilon \sum_{t \geq 1} t^2 \log(t+1) 2^{-t/2} \right),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1 + \frac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4} . The first 60 terms add up to 102.5864 to four

decimals (computed in 30-digit arithmetic by `verify_elephant_crossing.py`), so the series is below 102.59.

Step 3: the factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \leq 1 + \log N$, so $\kappa H_{N-1} \leq 2\varepsilon \log N + 2\varepsilon + 16\varepsilon^2(1 + \log N) \leq 2\varepsilon \log N + 3\varepsilon$. With $1 + y \leq e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. $\square$

A.1 The bridge and the lower bound

Fix $1 \leq t \leq h$, and start the Pólya(c) chain at state 1 at time $t + 1$. By Lemma A.1(b), given that it is at h at time N it moves up at time u from b with probability $\bar{r}_u(b) = \frac{h-b}{N-u}$, whatever c is. So its $h - 1$ up steps are placed uniformly at random among the $N - t - 1$ steps after time $t + 1$. We call this law of the path $(M_u)_{t+1 \leq u \leq N}$ the bridge, and we write $\mathbb{P}_t^{\text{br}}$ and $\mathbb{E}_t^{\text{br}}$ for it. For $1 \leq b \leq u - 1$ let

$$\varpi_u^+(b) = 1 + \varepsilon \frac{u - 2b}{b}, \quad \varpi_u^-(b) = 1 - \varepsilon \frac{u - 2b}{u - b},$$

$$\theta_u(b) = \frac{N(u - 2b)^2}{2(N - u)b(u - b)}, \quad Q_u(b) = (u - 2b)^2 \left[\frac{\bar{r}_u(b)}{b^2} + \frac{1 - \bar{r}_u(b)}{(u - b)^2} \right],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise. Put

$$A_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\text{br}} \theta_u(M_u), \quad B_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\text{br}} Q_u(M_u).$$

Both are nonnegative, and neither depends on ε .

Proposition A.5 (bridge formula). *Let $0 \leq \varepsilon \leq 1$ and $1 \leq t \leq h$. Then*

$$V_{t+1}(1) = \pi_t \mathbb{E}_t^{\text{br}} \left[\prod_{u=t+1}^{N-1} \varpi_u \right].$$

If $0 < \varepsilon \leq \frac{1}{2}$, then $\log V_{t+1}(1) \geq \log \pi_t + \varepsilon A_t - \varepsilon^2 B_t$.

Proof. For $u \geq t + 1$ and $b \in \mathcal{S}_u$ we have $\Phi_u^0(b) > 0$, and we put $R_u(b) = V_u(b)/\Phi_u^0(b)$. By Lemma A.1(a) and (b) with $c = 0$, $\frac{b}{u}\Phi_{u+1}^0(b+1) = \bar{r}_u(b)\Phi_u^0(b)$ and $(1 - \frac{b}{u})\Phi_{u+1}^0(b) = (1 - \bar{r}_u(b))\Phi_u^0(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi_u^0(b)$ gives

$$R_u(b) = (1 - \bar{r}_u(b)) \frac{1 - r_u(b)}{1 - b/u} R_{u+1}(b) + \bar{r}_u(b) \frac{r_u(b)}{b/u} R_{u+1}(b+1),$$

where a term with an infeasible state is absent, and its bridge probability is 0 (if $b = h$ then $\bar{r}_u(b) = 0$, and if $u - b = h$ then $\bar{r}_u(b) = 1$). Also $\frac{r_u(b)}{b/u} = \varpi_u^+(b)$ and $\frac{1 - r_u(b)}{1 - b/u} = \varpi_u^-(b)$. So $R_u(b) = \mathbb{E}_t^{\text{br}}[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_N(h) = 1$. Iterating from $u = t + 1$ and using $\Phi_{t+1}^0(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \leq \frac{1}{2}$. If $b \leq u/2$ then $\varpi_u^+(b) \geq 1$ and $\varpi_u^-(b) - 1 \in [-\varepsilon, 0]$, and if $b \geq u/2$ the same holds with $+$ and $-$ exchanged. So $\varpi_u \geq \frac{1}{2}$. By Jensen's inequality, $\mathbb{E} \prod_u \varpi_u \geq \exp \sum_u \mathbb{E} \log \varpi_u$. Also $\log(1 + y) \geq y - y^2$ for $y \geq -\frac{1}{2}$, since the difference vanishes at 0 and its derivative $y(1 + 2y)/(1 + y)$ has the sign of y . Write $\bar{r} = \bar{r}_u(b)$. Then $u\bar{r} - b = \frac{N(u-2b)}{2(N-u)}$, and so

$$\mathbb{E}_t^{\text{br}}[\varpi_u - 1 \mid M_u = b] = \varepsilon(u - 2b) \frac{u\bar{r} - b}{b(u - b)} = \varepsilon \theta_u(b), \quad \mathbb{E}_t^{\text{br}}[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

Summing over u gives the bound. $\square$

Lemma A.6 (the weights w_t). *For each fixed t , $w_t \rightarrow t2^{-t-1}$ as $N \rightarrow \infty$. For every even $N \geq 2$, $\sum_t w_t(t+1)^3 \leq 124$.*

Proof. Each fraction in $\pi_t = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{h-i}{N-i}$ tends to $\frac{1}{2}$, so $N\pi_t \rightarrow t2^{1-t}$, and $Nc_1 \rightarrow 4$. By the proof of Lemma 2.4, w_t is proportional to $\binom{t}{1} \binom{N-1-t}{h-1}$, and the Chu–Vandermonde identity gives $\sum_t \binom{t}{j} \binom{N-1-t}{h-1} = \binom{N}{h+j}$. Now $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, because both sides are polynomials of degree 4 that agree at $t = 0, 1, 2, 3, 4$. Since $\binom{N}{h+j} \leq \binom{N}{h+1}$ for $j \geq 1$, we get $\sum_t w_t(t+1)^3 \leq 8 + 38 + 54 + 24 = 124$. $\square$

Under $\mathbb{P}_t^{\text{br}}$ let $K = M_u - 1$ and $K' = m - K$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t + 1$ and u . So K is hypergeometric, with m draws from $N - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \leq M_u \leq 7u/8\}$.

Lemma A.7. *Let $N \geq 16$, $1 \leq t \leq h$ and $t + 1 \leq u \leq N - 1$.*

- (a) $\mathbb{E}_t^{\text{br}}(u - 2M_u)^2 = \frac{(t-1)^2(N-u)^2}{(N-t-1)^2} + \frac{4(u-t-1)(h-1)(h-t)(N-u)}{(N-t-1)^2(N-t-2)} \leq (t-1)^2 + u$.
- (b) If $t \leq \frac{N+2}{4}$, then $A_t \geq 2(H_{N-1} - H_t - 1 - \frac{1}{t}) \geq 2(\log N - \log t - 3)$.
- (c) If $t \leq \frac{N}{8}$ and $u \geq 8(t+1)$, then $\mathbb{P}_t^{\text{br}}(\Omega_u^c) \leq 2e^{-0.0943u}$.
- (d) If $t \leq \frac{N}{8}$, then $B_t \leq 84(t+1 + \log N) + 683(t+1)^3 + 19100$.

Proof. (a) Since $N - 2 = 2(h - 1)$, the mean $\mathbb{E}K = \frac{m(h-1)}{N-t-1}$ gives $\mathbb{E}(u - 2M_u) = \frac{(t-1)(N-u)}{N-t-1}$, and the second term is $4 \text{Var } K$. For the bound use $N - u \leq N - t - 1$, $4(h-1)(h-t) = (N-t-1)^2 - (t-1)^2$, and $m(N-u) \leq u(N-t-2)$ for $u \geq t+2$.

(b) Since $b(u-b) \leq u^2/4$, $\theta_u(b) \geq \frac{2N(u-2b)^2}{(N-u)u^2}$. Keeping only $4 \text{Var } K$ in (a) gives $A_t \geq P_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2}$ with $P_t = \frac{2N(N-2)(N-2t)}{(N-t-1)^2(N-t-2)}$. Here $P_1 = \frac{2N}{N-3} > 2$, and for real $t \in [1, \frac{N+2}{4}]$, $\frac{d}{dt} \log P_t = -\frac{2}{N-2t} + \frac{2}{N-t-1} + \frac{1}{N-t-2} \geq \frac{3}{N-t-1} - \frac{2}{N-2t} \geq 0$, since the last step is equivalent to $4t \leq N + 2$. So $P_t \geq 2$. The sum is nonnegative and at least $H_{N-1} - H_t - (t+1) \sum_{u>t} u^{-2} \geq H_{N-1} - H_t - \frac{t+1}{t}$. For the second inequality use $H_{N-1} - H_t \geq \log \frac{N}{t+1} \geq \log N - \log t - \frac{1}{t}$.

(c) Here $m \geq \frac{7}{8}u$, so $\frac{u}{8} \leq \frac{m}{7}$. Since $t \geq 1$, $\mathbb{E}K \geq \frac{m}{2}$, and since $t \leq \frac{N}{8}$, $\mathbb{E}K' = \frac{m(h-t)}{N-t-1} \geq \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [26, Section 6], so $\mathbb{P}(K - \mathbb{E}K \leq -\sigma m) \leq e^{-2\sigma^2 m}$ for $\sigma > 0$, and the same holds for K' . Now $\{M_u < \frac{u}{8}\} \subset \{K - \mathbb{E}K < -\frac{5m}{14}\}$, and since $u - M_u = t + K'$, also $\{u - M_u < \frac{u}{8}\} \subset \{K' - \mathbb{E}K' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \leq e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \leq 2e^{-0.10778m} \leq 2e^{-0.0943u}$.

(d) Since $\bar{r}(u-b)^2 + (1-\bar{r})b^2 \leq u^2$, we have $Q_u(b) \leq \frac{(u-2b)^2 u^2}{b^2(u-b)^2}$. On Ω_u , $b(u-b) \geq \frac{7u^2}{64}$, so $Q_u \leq \frac{4096}{49} \frac{(u-2b)^2}{u^2} \leq 83.6 \frac{(u-2b)^2}{u^2}$, and by (a) these parts of B_t add up to at most $83.6 \sum_{u>t} (\frac{(t-1)^2}{u^2} + \frac{1}{u}) \leq 83.6(t + H_{N-1}) \leq 84(t+1 + \log N)$. Off Ω_u , $b(u-b) \geq u-1 \geq \frac{u}{2}$ gives $Q_u \leq 4u^2$. For $u < 8(t+1)$ these parts add up to at most $\frac{4}{3}(8(t+1))^3 \leq 683(t+1)^3$. For $u \geq 8(t+1)$, (c) bounds them by $8u^2 e^{-0.0943u}$, and with $q = e^{-0.0943}$ we have $8 \sum_{u \geq 1} u^2 q^u = 8q(1+q)/(1-q)^3 < 19100$. $\square$

Proof of Proposition 2.9. The upper bound is Theorem 2.6(b). For the lower bound let $t_0 = \lfloor N/8 \rfloor \geq 8$. By Proposition 2.5(a), $C_N = \varepsilon c_1 \sum_t w_t(1-\varepsilon)^{t-1} V_{t+1}(1)/\pi_t$. We use Jensen's

inequality for the probability vector (w_t) , then Proposition A.5 for $t \leq t_0$ and Lemma A.1(d) for $t > t_0$, where $V_{t+1}(1)/\pi_t \geq 1$. This gives

$$\log C_N \geq \log(\varepsilon c_1) + \sum_t w_t(t-1) \log(1-\varepsilon) + \sum_{t \leq t_0} w_t(\varepsilon A_t - \varepsilon^2 B_t).$$

The first sum. By $\log(1-\varepsilon) \geq -\varepsilon - \varepsilon^2$ and Proposition 2.5(c), it is at least $-(\varepsilon + \varepsilon^2) \sum_t w_t(t-1) \geq -2\varepsilon - 2\varepsilon^2$.

The terms with A_t . Since $t_0 \leq \frac{N+2}{4}$, Lemma A.7(b) gives $A_t \geq 2(\log N - \log t - 3)$ for $t \leq t_0$. Let $w_{\leq} = \sum_{t \leq t_0} w_t$. By Jensen's inequality $\sum_t w_t \log t \leq \log \sum_t w_t t \leq \log 3$, so

$$\sum_{t \leq t_0} w_t \varepsilon A_t \geq 2\varepsilon(\log N - 3)w_{\leq} - 2\varepsilon \log 3 \geq 2\varepsilon \log N - 2\varepsilon(1 - w_{\leq}) \log N - 6\varepsilon - 2\varepsilon \log 3.$$

By Proposition 2.5(c), $1 - w_{\leq} \leq 2 \sum_{t > t_0} t 2^{-t} = 2(t_0 + 2)2^{-t_0}$, and $N < 8(t_0 + 1)$. As $(t_0 + 2)2^{-t_0} \log(8t_0 + 8)$ decreases for $t_0 \geq 8$, we get $2\varepsilon(1 - w_{\leq}) \log N \leq 4 \cdot 10 \cdot 2^{-8} \log 72 \varepsilon < 0.67 \varepsilon$.

The terms with B_t . Here $t \leq t_0 \leq \frac{N}{8}$, so Lemma A.7(d) applies. With $\sum_t w_t(t+1) \leq 4$ (Proposition 2.5(c)) and Lemma A.6 it gives $\sum_{t \leq t_0} w_t B_t \leq 84(3 + 1 + \log N) + 683 \cdot 124 + 19100 = 84 \log N + 104128$.

Collecting terms, and using $c_1 = \frac{4}{N+2}$,

$$\log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N \geq -(2 + 0.67 + 6 + 2 \log 3) \varepsilon - \varepsilon^2(84 \log N + 104130),$$

which gives the lower bound since $2 + 0.67 + 6 + 2 \log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84 \log N + 1.05 \cdot 10^5) \leq \varepsilon \frac{84 \log N + 1.05 \cdot 10^5}{100(1 + \log N)}$. The fraction decreases in N , and its value at $N = 64$ is below 205. Since $11 + 205 < 5200$, the two-sided bound follows. $\square$

A.2 The coefficient of ε^2

We prove Proposition 2.11 in three steps. Step 1 writes $k_2(N)$ through the sums A_t . Step 2 finds the limit of $A_t - 2 \log N$ for fixed t . Step 3 sums over t .

Lemma A.8. (a) For every even $N \geq 2$, $k_2(N) = \sum_{t=1}^h w_t[A_t - (t-1)]$.

(b) For each fixed $t \geq 1$, as $N \rightarrow \infty$ over even N , $A_t - 2 \log N \rightarrow A_t^\infty := 2(\gamma - 1 - \log 2) + (t - 2)H_{t-1}$.

(c) For even $N \geq 64$ and $1 \leq t \leq h$, $-2 \log(4t) - 6 \leq A_t - 2 \log N \leq 2 + 2(t+11) \log(t+1)$.

Proof. (a) By Proposition 2.5(a) and (b), $c_2 = \sum_t [-(t-1)\pi_t + \partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0}]$. In Proposition A.5 the bridge does not depend on ε , and each ϖ_u is affine in ε and equals 1 at $\varepsilon = 0$. So by the conditional mean computed in its proof, $\partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0} = \pi_t \sum_u \mathbb{E}_t^{\text{br}}[(\varpi_u - 1)/\varepsilon] = \pi_t A_t$. Divide by c_1 .

(c) Both sides of Proposition A.4(c) are differentiable in ε on $[0, \frac{1}{4})$ and equal π_t at $\varepsilon = 0$, and $\kappa = 2\varepsilon/(1 - 4\varepsilon)$ has derivative 2 at 0. Comparing the derivatives at 0 and using the proof of (a) gives $A_t \leq 2H_{N-1} + 2\mathcal{G}(t)$. By Step 1 in the proof of Theorem 2.6(b), $\mathcal{G}(t) \leq (t+11) \log(t+1)$, and $H_{N-1} \leq 1 + \log N$. This is the upper bound. For $t \leq \frac{N+2}{4}$ the lower bound follows from Lemma A.7(b). For $t > \frac{N+2}{4}$ we use $A_t \geq 0$ and $N < 4t$.

(b) *The limit exists.* Fix t and let $N \geq 16(t+1)$. For $1 \leq b \leq u-1$,

$$\theta_u(b) = \frac{N}{N-u} \left[\frac{2(u-2b)^2}{u^2} + \vartheta_u(b) \right], \quad \vartheta_u(b) = \frac{(u-2b)^4}{2b(u-b)u^2} \geq 0,$$

since $\frac{1}{2b(u-b)} = \frac{2}{u^2} + \frac{(u-2b)^2}{2b(u-b)u^2}$. Write $\varsigma_t = \sum_{u>t} u^{-2}$. By Lemma A.7(a), the first part of A_t is

$$\sum_{u=t+1}^{N-1} \frac{2N \mathbb{E}_t^{\text{br}}(u-2M_u)^2}{(N-u)u^2} = \sum_{u=t+1}^{N-1} \frac{2N(t-1)^2(N-u)}{(N-t-1)^2u^2} + P_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2},$$

with P_t as in the proof of Lemma A.7(b). The terms of the first sum on the right are at most $8(t-1)^2/u^2$ and tend to $2(t-1)^2/u^2$, so this sum tends to $2(t-1)^2\varsigma_t$. The second sum is $H_{N-1} - H_t - (t+1) \sum_{u=t+1}^{N-1} u^{-2} = \log N + \gamma - H_t - (t+1)\varsigma_t + o(1)$, and $P_t = 2 + O(1/N)$. So the first part of A_t is $2 \log N + 2\gamma - 2H_t + 2((t-1)^2 - t - 1)\varsigma_t + o(1)$.

The second part is $\sum_{u=t+1}^{N-1} \frac{N}{N-u} \mathbb{E}_t^{\text{br}} \vartheta_u(M_u)$. For fixed u the law of $K = M_u - 1$ tends to the binomial law with $m = u - t - 1$ trials and success probability $\frac{1}{2}$, and $\frac{N}{N-u} \rightarrow 1$. So each term tends to $\mathbb{E} \vartheta_u(\bar{M}_u)$, where $\bar{M}_u = 1 + \nu_u$ and ν_u has this binomial law. We now dominate the terms uniformly in N . We have $u - 2M_u - \mathbb{E}(u - 2M_u) = -2(K - \mathbb{E}K)$. The last $N - u$ steps of the bridge also form a uniform sample, so $h - M_u$ is hypergeometric with $N - u$ draws, and $u - 2M_u - \mathbb{E}(u - 2M_u) = 2(h - M_u - \mathbb{E}(h - M_u))$. By Hoeffding's comparison with sampling with replacement [26, Theorem 4], applied to the convex function $z \mapsto (z - a)^4$, the fourth central moment of a hypergeometric variable is at most that of the binomial variable with the same number k of draws and the same mean. The latter is $kp(1-p)(1+3(k-2)p(1-p)) \leq k^2/2$, where p is the success probability. With $|\mathbb{E}(u - 2M_u)| \leq t$ and $(y+z)^4 \leq 8(y^4 + z^4)$, this gives $\mathbb{E}_t^{\text{br}}(u - 2M_u)^4 \leq 64 \min(u, N-u)^2 + 8t^4$. Also $\frac{N}{N-u} \leq 2u$, because $u(N-u) \geq N-1 \geq \frac{N}{2}$, and $\frac{N}{N-u} \min(u, N-u)^2 \leq 2u^2$, which is clear for $u \leq \frac{N}{2}$ and follows from $N(N-u) \leq 2u^2$ for $u > \frac{N}{2}$. On Ω_u we have $2M_u(u - M_u)u^2 \geq \frac{7}{32}u^4$, so the part of the u -th term on Ω_u is at most $\frac{32}{7u^4}(128u^2 + 16t^4u)$. Off Ω_u , $M_u(u - M_u) \geq \frac{u}{2}$ gives $\vartheta_u \leq u$, so that part is at most $2u^2 \mathbb{P}_t^{\text{br}}(\Omega_u^c)$. This is at most $4u^2 e^{-0.0943u}$ for $u \geq 8(t+1)$ by Lemma A.7(c), and at most $2u^2$ for the finitely many smaller u . The bound is summable and does not depend on N . By dominated convergence,

$$A_t - 2 \log N \rightarrow \tilde{A}_t := 2\gamma - 2H_t + 2((t-1)^2 - t - 1)\varsigma_t + \sum_{u>t} \mathbb{E} \vartheta_u(\bar{M}_u).$$

Let $\bar{\theta}_u = \mathbb{E} \frac{(u-2\bar{M}_u)^2}{2\bar{M}_u(u-\bar{M}_u)} = \mathbb{E} \left[\frac{2(u-2\bar{M}_u)^2}{u^2} + \vartheta_u(\bar{M}_u) \right]$. Since $\mathbb{E}(u - 2\bar{M}_u)^2 = (t-1)^2 + (u-t-1)$, the computation of the first part gives $\sum_{u=t+1}^{u_1} \mathbb{E} \frac{2(u-2\bar{M}_u)^2}{u^2} = 2 \log u_1 + 2\gamma - 2H_t + 2((t-1)^2 - t - 1)\varsigma_t + o(1)$ as $u_1 \rightarrow \infty$. Hence

$$\tilde{A}_t = \lim_{u_1 \rightarrow \infty} \left(\sum_{u=t+1}^{u_1} \bar{\theta}_u - 2 \log u_1 \right). \quad (11)$$

The closed form. Let $m = u - t - 1$ and $\nu = \nu_u$. For $0 \leq q \leq 1$ let $y = \frac{1+q}{2}$ and $\sigma(q) = \sum_{j=0}^{t-2} q^j$, with $\sigma = 0$ for $t = 1$. Since $(u-2b)^2 = u^2 - 4b(u-b)$ and $\frac{1}{b(u-b)} = \frac{1}{u} \left(\frac{1}{b} + \frac{1}{u-b} \right)$, we have $\bar{\theta}_u = \frac{u}{2} \mathbb{E} \left[\frac{1}{1+\nu} + \frac{1}{u-1-\nu} \right] - 2$, and $u-1-\nu$ has the law of $t+\nu$ because $m-\nu$ has the law of ν . For $b \geq 1$, $\mathbb{E} \frac{1}{b+\nu} = \int_0^1 q^{b-1} \mathbb{E} q^\nu dq = \int_0^1 q^{b-1} y^m dq$. Hence

$$\bar{\theta}_u = u \int_0^1 \frac{1+q^{t-1}}{2} y^m dq - 2 = \left[\frac{2t}{m+1} - \left(1 + \frac{t}{m+1} \right) 2^{-m} \right] - \mathcal{T}_m, \quad \mathcal{T}_m = u \int_0^1 (1-y) \sigma(q) y^m dq,$$

using $\frac{1+q^{t-1}}{2} = 1 - (1-y)\sigma(q)$, $\int_0^1 y^m dq = \frac{2(1-2^{-m-1})}{m+1}$ and $u = m+1+t$. Split $u = (m+1) + t$ in $\mathcal{T}_m$. Since $\frac{d}{dq}y^{m+1} = \frac{m+1}{2}y^m$, integration by parts gives $(m+1)\int_0^1 q^j \frac{1-q}{2} y^m dq = -\delta_{j0}2^{-m-1} + \int_0^1 y^{m+1}((j+1)q^j - jq^{j-1})dq$ for $j \geq 0$, and the integrands add up to $(t-1)q^{t-2}y^{m+1}$ over $0 \leq j \leq t-2$. With $I(j, k) = \int_0^1 q^j y^k dq$ and $(1-y)\sigma(q) = \frac{1-q^{t-1}}{2}$,

$$\mathcal{T}_m = -[t \geq 2]2^{-m-1} + (t-1)I(t-2, m+1) + \frac{t}{2} \int_0^1 (1-q^{t-1})y^m dq,$$

where $[t \geq 2]$ is 1 if $t \geq 2$ and 0 if $t = 1$, and the term with I is absent for $t = 1$. Sum over $0 \leq m \leq m_0$, so that $u_1 = m_0 + t + 1$. Since $\sum_{m \leq m_0} y^m = \frac{2(1-y^{m_0+1})}{1-q}$,

$$\begin{aligned} \sum_{u=t+1}^{u_1} \bar{\theta}_u &= 2tH_{m_0+1} - \sum_{m=0}^{m_0} \left(1 + \frac{t}{m+1}\right)2^{-m} + [t \geq 2](1 - 2^{-m_0-1}) \\ &\quad - (t-1) \sum_{k=0}^{m_0} I(t-2, k+1) - t \int_0^1 \sigma(q)(1-y^{m_0+1})dq. \end{aligned}$$

Let $m_0 \rightarrow \infty$. First $\sum_{m \geq 0} (1 + \frac{t}{m+1})2^{-m} = 2 + 2t \log 2$, and $\int_0^1 \sigma(1-y^{m_0+1})dq \rightarrow \int_0^1 \sigma dq = H_{t-1}$. For $t \geq 2$ write $\sum_{k \leq m_0} I(t-2, k+1) = 2 \int_0^1 q^{t-2} y^{\frac{1-y^{m_0+1}}{1-q}} dq$ and $q^{t-2} = 1 - (1-q^{t-2})$. The substitution $q = 2y - 1$ gives $\int_0^1 y^{\frac{1-y^{m_0+1}}{1-q}} dq = \sum_{j=2}^{m_0+2} \frac{1-2^{-j}}{j} = H_{m_0+2} - \frac{1}{2} - \log 2 + o(1)$, and by dominated convergence $\int_0^1 (1-q^{t-2})y^{\frac{1-y^{m_0+1}}{1-q}} dq \rightarrow \int_0^1 \sum_{j=0}^{t-3} q^j \frac{1+q}{2} dq = \frac{1}{2}(H_{t-2} + H_{t-1} - 1)$. So $\sum_{k \leq m_0} I(t-2, k+1) = 2H_{m_0+2} - 2 \log 2 - H_{t-2} - H_{t-1} + o(1)$. Collecting terms, with $2tH_{m_0+1} - 2(t-1)H_{m_0+2} = 2H_{m_0+1} + o(1)$,

$$\sum_{u=t+1}^{u_1} \bar{\theta}_u = 2H_{m_0+1} - 2 - 2 \log 2 + [t \geq 2] + (t-1)H_{t-2} - H_{t-1} + o(1).$$

Here $[t \geq 2] + (t-1)H_{t-2} - H_{t-1} = (t-2)H_{t-1}$ for all $t \geq 1$, and $H_{m_0+1} = \log u_1 + \gamma + o(1)$. So by (11), $\tilde{A}_t = 2(\gamma - 1 - \log 2) + (t-2)H_{t-1} = A_t^\infty$. $\square$

Proof of Proposition 2.11. Since $\sum_t w_t = 1$, Lemma A.8(a) gives $k_2(N) - 2 \log N = \sum_{t \leq h} w_t [A_t - 2 \log N - (t-1)]$. By Lemma A.8(c) and $w_t \leq 2t2^{-t}$, for $N \geq 64$ the t -th term is at most $C(1+t \log(t+1))t2^{-t}$ in absolute value, with C independent of N . By Lemma A.8(b), Lemma A.6 and dominated convergence,

$$k_2(N) - 2 \log N \rightarrow \sum_{t \geq 1} t2^{-t-1} [A_t^\infty - (t-1)].$$

To sum the series, note that $\sum_t t2^{-t-1} = 1$ and $\sum_t t(t-1)2^{-t-1} = 2$. Let $\mathcal{H}(q) = \sum_{k \geq 0} H_k q^k = \frac{\ell}{1-q}$ with $\ell = -\log(1-q)$, and let $D = \frac{d}{dq}$. With $k = t-1$,

$$\sum_{t \geq 1} t(t-2)2^{-t-1} H_{t-1} = \frac{1}{4} \sum_{k \geq 0} (k^2 - 1) H_k 2^{-k} = \frac{1}{4} [(qD)^2 \mathcal{H} - \mathcal{H}]_{q=1/2}.$$

Here $(qD)^2 \mathcal{H} = q \left[\frac{1+\ell}{(1-q)^2} + \frac{q}{(1-q)^3} + \frac{2q(1+\ell)}{(1-q)^3} \right]$ equals $8 + 6 \log 2$ at $q = \frac{1}{2}$, and $\mathcal{H}(\frac{1}{2}) = 2 \log 2$. So the sum is $2 + \log 2$, and the limit is $2(\gamma - 1 - \log 2) + (2 + \log 2) - 2 = 2\gamma - 2 - \log 2$. $\square$

B Proofs for Section 3

We keep the conventions of Section 3. We write C for an absolute constant whose value may change from line to line, and $\bar{c}$, A_0 , A_1 for fixed absolute constants. Since M is independent of X_1 , $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The family $\mathcal{T}_v$ has the law β_v of Lemma 2.4. Let $\bar{\beta}_v(s) = \mathbb{P}(|\mathcal{T}_v| \geq s)$, with $\bar{\beta}_v(s) = 1$ for $s \leq 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \leq s \leq N-1$,

$$\bar{\beta}_v(s) = \binom{N-s}{v-1} / \binom{N-1}{v-1} = \prod_{i=1}^{v-1} \left(1 - \frac{s-1}{N-i}\right) \leq e^{-(v-1)(s-1)/(N-1)}, \quad \beta_v(s) = \frac{v-1}{N-s} \bar{\beta}_v(s), \quad (12)$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \quad \frac{\bar{\beta}_v(s-1)}{\bar{\beta}_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \quad (13)$$

Lemma B.1 (monotone sandwich). *Let $0 \leq a \leq b \leq N$. If $m_h \leq a$, then $(b-a+1)f_N(b) \leq \mathbb{P}(a \leq M_N \leq b)$. If $b \leq m_h$, then $(b-a+1)f_N(a) \leq \mathbb{P}(a \leq M_N \leq b)$.*

Proof. By Lemma 3.4, $f_N(\ell) \geq f_N(b)$ for $a \leq \ell \leq b$ in the first case, and $f_N(\ell) \geq f_N(a)$ in the second. Sum over ℓ . $\square$

Lemma B.2 (tail). *Let $0 < \varepsilon < 1$, $k_0 \geq 1$ and $t \geq k_0$. For every integer $z \geq 1$,*

$$\mathbb{P}(M_t^{[k_0]} \geq z) \leq \frac{\varepsilon t H_z}{z}, \quad \mathbb{E}M_t^{[k_0]} \leq \varepsilon t \log \frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \geq z) \leq x H_z / z$, and $\mathbb{P}(Z_s^{(j)} \geq z) \leq x H_z / z$ for $j-1 \leq s \leq N$.

Proof. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \leq \sum_{v \text{ marked}} |\mathcal{T}_v|$. So $M_t^{[k_0]} \geq z$ needs a marked family of size at least z , or $\sum_{v \text{ marked}} |\mathcal{T}_v| \mathbf{1}\{|\mathcal{T}_v| < z\} \geq z$. By the union bound and Markov's inequality, and since marks and tree are independent, the probability is at most $\varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \geq z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}[|\mathcal{T}_v|; |\mathcal{T}_v| < z]$. By the first sum in Lemma 2.4, with t in place of N , this is at most $\varepsilon \frac{t}{z} + \frac{\varepsilon}{z} \sum_{d=1}^{z-1} \frac{t}{d+1} = \frac{\varepsilon t H_z}{z}$. For the mean, $\mathbb{E}M_t^{[k_0]} \leq \varepsilon \sum_{v=k_0+1}^t \frac{t}{v} \leq \varepsilon t \log \frac{t}{k_0}$. $\square$

Lemma B.3 (late start). *Let $j \geq 2$, $t \geq j-1$ and $z \geq 0$. Then $\mathbb{P}(M_t^{[j-1]} \geq z) \leq \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon \log \frac{2t}{j-1})\}$.*

Proof. Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k > \nu$ let $\theta_k = \log \frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. Since $r_k(c) \leq \varepsilon + c/k$ and $1 + \nu \leq e^\nu$, $\mathbb{E}[e^{\theta_{k+1} G_{k+1}} | G_k = c] \leq \exp\{c(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}) + \frac{\varepsilon \nu}{k+1-\nu}\}$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \geq \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. So $\mathbb{E}e^{\theta_{k+1} G_{k+1}} \leq e^{\varepsilon \nu / (k+1-\nu)} \mathbb{E}e^{\theta_k G_k}$ for $k \geq j-1$. Starting from $G_{j-1} = 0$ and using $\sum_{k=j}^t \frac{1}{k-\nu} \leq \log \frac{t-\nu}{\nu} \leq \log \frac{2t}{j-1}$, we get $\mathbb{E}e^{\theta_t G_t} \leq \exp\{\varepsilon \nu \log \frac{2t}{j-1}\}$. Markov's inequality with $\theta_t \geq \nu/t$ gives the claim. $\square$

Lemma B.4 (the first flipped family). *For $1 \leq d \leq N-1$,*

$$\frac{x}{d(d+1)} \left(1 - \frac{2x(N-d-1)}{N(d+2)}\right) \leq \mathbb{P}(D = d) \leq \frac{x}{d(d+1)}, \quad \mathbb{P}(D \geq d) \leq x \left(\frac{1}{d} - \frac{1}{N}\right).$$

Proof. By Lemma 3.3, $\mathbb{P}(D = d) = \sum_j \varepsilon(1 - \varepsilon)^{j-2} \beta_j(d)$, and $1 - (j - 2)\varepsilon \leq (1 - \varepsilon)^{j-2} \leq 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \geq d$. $\square$

Proposition B.5 (local lower bound). *There is an absolute constant C such that the following hold.*

- (a) For $2 \leq m \leq N/4$, $f_N(m) \geq \frac{x}{m(m+1)}(1 - e_1(m))$ with $e_1(m) = C(\frac{x \log(m+1)}{m} + \varepsilon \log N + me^{-m/9})$.
- (b) For $N/4 < m \leq N - 16$ and $r = N - m$, $f_N(m) \geq \frac{x}{m(m+1)}(1 - e_2(m))$ with $e_2(m) = C(\frac{x \log N}{r} + \frac{x \log N}{m} + Ne^{-r/40})$.

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z_{N-d}^{(j)}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. The law β_j of d is nearly flat near m , and this gives almost the full weight $\beta_j(m)$. Then we sum over j .

Proof. (a) *Step 1: one value of j .* Fix $2 \leq j \leq J := \lfloor N/8 \rfloor$ and put $\Gamma_d = d - Y_d + Z_{N-d}^{(j)}$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E} \sum_d \beta_j(d) \mathbf{1}\{\Gamma_d = m\}$, and every step of Γ lies in $\{-1, 0, 1\}$. On the event $\{Z_{N-1}^{(j)} \leq m - 1, Y_{2m} \leq m\}$ we have $\Gamma_1 = 1 + Z_{N-1}^{(j)} \leq m$, and at $d^\circ = m + Y_{2m} \leq 2m$ we have $\Gamma_{d^\circ} \geq d^\circ - Y_{2m} = m$, since Y is nondecreasing and $Z \geq 0$ (and $d^\circ \leq N - j + 1$ because $2m + j \leq N$). So Γ hits m at some $d \leq d^\circ$, and as β_j is nonincreasing the sum is at least $\beta_j(m + Y_{2m})$. By (13), $\beta_j(m + y)/\beta_j(m) = \prod_{i=1}^y (1 - \frac{j-2}{N-m-i}) \geq 1 - \frac{2y(j-2)}{N}$ for $y \leq m$, since each fraction is at most $\frac{2(j-2)}{N} \leq \frac{1}{4}$. Let $\mathbf{1}_\mathcal{E}$ be the indicator of the event. Then the sum is at least $\beta_j(m)(\mathbf{1}_\mathcal{E} - \frac{2(j-2)}{N}Y_{2m})$ in every case, and $\mathbf{1}_\mathcal{E} \geq 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z_{N-1}^{(j)} \geq m\}$. Taking means gives $Q_j(m) \geq \beta_j(m)[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z_{N-1}^{(j)} \geq m)] - \frac{2(j-2)}{N}\mathbb{E}Y_{2m}\beta_j(m)$.

Step 2: sum over j . Lemma B.2 gives $\mathbb{P}(Y_{2m} > m) \leq 2\varepsilon H_{m+1}$, $\mathbb{P}(Z_{N-1}^{(j)} \geq m) \leq xH_m/m$ and $\mathbb{E}Y_{2m} \leq 2\varepsilon m \log N$. If $2\varepsilon H_{m+1} + xH_m/m > 1$, then $e_1(m) > 1$ (once C is large) and there is nothing to prove. Otherwise the bracket is nonnegative. We multiply the bound of Step 1 by $\mathbb{P}(V = j)$ and sum over $2 \leq j \leq J$, which bounds $f_N(m)$ from below by Lemma 3.3. In the terms with $\mathbb{E}Y_{2m}$ we use $\mathbb{P}(V = j) \leq \varepsilon$. By the second sum in Lemma 2.4, the terms with $\mathbb{E}Y_{2m}$ add up to at most $\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \leq \frac{x}{m(m+1)} 8\varepsilon \log N$. By Lemma B.4, $\sum_{j \leq J} \mathbb{P}(V = j)\beta_j(m) \geq \frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j > J} \beta_j(m)$.

Step 3: the late first flips $j > J$. By (12), $\beta_j(m) \leq \frac{j-1}{N-m}e^{-(j-1)a}$ with $a = \frac{m-1}{N-1} \leq \frac{1}{4}$, and $aJ \geq \frac{m-1}{9}$ because $\lfloor N/8 \rfloor \geq \frac{N-1}{9}$ for $N \geq 64$. Summing the series, with $1 - e^{-a} \geq a/2$ and $N - m \geq \frac{3}{4}N$, gives $\varepsilon \sum_{j > J} \beta_j(m) \leq \frac{C\varepsilon}{N}(\frac{J}{a} + \frac{1}{a^2})e^{-(m-1)/9} \leq \frac{x}{m(m+1)}Cme^{-m/9}$. Collecting the terms gives (a).

(b) The same argument works with $J = w = \lfloor r/8 \rfloor \geq 2$, the event $\{Z_{N-1}^{(j)} \leq m - 1, Y_{m+w} \leq w\}$ and $d^\circ = m + Y_{m+w} \leq m + w$. Note that $m + w + j \leq N$. Only three estimates change. For $y \leq w$ and $j \leq J$ we have $N - m - i \geq \frac{7}{8}r$, so $\beta_j(m + y) \geq \beta_j(m)(1 - \frac{8y(j-2)}{7r})$. Since $w + 1 \geq r/8$, Lemma B.2 gives $\mathbb{P}(Y_{m+w} > w) \leq 8xH_N/r$ and $\mathbb{E}Y_{m+w} \leq x \log N$, and the terms with $\mathbb{E}Y_{m+w}$ add up to at most $\frac{x}{m(m+1)} \cdot \frac{Cx \log N}{m}$. For the dropped terms, $a \geq \frac{1}{5}$ and $J \geq \frac{r-7}{8}$, so $\varepsilon \sum_{j > J} \beta_j(m) \leq \frac{C\varepsilon(J+1)}{r}e^{-J/5} \leq \frac{x}{m(m+1)}CNe^{-r/40}$. $\square$

Proposition B.6 (cumulative upper bound). *There is an absolute constant C such that for $6 \leq m \leq N/2$ with $m \geq 16x \log m$ we have $\mathbb{P}(M_N \geq m) \leq \frac{x}{m}(1 + C \frac{x \log^2 m}{m})$.*

Idea of the proof. Given the first flip at time j , the count M_N is at most $D + Z_j$, the size of the first flipped family plus the count from the rest of the tree. The event $D \geq m$ gives the main term x/m . The other terms come from a shift of D by Z_j , from a large Z_j , and from a late first flip, and each is smaller by a factor of order $x \log^2 m/m$.

Proof. Step 1: split by the first flip. Let $J = \lfloor N/4 \rfloor$ and $Z_j = Z_{N-1}^{(j)}$. By Lemma 3.3, given $V = j$ the count M_N has the law of $D - Y_D + Z_{N-D}^{(j)} \leq D + Z_j$, with D of law β_j independent of Z_j . On $\{V > J\}$ there is no mark on $2, \dots, J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V > J\}$. For $0 \leq z \leq m/8$, (13) gives $\bar{\beta}_j(m-z) \leq \bar{\beta}_j(m) \varrho_j^z$ with $\varrho_j = 1 + \frac{j-1}{N-m-j+2} \leq 1 + \frac{4(j-1)}{N}$, and $\varrho^z - 1 \leq z(\varrho - 1)\varrho^z$. Hence

$$\mathbb{P}(M_N \geq m) \leq \sum_{j=2}^J \mathbb{P}(V = j) \left[\bar{\beta}_j(m) (1 + (\varrho_j - 1) \varrho_j^{m/8} \mathbb{E} Z_j) + \mathbb{P}\left(Z_j > \frac{m}{8}\right) \right] + \mathbb{P}(V > J) \mathbb{P}(M_N^{[J]} \geq m).$$

This gives four terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V = j) \bar{\beta}_j(m) \leq \mathbb{P}(D \geq m) \leq \frac{x}{m}$ by Lemma B.4.

Step 2: the shift term. We use $\mathbb{P}(V = j) \leq \varepsilon$, $\bar{\beta}_j(m) \leq e^{-(j-1)(m-1)/(N-1)}$, $\varrho_j^{m/8} \leq e^{(j-1)m/(2N)}$ and $\mathbb{E} Z_j \leq x \log \frac{N}{j-1}$ (Lemma B.2). As $m \geq 6$, the exponents add up to at most $-(j-1)a$ with $a = \frac{m}{3N}$. With $i = j-1$ and $\log \frac{N}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$, the shift term is at most $\frac{4\varepsilon x}{N} [\log \frac{m}{3} \sum_{i \geq 1} i e^{-ai} + \sum_{i \leq 1/a} i \log \frac{1}{ai}] \leq \frac{4\varepsilon x}{N} \cdot \frac{\log m + C}{a^2} \leq \frac{x}{m} \cdot \frac{Cx \log m}{m}$, since $\sum_i i e^{-ai} \leq a^{-2}$ and $t \log \frac{1}{at} \leq \frac{1}{ea}$ for $0 < t \leq 1/a$.

Step 3: the big- Z term. Let $\mathcal{K} = 32 \log(m/x)$ and $j_{\mathcal{K}} = \lfloor \mathcal{K}N/m \rfloor + 1$. Note that $m \geq 16x \log 6 > 28x$, so $\mathcal{K} \geq 2$. For $j \leq j_{\mathcal{K}}$, Lemma B.2 gives $\mathbb{P}(Z_j > m/8) \leq 8xH_m/m$, and these j contribute at most $\varepsilon \frac{\mathcal{K}N}{m} \cdot \frac{8xH_m}{m} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$. For $j > j_{\mathcal{K}}$ we have $\frac{2(N-1)}{j-1} \leq \frac{2m}{\mathcal{K}} \leq m$, so $\varepsilon \log \frac{2(N-1)}{j-1} \leq \frac{x \log m}{N} \leq \frac{m}{16N}$, and Lemma B.3 with $t = N-1$ gives $\mathbb{P}(Z_j \geq m/8) \leq e^{-(j-1)m/(32N)}$. Summing over $j-1 \geq j_{\mathcal{K}} > \mathcal{K}N/m$ gives at most $\varepsilon \frac{e^{-\mathcal{K}/32}}{1-e^{-m/(32N)}} \leq \varepsilon \left(\frac{32N}{m} + 1 \right) \frac{x}{m} \leq \frac{x}{m} \cdot \frac{33x}{m}$.

Step 4: no early flip. Finally $\mathbb{P}(V > J) = (1 - \varepsilon)^{J-1} \leq e^{\varepsilon - 3x/16}$, since $J \geq 3N/16$, and Lemma B.3 with $j-1 = J \in [\frac{3N}{16}, \frac{N}{4}]$ and $t = N$ gives $\mathbb{P}(M_N^{[J]} \geq m) \leq \exp\{-\frac{3m}{32} + \frac{x}{8} \log \frac{32}{3}\}$. As $x \leq m/28$, the last term is at most $e^{1/4 - m/13} \leq \frac{x}{m} \cdot \frac{Cx \log^2 m}{m}$. $\square$

Lemma B.7 (mode bound). *There are absolute constants $\bar{c} > 0$ and A_0 such that if $\varepsilon \log N \leq \bar{c}$ and $A_0 x \log(x+2) \leq N/4$, then $m_h \leq A_0 x \log^2(x+1)$.*

Proof. Let $\ell = \lceil Ax \log(x+2) \rceil$. Since $\log(\ell+1) \leq \log(A+2) + 2 \log(x+2)$, we have $\frac{x \log(\ell+1)}{\ell} \leq \frac{\log(A+2)+2}{A}$, and $\ell e^{-\ell/9}$ is small when A is large. So we can fix the absolute constants A and $\bar{c}$ such that $e_1(\ell) \leq \frac{1}{2}$ in Proposition B.5(a) whenever $\varepsilon \log N \leq \bar{c}$. We take $A_0 \geq A+1$, so that $2 \leq \ell \leq N/4$. Then $f_N(\ell) \geq \frac{x}{2\ell(\ell+1)}$. If $m_h \geq \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1)f_N(\ell) \leq \mathbb{P}(M_N \geq \ell) \leq xH_\ell/\ell$. So $m_h \leq (\ell+1)(1+2H_\ell)$ in every case. As $\log \ell \leq C \log(x+2)$ and $\log(x+2) \leq 2 \log(x+1)$, this is at most $A_0 x \log^2(x+1)$ once A_0 is large. $\square$

Proposition B.8 (local upper bound). *There are absolute constants A_1 and C such that if $\varepsilon \log N \leq \bar{c}$ and $A_1 x \log^2(x+1) \leq m \leq N/8$, then*

$$f_N(m) \leq \frac{x}{m(m+1)} (1 + C \Delta(m)^{1/2}), \quad \Delta(m) = \frac{x \log^2 m}{m} + \varepsilon \log N + \frac{m}{N}.$$

Idea of the proof. To the right of its mode f_N is nonincreasing, so $(k+1)f_N(m)$ is at most the probability of $\{m-k \leq M_N \leq m\}$, which is a difference of two tails. We bound the first tail from above by Proposition B.6 and the second from below by summing Proposition B.5. Then we choose k to balance the errors.

Proof. We take $A_1 \geq 2A_0$ so large that Lemma B.7 applies and every $m' \geq m/2$ satisfies $m' \geq 6$ and $m' \geq 16x \log m'$ (as $t/\log t$ increases for $t \geq e$).

Step 1: a difference of tails. Let $1 \leq k \leq m/2$. Then $m-k \geq m/2 \geq m_h$, and Lemma B.1 with $a = m-k$ and $b = m$ gives $(k+1)f_N(m) \leq \mathbb{P}(M_N \geq m-k) - \mathbb{P}(M_N \geq m+1)$.

Step 2: bound each tail. Proposition B.6 at $m-k$ bounds the first term by $\frac{x}{m-k} + \frac{Cx^2 \log^2 m}{m^2}$. For the second we sum Proposition B.5(a) over $m < \ell \leq N/4$. Since $\sum_{m < \ell \leq N/4} \frac{1}{\ell(\ell+1)} \geq \frac{1}{m+1} - \frac{4}{N}$, $\frac{\log(\ell+1)}{\ell}$ and $\ell e^{-\ell/9}$ decrease for $\ell \geq 9$, and $me^{-m/9} \leq C/m \leq C\Delta(m)$, we get $\mathbb{P}(M_N \geq m+1) \geq \frac{x}{m+1}(1 - C\Delta(m))$. As $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$, this gives $f_N(m) \leq \frac{x}{(m-k)(m+1)} + \frac{Cx\Delta(m)}{(k+1)m}$.

Step 3: choose k . If $\Delta(m) \leq \frac{1}{16}$, take $k = \lceil m\Delta(m)^{1/2} \rceil \leq \frac{m}{2}$. Then $\frac{m}{m-k} \leq 1 + \frac{2k}{m} \leq 1 + 2\Delta^{1/2} + \frac{2}{m}$ and $\frac{\Delta}{k+1} \leq \frac{\Delta^{1/2}}{m}$, which gives the claim, as $\frac{1}{m} \leq \Delta(m) \leq \Delta(m)^{1/2}$. If $\Delta(m) > \frac{1}{16}$, take $k = \lfloor m/2 \rfloor$. Then $f_N(m) \leq \frac{x}{m(m+1)}(2 + C\Delta(m))$, and the claim follows after enlarging C , since $\Delta(m)$ is bounded by an absolute constant ($\frac{x \log^2 m}{m}$ decreases for $m \geq e^2$ and is bounded at $m = A_1 x \log^2(x+1)$). $\square$

Proof of Theorem 3.14. Step 1: the hypotheses hold. Put $\ell_x = \log(x+1) \geq \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x \ell_x^2 \leq m \leq \delta'_N N$. Since $x \log^2 N/N \rightarrow 0$, for large N we have $\varepsilon(1 + \log N) \leq \min(\bar{c}, \frac{1}{100})$, $\omega_N \geq \max(A_1, 16)$, $\delta'_N \leq \frac{1}{8}$ and $A_0 x \log(x+2) \leq N/4$. The standing assumptions $N \geq 64$ and $\varepsilon \leq \frac{1}{4}$ hold as well.

Step 2: the errors are small on $\mathcal{R}_N$. The function $\frac{x \log^2 m}{m}$ decreases for $m \geq e^2$. At $m = \omega_N x \ell_x^2$ we have $m \geq e^2$ and $\log m \leq \log \omega_N + 3\ell_x$, as $\log x$ and $\log \ell_x$ are at most ℓ_x . So on $\mathcal{R}_N$, uniformly in $x \geq 1$, $\frac{x \log^2 m}{m} \leq \frac{(\log \omega_N + 3\ell_x)^2}{\omega_N \ell_x^2} \leq \frac{1}{\omega_N} (\frac{\log \omega_N}{\log 2} + 3)^2 \rightarrow 0$. Also $m \geq \omega_N \log^2 2 \rightarrow \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \leq 2 \log^2 m$ for $m \geq 3$, Proposition B.5(a) has $e_1(m) \leq C(\frac{x \log^2 m}{m} + \varepsilon \log N + m e^{-m/9}) \rightarrow 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \leq \frac{x \log^2 m}{m} + \varepsilon \log N + \delta'_N \rightarrow 0$ there, and Proposition B.8 applies. So $f_N(m) = \frac{x}{m(m+1)}(1 + o(1))$ uniformly on $\mathcal{R}_N$.

Step 3: the mirror term. Lemma B.7 gives $m_h \leq A_0 x \ell_x^2 \leq h$ for large N , since $x \ell_x^2 \leq x \log^2(N+1) = o(N)$. As $N-m \geq h$, Lemma 3.4 gives $0 \leq f_N(N-m) \leq f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon \log N \rightarrow 0$, $C_N \leq \frac{4x}{N(N+2)} N^{2\varepsilon} e^{5200\varepsilon} = \frac{4x}{N^2} (1 + o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)} \cdot 8\delta_N'^2 (1 + o(1))$, since $m(m+1) \leq 2\delta_N'^2 N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)] = \frac{x}{2m(m+1)}(1 + o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1 + o(1))$ because $m \rightarrow \infty$. $\square$

C Proofs for the bulk

We prove Theorems 3.9 and 3.10, Lemma 3.11 and Corollary 3.12. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v , and μ_v is the mark of v . By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v , with mean N/v , and $\sum_{v=2}^N \beta_v(d) = N/(d(d+1))$ for $1 \leq d \leq N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d} - 1}{d(d+1)}, \quad \Psi_\infty(\theta) = \sum_{d \geq 1} \frac{e^{i\theta d} - 1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\text{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.9

Here we assume only $N \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$. Let $Z_+ = \sum_{v=2}^N \mu_v |\mathcal{T}_v|$ be the total size of the flipped families.

Step 1: $M_N = Z_+$ with high probability. Two families are nested or disjoint. Suppose that no marked vertex has a marked strict descendant. Then the marked families are disjoint, each of their vertices has exactly one mark on its path to the root, and the other vertices have none. So $M_N = Z_+$ by Lemma 3.1. The number of pairs of a marked vertex and a marked strict descendant has mean $\varepsilon^2 \mathbb{E} \sum_v (|\mathcal{T}_v| - 1)$, since marks and tree are independent. Hence

$$\mathbb{P}(M_N \neq Z_+) \leq \varepsilon^2 \sum_{v=2}^N \left(\frac{N}{v} - 1 \right) \leq \varepsilon^2 N (H_N - 1) \leq \frac{x^2 \log N}{N}.$$

Step 2: given the tree, a product. Fix $\theta \in [-\pi, \pi]$. Given $\mathcal{T}$ the marks are independent, so $\mathbb{E}[e^{i\theta Z_+} \mid \mathcal{T}] = \prod_{v=2}^N (1 + a_v)$ with $a_v = \varepsilon(e^{i\theta|\mathcal{T}_v|} - 1)$. Here $|a_v| \leq 2\varepsilon \leq \frac{1}{2}$ and $\text{Re } a_v \leq 0$. Put $W = \sum_v a_v$, so that $\text{Re } W \leq 0$. For complex a with $|a| \leq \frac{1}{2}$ the series of the logarithm gives $|\log(1 + a) - a| \leq |a|^2$. So $\prod_v (1 + a_v) = e^{W + \varrho}$ with $|\varrho| \leq \sum_v |a_v|^2 \leq 4\varepsilon^2(N - 1)$. Since $|e^w - 1| \leq |w|e^{|w|}$,

$$\left| \prod_v (1 + a_v) - e^W \right| \leq e^{\text{Re } W} 4\varepsilon^2(N - 1) e^{4\varepsilon^2(N - 1)} \leq \frac{4x^2}{N} e^{4x^2/N}.$$

Step 3: replace W by its mean. Write $W = \varepsilon F$ with $F = \sum_{v=2}^N (e^{i\theta|\mathcal{T}_v|} - 1)$. Then $\mathbb{E}W = \varepsilon \sum_d (e^{i\theta d} - 1) \sum_v \beta_v(d) = x\Psi_N(\theta)$, which is $\log \widehat{\text{CP}}_N(\theta)$. Put $\delta = W - \mathbb{E}W$. Taylor's formula for $s \mapsto e^{\mathbb{E}W + s\delta}$ on $[0, 1]$ gives

$$e^W - e^{\mathbb{E}W} - e^{\mathbb{E}W} \delta = \delta^2 \int_0^1 (1 - s) e^{\mathbb{E}W + s\delta} ds.$$

The exponent $\mathbb{E}W + s\delta = (1 - s)\mathbb{E}W + sW$ has real part at most 0, so the right side has modulus at most $\frac{1}{2}|\delta|^2$. As $\mathbb{E}\delta = 0$, we get $|\mathbb{E}e^W - e^{\mathbb{E}W}| \leq \frac{1}{2}\varepsilon^2 \text{Var } F$, where $\text{Var } F = \mathbb{E}|F - \mathbb{E}F|^2$.

Step 4: the variance of F . The variable F is a function of the independent parents $U_1, \dots, U_{N-1}$. Let $F^{(k)}$ be F with U_k replaced by an independent copy U'_k . The Efron–Stein inequality [25] says that $\text{Var } F \leq \frac{1}{2} \sum_k \mathbb{E}|F - F^{(k)}|^2$. We use it for a complex function of independent variables that are not identically distributed, so we recall the proof. Let $\mathbb{E}_k$ be the average over U_k alone, and let $\zeta_k = \mathbb{E}[F - \mathbb{E}_k F \mid U_1, \dots, U_k]$. Then $\zeta_k = \mathbb{E}[F \mid U_1, \dots, U_k] - \mathbb{E}[F \mid U_1, \dots, U_{k-1}]$, so $F - \mathbb{E}F = \sum_k \zeta_k$ and the ζ_k are orthogonal. Hence $\text{Var } F = \sum_k \mathbb{E}|\zeta_k|^2 \leq \sum_k \mathbb{E}|F - \mathbb{E}_k F|^2$, and $\mathbb{E}|F - \mathbb{E}_k F|^2 = \frac{1}{2} \mathbb{E}|F - F^{(k)}|^2$, because given the other parents F and $F^{(k)}$ are independent with the same law.

Replacing U_k by U'_k moves the family of $k + 1$ from U_k to U'_k . So $|\mathcal{T}_v|$ changes only for vertices $v \geq 2$ on the path from U_k to the root or on the path from U'_k to the root. These paths do not depend on the choice at step k , since $U_k, U'_k \leq k$. Let $\text{dp}(u)$ be the depth of u , namely the number of edges between u and the root. There are at most $\text{dp}(U_k) + \text{dp}(U'_k)$ such vertices, and each term of F moves by at most 2. Hence $\mathbb{E}|F - F^{(k)}|^2 \leq 8 \mathbb{E}[\text{dp}(U_k)^2 + \text{dp}(U'_k)^2] = 16 \mathbb{E} \text{dp}(U_k)^2$.

Since $\text{dp}(u + 1) = 1 + \text{dp}(U_u)$, the generating function $g_u(s) = \mathbb{E}s^{\text{dp}(u)}$ satisfies $g_1 = 1$ and $u g_{u+1}(s) = s \sum_{j \leq u} g_j(s)$. Subtracting the same identity for $u - 1$ gives $u g_{u+1}(s) = (u - 1 + s) g_u(s)$

for $u \geq 2$, and $g_2(s) = s$. So $g_u(s) = \prod_{i=1}^{u-1} \frac{i-1+s}{i}$. Thus $\text{dp}(u)$ is a sum of independent Bernoulli($1/i$) variables with $1 \leq i < u$, and $\mathbb{E} \text{dp}(u)^2 \leq H_{u-1}^2 + H_{u-1}$. The parent U_k is independent of the depths of $1, \dots, k$, so $\mathbb{E} \text{dp}(U_k)^2 \leq H_N^2 + H_N$. We get $\text{Var } F \leq \frac{1}{2}(N-1) \cdot 16(H_N^2 + H_N) \leq 8N(H_N^2 + H_N)$, and the bound of Step 3 is at most $4x^2(H_N^2 + H_N)/N$.

Step 5: Fourier inversion. For every integer m ,

$$\mathbb{P}(Z_+ = m) - \text{CP}_N(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-im\theta} (\mathbb{E} e^{i\theta Z_+} - \widehat{\text{CP}}_N(\theta)) d\theta,$$

so the left side is at most $\sup_{\theta} |\mathbb{E} e^{i\theta Z_+} - \widehat{\text{CP}}_N(\theta)|$ in absolute value. By Steps 2 to 4 this is at most $\frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$. Finally $|f_N(m) - \mathbb{P}(Z_+ = m)| \leq \mathbb{P}(M_N \neq Z_+)$, and Step 1 bounds it. This proves Theorem 3.9. $\square$

C.2 Proof of Theorem 3.10

We write $B(t) = -it \log|t| - \frac{\pi}{2}|t|$ for real $t \neq 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u \log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_{∞} .

Fact 1. For real θ , $|\Psi_N(\theta) - \Psi_{\infty}(\theta)| \leq \sum_{d \geq N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \leq \pi$ we have $\Psi_{\infty}(\theta) = -i\theta \log(-i\theta) + R(\theta)$ with $|R(\theta)| \leq \theta^2(0.631|\log|\theta|| + 1.53)$. To see this, note first that for $0 < |s| \leq 1$, $s \neq 1$,

$$\sum_{d \geq 1} \frac{s^d - 1}{d(d+1)} = \chi(s) := \frac{(1-s) \log(1-s)}{s},$$

because $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s) - s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_{\infty}(\theta) = \chi(e^{i\theta})$. Write $1 - e^{i\theta} = (-i\theta)u$ with $u = \text{sinc}(\theta/2) e^{i\theta/2}$, where $\text{sinc } y = \sin y/y$ is positive for $|y| \leq \frac{\pi}{2}$. The arguments of $-i\theta$ and of u add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1 - e^{i\theta}) = \log(-i\theta) + \log u$ for principal logarithms. Hence

$$R(\theta) = -i\theta \log(-i\theta)(ue^{-i\theta} - 1) - i\theta u e^{-i\theta} \log u.$$

With $y = \theta/2$ we have $|ue^{-i\theta} - 1| = |\text{sinc } y - e^{iy}| \leq (1 - \text{sinc } y) + |1 - e^{iy}| \leq \frac{y^2}{6} + |y| \leq (\frac{1}{2} + \frac{\pi}{24})|\theta|$. Next $|\log u|^2 = \log^2 \text{sinc}(\theta/2) + \theta^2/4$, and $-\log \text{sinc } y = \sum_{k \geq 1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log \text{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log u| \leq (\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2})^{1/2}|\theta| \leq 0.53|\theta|$. Finally $|\log(-i\theta)| \leq |\log|\theta|| + \frac{\pi}{2}$ and $|u| \leq 1$. Since $\frac{1}{2} + \frac{\pi}{24} \leq 0.631$ and $0.631 \cdot \frac{\pi}{2} + 0.53 \leq 1.53$, the bound follows.

Fact 3. For $|\theta| \leq \pi$, $-\text{Re } \Psi_{\infty}(\theta) = \sum_d \frac{1 - \cos \theta d}{d(d+1)} \geq \frac{|\theta|}{2\pi}$. We may take $\theta \neq 0$. Keep only the terms with $d \leq D := \lfloor \pi/|\theta| \rfloor$ and use $1 - \cos v \geq 2v^2/\pi^2$ for $|v| \leq \pi$. This gives at least $\frac{2\theta^2}{\pi^2} \sum_{d \leq D} \frac{d}{d+1} \geq \frac{\theta^2 D}{\pi^2} \geq \frac{|\theta|}{2\pi}$, since $D \geq \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_{∞} with parameter x , namely the law of $\sum_{d \geq 1} d \mathcal{N}_d$. Its characteristic function is $e^{x\Psi_{\infty}(\theta)}$. By Fact 2, for $t \neq 0$ and $x \geq \max(1, |t|/\pi)$,

$$x\Psi_{\infty}(t/x) - it \log x = B(t) + xR(t/x), \quad |xR(t/x)| \leq \frac{t^2}{x} (0.631(|\log|t|| + \log x) + 1.53). \quad (14)$$

So the characteristic function of $(S_x - x \log x)/x$ tends to $e^{B(t)}$ as $x \rightarrow \infty$, for every real t . The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.10. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict \log c$, and hence

$$e^{B(t)} = e^{B(ct)} \cdot e^{B((1-c)t)} e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [24].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ and the substitution $\theta = t/x$,

$$x \text{CP}_N(m) = \frac{1}{2\pi} \int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)} dt, \quad A(t) = x\Psi_N(t/x) - it \log x.$$

Let $|t| \leq \pi x$. By (14) and Fact 1,

$$|A(t) - B(t)| \leq \frac{t^2}{x} (0.631(|\log|t|| + \log x) + 1.53) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\text{Re } A(t), \text{Re } B(t)) \leq -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \leq |A - B| e^{\max(\text{Re } A, \text{Re } B)}$, and $\frac{2x}{N} \leq \frac{1}{2}$. The integrals of $t^2 e^{-|t|/2\pi}$, of $t^2 |\log|t|| e^{-|t|/2\pi}$ and of $e^{-|t|/2\pi}$ over the real line are finite. So

$$\frac{1}{2\pi} \int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}| dt \leq C \left(\frac{1 + \log x}{x} + \frac{x}{N} \right).$$

Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi} \int_{|t| > \pi x} |e^{B(t)}| dt = \frac{2}{\pi^2} e^{-\pi^2 x/2} \leq \frac{C}{x}$. Since $f_L(\lambda_m) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)} dt$, this proves (a).

Proof of (b). By Theorem 3.9, $|x f_N(m) - x \text{CP}_N(m)| \leq x \delta_B$. Add (a).

Proof of (c). Let $x \rightarrow \infty$ and $x^3 \log^2 N/N \rightarrow 0$. Then $x \delta_B \rightarrow 0$ and $x/N \rightarrow 0$, so the bound of (b) tends to 0. Let U be uniform on $[0, 1)$ and independent of M_N . Then $(M_N + U - x \log x)/x$ has the density g_N with $g_N(\lambda) = x f_N(m)$ for $\lambda_m \leq \lambda < \lambda_m + \frac{1}{x}$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $g_N \rightarrow f_L$ uniformly on compact sets. By Scheffé's lemma $g_N \rightarrow f_L$ in L^1 , and since $U/x \rightarrow 0$ this gives the convergence in law. At the crossing $x_N \sim 2 \log N$ by Theorem 2.7, so both conditions hold. $\square$

C.3 Proof of Lemma 3.11

Step 1: a real integral. We claim that for every real λ and $k \geq 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi} \int_0^\infty (-s)^k e^{-s \log s - \lambda s} \sin(\pi s) ds. \quad (15)$$

Since $B(-t) = \overline{B(t)}$, the definition of f_L gives $f_L(\lambda) = \frac{1}{\pi} \text{Re} \int_0^\infty \Phi(-it) dt$ with $\Phi(u) = \exp(u \log u + \lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u = \rho e^{i\varphi}$ with $-\pi < \varphi \leq -\frac{\pi}{2}$. Then $\text{Re}(u \log u) = \rho(\log \rho \cos \varphi - \varphi \sin \varphi)$, where $\cos \varphi \leq 0$ and $\varphi \sin \varphi \geq 0$. For $\log \rho \geq |\lambda| + 1$ this gives $\text{Re}(u \log u + \lambda u) \leq -\rho(|\cos \varphi| + \varphi \sin \varphi) \leq -\rho$, because $|\cos \varphi| + \varphi \sin \varphi \geq 1$ on $[-\pi, -\frac{\pi}{2}]$ (for $\psi = \varphi + \pi \in [0, \frac{\pi}{2}]$ it equals $\cos \psi + (\pi - \psi) \sin \psi \geq \cos \psi + \sin \psi \geq 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u = -it$ to the lower side of the negative real axis, where $u = -s$ and $\log u = \log s - i\pi$. There $u \log u = -s \log s + i\pi s$, and $dt = i du = -i ds$. Hence

$$f_L(\lambda) = \frac{1}{\pi} \text{Re} \left[-i \int_0^\infty e^{-s \log s - \lambda s} e^{i\pi s} ds \right] = \frac{1}{\pi} \int_0^\infty e^{-s \log s - \lambda s} \sin(\pi s) ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (15). Also f_L'' is bounded on the real line, because $t^2 e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2: certified enclosures. Let $-0.223 \leq \lambda \leq 0$ and $k \in \{1, 2\}$, and split the integral in (15) at $\delta = 2^{-20}$ and at $T = 30$. On $[0, \delta]$ we have $-s \log s \leq 1/e$, $-\lambda s \leq 0.223 s$ and $|\sin \pi s| \leq \pi s$, so this part is at most $\pi e^{1/e+0.223\delta} \delta^{k+2}/(k+2)$ in absolute value. On $[T, \infty)$ we have $s \log s + \lambda s \geq cs$ with $c = \log 30 - 0.223 > 3$, so this part is at most $\int_T^\infty s^2 e^{-cs} ds \leq T^2 e^{-cT}$. Both are below 10^{-17} . On $[\delta, T]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [28], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229) \in [9.232 \cdot 10^{-6}, 9.234 \cdot 10^{-6}], \quad f_L'(-0.2227) \in [-6.548 \cdot 10^{-6}, -6.546 \cdot 10^{-6}],$$

and $f_L''(\lambda) \in [-0.0792, -0.0786]$ for every $\lambda \in [-0.2229, -0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297 .

Step 3: the mode. Write $[a, b] = [-0.2229, -0.2227]$. By Step 2, f_L' is strictly decreasing on $[a, b]$ with $f_L'(a) > 0 > f_L'(b)$. So f_L' has exactly one zero λ_0 in $[a, b]$, and f_L is strictly increasing on $[a, \lambda_0]$ and strictly decreasing on $[\lambda_0, b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty, \lambda_*)$ and nonincreasing on $[\lambda_*, \infty)$. If $\lambda_* > \lambda_0$, then f_L would be nondecreasing just to the right of λ_0 , which is false. In the same way $\lambda_* < \lambda_0$ is impossible. So $\lambda_* = \lambda_0$. By Step 2 again, $\lambda_0 \in (-0.22278299, -0.22278297)$. This proves (a) and (b).

Step 4: the constant κ_L . Let $\delta_0 = \min(\lambda_0 - a, b - \lambda_0) > 0$. For $|\lambda - \lambda_0| \leq \delta_0$, Taylor's formula with $f_L'(\lambda_0) = 0$ and $f_L'' \leq -0.078$ on $[a, b]$ gives $f_L(\lambda_0) - f_L(\lambda) \geq 0.039(\lambda - \lambda_0)^2$. For $|\lambda - \lambda_0| > \delta_0$ the monotonicity of Step 3 gives $f_L(\lambda) \leq f_L(\lambda_0 \pm \delta_0) \leq f_L(\lambda_0) - 0.039\delta_0^2$. Since $\delta_0 < 1$, (c) holds with $\kappa_L = 0.039\delta_0^2$. $\square$

C.4 Proof of Corollary 3.12

We first prove a statement at finite N . Assume the standing assumptions of Section 3, let κ_L be as in Lemma 3.11(c), and let E_1 be the bound of Theorem 3.10(b). Put

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

We claim that if $E' < \kappa_L/2$ and $m_h \leq N/4$, then every mode m of f_N satisfies $|m - x \log x - \lambda_0 x| \leq x\sqrt{2E'/\kappa_L}$, and so does m^* for every mode $a^* \geq N/2$ of P_N .

Let m' be the integer nearest to $x \log x + \lambda_0 x$. This number increases in $x \geq 1$ and is at least $\lambda_0 > -\frac{1}{2}$, so $m' \geq 0$. Also $|\lambda_{m'} - \lambda_0| \leq \frac{1}{2x}$, and since $f_L'(\lambda_0) = 0$, $f_L(\lambda_{m'}) \geq f_L(\lambda_0) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde{m}$ be a mode of f_N or $\tilde{m} = m^*$. We claim that $f_N(\tilde{m}) \geq f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N - m^*) = 2P_N(N - m^*) \geq 2P_N(N - m') \geq f_N(m')$. Here $N - m^* \geq \lceil N/2 \rceil \geq \lceil N/4 \rceil \geq m_h$. So f_N is nonincreasing on $\{\lceil N/4 \rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4 \rceil$ to $\lceil N/2 \rceil$, so by Lemma B.2

$$f_N(N - m^*) \leq f_N(\lceil N/2 \rceil) \leq \frac{\mathbb{P}(M_N \geq \lceil N/4 \rceil)}{N/4} \leq \frac{16xH_N}{N^2}.$$

Now Theorem 3.10(b), used at $\tilde{m}$ and at m' , gives

$$f_L(\lambda_{\tilde{m}}) \geq x f_N(\tilde{m}) - E_1 \geq x f_N(m') - \frac{16x^2 H_N}{N^2} - E_1 \geq f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2 H_N}{N^2} \geq f_L(\lambda_0) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.11(c) gives $(\lambda_{\tilde{m}} - \lambda_0)^2 \leq 2E'/\kappa_L$, which is the claim.

Now take $x = x_N$ at the crossing. By Theorem 2.7, $x \sim 2 \log N$. So $x\delta_B$, x/N , $16x^2 H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all $O(\log x/x)$, and $E' = O(\log x/x)$. For large N the standing assumptions hold, $E' < \kappa_L/2$, and Lemma B.7 gives $m_h \leq A_0 x \log^2(x+1) \leq N/4$. Hence $|m^* - x \log x - \lambda_0 x| \leq x\sqrt{2E'/\kappa_L} = O(\sqrt{x \log x})$, and the same holds for m_h . $\square$

D Proofs for Section 4

We use the notation of Section 4, with $\tau_0 = 0$. For a first run D let $\tau_m^D = D + \tau'_{m-1}$, where (τ'_m) is an independent copy of (τ_m) , and $\tau_0^D = 0$. Then $u_m^D(k) = \mathbb{P}(\tau_m^D = k)$ and $q_m^D(k) = \mathbb{P}(\tau_{m-1}^D < k \leq \tau_m^D)$. Conditioning on the last run gives, for $m, k \geq 1$,

$$\begin{aligned} q_m(k) &= \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j), & q_{m+1}^D(k) &= \sum_{j \geq 1} \bar{F}(j) u_m^D(k-j), \\ u_m^D(k) &= \sum_{d \geq 1} \mathbb{P}(D = d) u_{m-1}(k-d). \end{aligned} \tag{16}$$

Since τ_m and τ_m^D increase strictly, $\sum_m q_m(k) = \sum_m q_m^D(k) = 1$, $\sum_m u_m^D(k) \leq 1$ and $\sum_{m \leq m'} q_m(k) = \mathbb{P}(\tau_{m'} \geq k)$.

Sections D.1 to D.4 treat $\alpha > 1$ and prove Theorems 4.4 and 4.7 together. Section D.5 treats $0 < \alpha < 1$ and proves Theorem 4.8.

D.1 The setting for $\alpha > 1$

Let $\alpha > 1$, so that $\mu = \zeta(\alpha) < \infty$. The proof of the center asymptotics uses the exponent only through the following three facts about a norming sequence (a_m) of positive numbers.

- (F1) For $m \geq 3$ we have $a_m = a(m)$, where $a(x) = c_a x^\kappa (\log x)^\beta$ on $[3, \infty)$ with constants $c_a > 0$, $0 < \kappa < 1$ and $\beta \geq 0$.
- (F2) The variables $(\tau_m - m\mu)/a_m$ converge in law to a variable G that has a bounded, uniformly continuous density g with $g(z) \rightarrow 0$ as $|z| \rightarrow \infty$. With $z_{m,k} = (k - m\mu)/a_m$, the numbers $\delta_m = \sup_k |a_m u_m(k) - g(z_{m,k})|$ tend to 0, and $\bar{u} = \sup_{m \geq 1} \sup_k a_m u_m(k)$ is finite.
- (F3) There is a constant C_B such that $u_m(k) \leq C_B m k^{-\alpha-1}$ for all $m \geq 1$ and $k \geq 2m\mu$.

The function a in (F1) is increasing, and $a(\lambda x)/a(x) \rightarrow \lambda^\kappa$ as $x \rightarrow \infty$, uniformly for λ in a compact subset of $(0, \infty)$. So $a_m \rightarrow \infty$ and $a_m/a_{m-1} \rightarrow 1$. For $1 < \alpha < 2$ we take $a_m = m^{1/\alpha}$. Then (F1) holds with $\kappa = 1/\alpha$ and $\beta = 0$, (F2) is Lemma 4.3(ii) and (iii), and (F3) is Lemma D.1 below. We write ω for the modulus of continuity of g and $\bar{g}(x) = \sup_{|y| \geq x} g(y)$. So $\omega(\eta) \rightarrow 0$ as $\eta \rightarrow 0$ and $\bar{g}(x) \rightarrow 0$ as $x \rightarrow \infty$.

D.2 Two lemmas on renewal sums

A sum of m runs that is much larger than its mean is most likely made by a single long run. The first lemma turns this into a local bound.

Lemma D.1. *Let $\alpha > 1$. There is a constant $C_B = C_B(\alpha)$ such that $u_m(k) \leq C_B m k^{-\alpha-1}$ for all $m \geq 1$ and $k \geq 2m\mu$.*

Proof. Here C depends only on α and may change from line to line. Since $\mu \geq 1$ we have $m \leq k/2$. Let $r > 2$ be a constant chosen below, and put $y = k/r$. We split according to whether some run is at least y .

One big jump. By the mean value theorem $\mathbb{P}(\xi = \ell) \leq \alpha \ell^{-\alpha-1}$. So for each i the probability $\mathbb{P}(\tau_m = k, \xi_i \geq y) = \sum_{\ell \geq y} \mathbb{P}(\xi = \ell) \mathbb{P}(\tau_m - \xi_i = k - \ell)$ is at most $\alpha r^{\alpha+1} k^{-\alpha-1}$, and the m indices give at most $\alpha r^{\alpha+1} m k^{-\alpha-1}$.

No big jump. We may take $y \geq 1$, since otherwise $\max_i \xi_i < y$ is impossible. Let $x_0 > 0$ and $s \geq 0$. Markov's inequality for $e^{s\tau_m/y}$ and $e^x \leq 1 + x + \frac{x^2}{2}e^x$ for $x \geq 0$ give

$$\mathbb{P}(\tau_m \geq x_0, \max_{i \leq m} \xi_i < y) \leq \exp\left(-\frac{sx_0}{y} + m\left(\frac{s}{y} \mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2} \mathbb{E}[\xi^2; \xi < y]\right)\right). \quad (17)$$

We take $x_0 = k$ and use $\mathbb{E}[\xi; \xi < y] \leq \mu$ and $k - m\mu \geq k/2$. Then the exponent is at most $-sr/2 + \frac{1}{2}r^2 s^2 e^s m k^{-2} \mathbb{E}[\xi^2; \xi < y]$. There are three cases.

Let $1 < \alpha < 2$. Then $\mathbb{E}[\xi^2; \xi < y] \leq \alpha \sum_{\ell < y} \ell^{1-\alpha} \leq Cy^{2-\alpha}$, so the exponent is at most $-sr/2 + Cr^\alpha s^2 e^s m/k^\alpha$. Take $s = \frac{1}{2} \log(k^\alpha/m)$, which is positive since $m \leq k/2 < k^\alpha$. Then $e^s m/k^\alpha = e^{-s}$ and $s^2 e^{-s} \leq 1$, so the bound is $C(m/k^\alpha)^{r/4}$. With $r = 4 + 4/(\alpha - 1)$ and $m \leq k$ this is at most $C(m/k^\alpha)(k^{1-\alpha})^{1/(\alpha-1)} = Cmk^{-\alpha-1}$.

Let $\alpha > 2$. Then $\mathbb{E}[\xi^2; \xi < y] \leq \mathbb{E}\xi^2 < \infty$, so the exponent is at most $-sr/2 + Cr^2 s^2 e^s m/k^2$. Take $s = \frac{1}{2} \log(k^2/m)$, which is positive since $m < k^2$. As before the bound is $C(m/k^2)^{r/4}$. With $r = 4\alpha$ and $m \leq k$ this is $Cm m^{\alpha-1} k^{-2\alpha} \leq Cmk^{-\alpha-1}$.

Let $\alpha = 2$. Then $\mathbb{E}[\xi^2; \xi < y] \leq 2 \sum_{\ell < y} \ell^{-1} \leq 2 \log(ek)$, so the exponent is at most $-sr/2 + r^2 s^2 e^s m \log(ek)/k^2$. Take $s = \frac{1}{2} \log \frac{k^2}{m \log(ek)}$, which is positive since $m \log(ek) \leq \frac{1}{2} k \log(ek) < k^2$. The bound is $C(m \log(ek)/k^2)^{r/4}$. With $r = 20$ and $m \leq k$ this is at most $Cmk^{-6} \log^5(ek) \leq Cmk^{-3}$. $\square$

Lemma D.2. *Let $\alpha > 1$ and assume (F1) and (F2). Let D be any a.s. finite first run. Then*

$$\sup_k \left| \frac{a_m}{\mu} q_m(k) - g(z_{m,k}) \right|, \quad \sup_k |a_m u_m^D(k) - g(z_{m,k})|, \quad \sup_k \left| \frac{a_m}{\mu} q_{m+1}^D(k) - g(z_{m,k}) \right|$$

tend to 0 as $m \rightarrow \infty$. For $m \geq 2$, $\sup_k q_m(k) \leq \mu \bar{u}/a_{m-1}$ and $\sup_k u_m^D(k) \leq \bar{u}/a_{m-1}$.

Proof. By (16), q_m and u_m^D are mixtures of shifts of u_{m-1} with total weights μ and 1. This gives the bounds for $m \geq 2$.

Fix $J \geq 1$ and $Z > 0$, put $\sigma_J = \sum_{j > J} \bar{F}(j)$ and write $z = z_{m,k}$. In the sum for $q_m(k)$ the terms with $j > J$ add up to at most $\bar{u} \sigma_J / a_{m-1}$. For $j \leq J$ we have $|a_{m-1} u_{m-1}(k-j) - g(z_{m-1,k-j})| \leq \delta_{m-1}$ and $z_{m-1,k-j} = z a_m / a_{m-1} + (\mu - j) / a_{m-1}$. If $|z| \leq Z$, then $|z_{m-1,k-j} - z| \leq \eta_m = Z |a_m / a_{m-1} - 1| + (J + \mu) / a_{m-1}$. If $|z| > Z$, then for large m both $g(z)$ and $g(z_{m-1,k-j})$ lie in $[0, \bar{g}(Z/2)]$. Since $\mu - \sum_{j \leq J} \bar{F}(j) = \sigma_J$, we get, uniformly in k for large m ,

$$|a_{m-1} q_m(k) - \mu g(z)| \leq \mu (\delta_{m-1} + \omega(\eta_m) + \bar{g}(Z/2)) + (\bar{u} + \|g\|_\infty) \sigma_J.$$

We let $m \rightarrow \infty$, then $J, Z \rightarrow \infty$, and use $a_m / a_{m-1} \rightarrow 1$. This gives the first limit. The second follows in the same way from the sum for u_m^D , with the weights $\mathbb{P}(D = d)$ and with $\mathbb{P}(D > J)$ in place of σ_J . The third follows from the sum for q_{m+1}^D , with $a_m u_m^D$ in place of $a_{m-1} u_{m-1}$, the second limit in place of $\delta_{m-1} \rightarrow 0$, and the shift $z_{m,k-j} - z_{m,k} = -j/a_m$. $\square$

D.3 The center for $\alpha > 1$

Proposition D.3. *Let $\alpha > 1$ and assume (F1), (F2) and (F3). Let D be any a.s. finite first run and put $b_h = a(h/\mu)$. As $N = 2h \rightarrow \infty$,*

$$C_N \sim \frac{2}{b_h} \int_{\mathbb{R}} g(z)^2 dz.$$

Idea of the proof. We split the sum in Proposition 4.2 by the number m of runs into four regions. In the main region $|m\mu - h| \leq Mb_h$ both factors are given by the local limit theorem, and the terms form a Riemann sum for $2b_h^{-1} \int_{-M}^M g^2$. For $m \leq h/(4\mu)$ the level h is far above the mean of τ_m , and (F3) makes these terms $O(h^{-\alpha})$. In the other two regions (the m between these two, and the m above the main region) one factor of each term is $O(1/b_h)$, and the other factors add up to a tail probability of G , which is small when M is large.

Proof. By (F1), $b_h = o(h)$ and $h^{-\alpha} = o(1/b_h)$, since $\kappa < 1 < \alpha$. Also $a(h/(8\mu)) \geq b_h/9$ and $a(h/(2\mu)) \geq b_h/3$ for large h , since $8^{-\kappa} > \frac{1}{9}$ and $2^{-\kappa} > \frac{1}{3}$. And $a_m/b_h \rightarrow 1$ uniformly in m with $|m\mu - h| \leq 2Mb_h$, for each fixed M . By Proposition 4.2, $C_N = \sum_{m \geq 1} [u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h)]$. Fix $M > 1$, put $m_{\pm} = (h \pm Mb_h)/\mu$, and let $\Sigma_1, \dots, \Sigma_4$ be the parts of this sum over the regions $\mathcal{R}_1 = \{m \leq h/(4\mu)\}$, $\mathcal{R}_2 = \{h/(4\mu) < m < m_{-}\}$, $\mathcal{R}_3 = \{m_{-} \leq m \leq m_{+}\}$ and $\mathcal{R}_4 = \{m > m_{+}\}$.

Region $\mathcal{R}_1$. Here $h \geq 4m\mu$, so $u_m(h) \leq C_B m h^{-\alpha-1} \leq C_B h^{-\alpha}/(4\mu)$ by (F3). Also $q_1(h) = h^{-\alpha}$. For $m \geq 2$ we split the sum for $q_m(h)$ at $j = h/2$. The terms with $j \geq h/2$ add up to at most $(h/2)^{-\alpha}$. For $j < h/2$ we have $h - j > h/2 \geq 2\mu(m-1)$, so the other terms add up to at most $\mu C_B (m-1)(h/2)^{-\alpha-1} \leq 2^{\alpha-1} C_B h^{-\alpha}$. Since $\sum_m q_{m+1}^D(h) \leq 1$ and $\sum_m u_m^D(h) \leq 1$, we get $\Sigma_1 = O(h^{-\alpha})$, which is $o(1/b_h)$.

Region $\mathcal{R}_2$. For large h we have $m-1 \geq h/(8\mu)$ here, so $a_m \geq a_{m-1} \geq b_h/9$ and $u_m(h), u_m^D(h) \leq 9\bar{u}/b_h$ by (F2) and Lemma D.2. Let $m' = \lfloor m_{-} \rfloor$. Then $\sum_{m \in \mathcal{R}_2} q_m(h) \leq \mathbb{P}(\tau_{m'} \geq h)$ and $\sum_{m \in \mathcal{R}_2} q_{m+1}^D(h) \leq \mathbb{P}(D + \tau_{m'}' \geq h)$, which is at most $\mathbb{P}(D \geq b_h) + \mathbb{P}(\tau_{m'}' \geq h - b_h)$. Since $m'\mu \leq h - Mb_h$ and $a_{m'} \leq b_h$, the event $\{\tau_{m'}' \geq h - b_h\}$ implies $(\tau_{m'}' - m'\mu)/a_{m'} \geq M - 1$. As D is a.s. finite and $(\tau_{m'}' - m'\mu)/a_{m'}$ converges in law to G , which has a continuous law, we get $\limsup_h b_h \Sigma_2 \leq 18\bar{u} \mathbb{P}(G \geq M - 1)$.

Region $\mathcal{R}_4$. For large h we have $m-1 \geq h/(2\mu)$ here, so $u_m(h), u_m^D(h) \leq 3\bar{u}/b_h$. Let $k = \lfloor m_{+} \rfloor$. Then $\sum_{m \in \mathcal{R}_4} q_m(h) = \mathbb{P}(\tau_k < h)$ and, since $D \geq 1$, $\sum_{m \in \mathcal{R}_4} q_{m+1}^D(h) = \mathbb{P}(D + \tau_k' < h) \leq \mathbb{P}(\tau_k < h)$. Now $(h - k\mu)/a_k \leq (\mu - Mb_h)/a_k \rightarrow -M$, so $\limsup_h b_h \Sigma_4 \leq 6\bar{u} \mathbb{P}(G \leq -M)$.

Region $\mathcal{R}_3$. Here $m \rightarrow \infty$ and $a_m = b_h(1 + o(1))$ uniformly. By Lemma D.2 and $\delta_m \rightarrow 0$, uniformly in $m \in \mathcal{R}_3$,

$$u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h) = \frac{2\mu}{a_m^2} (g(z_{m,h})^2 + o(1)).$$

Put $y_m = (h - m\mu)/b_h$. Then $|z_{m,h} - y_m| \leq M|b_h/a_m - 1| \rightarrow 0$, so $g(z_{m,h})^2 = g(y_m)^2 + o(1)$. The y_m with $m \in \mathcal{R}_3$ are the points of an arithmetic progression with step μ/b_h that lie in $[-M, M]$. Hence

$$b_h \Sigma_3 = 2(1 + o(1)) \frac{\mu}{b_h} \sum_{m \in \mathcal{R}_3} (g(y_m)^2 + o(1)) \rightarrow 2 \int_{-M}^M g^2.$$

Adding the four regions,

$$\limsup_{h \rightarrow \infty} \left| b_h C_N - 2 \int g^2 \right| \leq 2 \int_{|z| > M} g^2 + 18\bar{u} \mathbb{P}(G \geq M - 1) + 6\bar{u} \mathbb{P}(G \leq -M),$$

and letting $M \rightarrow \infty$ gives $C_N \sim 2b_h^{-1} \int g^2$. $\square$

Proof of Theorem 4.4. Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. As noted in Section D.1, (F1) to (F3) hold, with $b_h = (h/\mu)^{1/\alpha}$. So Proposition D.3 gives $C_N \sim 2I_\alpha(\mu/h)^{1/\alpha}$, and the second form follows from Lemma 4.3(iv) and $h = N/2$. $\square$

D.4 Proof of Theorem 4.7

Let $\alpha \geq 2$ and let g be the standard normal density. It is bounded and uniformly continuous, it vanishes at $\pm\infty$, and $\int g^2 = 1/(2\sqrt{\pi})$. Every $\mathbb{P}(\xi = k)$ is positive, so ξ has maximal span 1. Fact (F3) is Lemma D.1. So by Proposition D.3 it is enough to find (a_m) with (F1) such that $(\tau_m - m\mu)/a_m$ converges in law to the standard normal law. Then Gnedenko's local limit theorem [21, Theorem 4.2.1] gives $\delta_m \rightarrow 0$, and $\bar{u} < \infty$ follows as in the proof of Lemma 4.3(iii). This is (F2), and we get $C_N \sim 1/(\sqrt{\pi} b_h)$.

(a) Let $\alpha > 2$. Summation by parts gives $\mathbb{E}\xi^2 = \sum_{k \geq 1} (2k-1)\bar{F}(k) = 2\zeta(\alpha-1) - \zeta(\alpha)$, so v is as stated and finite. It is positive because ξ is not constant. We take $a_m = \sqrt{vm}$, which has (F1) with $\kappa = \frac{1}{2}$ and $\beta = 0$, and the central limit theorem gives the normal limit. So $b_h = \sqrt{vh/\mu}$ and $C_N \sim (\pi vh/\mu)^{-1/2} = \sqrt{2\mu/(\pi vN)}$. For the stationary start Proposition 4.1 gives $E_N = N^{1-\alpha}(1 + O(1/N))/(2(\alpha-1)\mu)$, and dividing gives $R_N \sim C_2(\alpha)N^{3/2-\alpha}$.

(b) Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}\xi^2 = \infty$. The truncated second moment is

$$U(x) = \mathbb{E}[\xi^2; \xi \leq x] = \sum_{k \leq x} k^2(k^{-2} - (k+1)^{-2}) = \sum_{k \leq x} \frac{2k+1}{(k+1)^2} = 2 \log x + O(1),$$

which varies slowly. So ξ is in the domain of attraction of the normal law, and $(\tau_m - m\mu)/a_m$ converges in law to the standard normal law for every sequence (a_m) with $mU(a_m)/a_m^2 \rightarrow 1$ [22, Theorem XVII.5.3 and eq. (XVII.5.23)]. For $a(x) = \sqrt{x \log x}$ we have $mU(a(m))/a(m)^2 = (\log m + \log \log m + O(1))/\log m \rightarrow 1$. We take $a_m = a(m)$ for $m \geq 3$ and $a_1 = a_2 = 1$. This has (F1) with $\kappa = \beta = \frac{1}{2}$. So $b_h^2 = (h/\mu) \log(h/\mu) \sim N \log N/(2\mu)$ and $C_N \sim \sqrt{2\mu/(\pi N \log N)}$. For the stationary start $E_N = N^{-1}(1 + O(1/N))/(2\mu)$ by Proposition 4.1, so $R_N \sim \sqrt{\pi \log N/(8\mu^3 N)}$.

In both cases the fresh start has the edge $E_N = \frac{1}{2}N^{-\alpha}$ and the same center asymptotics. So R_N is of order $N^{1/2-\alpha}$ for $\alpha > 2$ and of order $N^{-3/2}(\log N)^{1/2}$ for $\alpha = 2$, and it tends to 0. $\square$

D.5 Proof of Theorem 4.8

Let $0 < \alpha < 1$ with a fresh start, $a_m = m^{1/\alpha}$ and $z_{m,k} = k/a_m$. There is no centering now. In this section μ is not used, and constants may depend on α .

Lemma D.4. *Let $0 < \alpha < 1$.*

- (i) τ_m/a_m converges in law to a positive variable G with $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ for $s \geq 0$. The law of G has a bounded, uniformly continuous density g that vanishes on $(-\infty, 0]$ and at $+\infty$.
- (ii) (Gnedenko) The constant $\bar{u} = \sup_{m \geq 1} \sup_k a_m u_m(k)$ is finite, and $\delta_m = \sup_k |a_m u_m(k) - g(k/a_m)| \rightarrow 0$ as $m \rightarrow \infty$.
- (iii) There is a constant C_B such that $u_m(k) \leq C_B m k^{-\alpha-1}$ for all $m \geq 1$ and $k \geq a_m$.
- (iv) $g(y) \leq C_B y^{-1-\alpha}$ for $y > 1$. So $\int_0^\infty t g(t)^2 dt < \infty$.

Proof. (i) Summation by parts gives $1 - \mathbb{E}x^\xi = (1-x) \sum_{k \geq 1} \bar{F}(k)x^{k-1}$ for $0 \leq x < 1$. With $x = e^{-v}$ this is $1 - \mathbb{E}e^{-v\xi} = (e^v - 1)\text{Li}_\alpha(e^{-v})$, and [29, 25.12.12] gives $\text{Li}_\alpha(e^{-v}) = \Gamma(1-\alpha)v^{\alpha-1} + \zeta(\alpha) + O(v)$ as $v \downarrow 0$. So $1 - \mathbb{E}e^{-v\xi} = \Gamma(1-\alpha)v^\alpha + O(v)$. As $1/\alpha > 1$, this gives $m(1 - \mathbb{E}e^{-s\xi/a_m}) \rightarrow \Gamma(1-\alpha)s^\alpha$ for each $s \geq 0$. Hence $\mathbb{E}e^{-s\tau_m/a_m} \rightarrow e^{-\Gamma(1-\alpha)s^\alpha}$, and the continuity theorem for Laplace transforms [22, Section XIII.1] gives the limit $G \geq 0$. Both sides of $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ are continuous on $\text{Re } s \geq 0$ and analytic on $\text{Re } s > 0$, with the principal branch on the right. So the identity holds at $s = -i\theta$, and $|\mathbb{E}e^{i\theta G}| = \exp(-\Gamma(1-\alpha)\cos(\frac{\pi\alpha}{2})|\theta|^\alpha)$ is integrable. Fourier inversion and the Riemann–Lebesgue lemma give g . It vanishes on $(-\infty, 0)$ because $G \geq 0$, and at 0 by continuity.

(ii) As in Lemma 4.3(iii), ξ has maximal span 1, and Gnedenko's local limit theorem [21, Theorem 4.2.1] gives $\delta_m \rightarrow 0$. Then $a_m u_m(k) \leq \|g\|_\infty + 1$ for large m and $a_m u_m(k) \leq a_m$ for the others.

(iii) We argue as in Lemma D.1, with $r = 4 + 4/\alpha$, $y = k/r$ and $t = k^\alpha/m \geq 1$. For $m = 1$ the claim is $\mathbb{P}(\xi = k) \leq \alpha k^{-\alpha-1}$. Let $m \geq 2$. The bound $\alpha r^{\alpha+1} m k^{-\alpha-1}$ for the event that some run is at least y holds as before. For the other event we may take $y \geq 1$. Split the indices into $I_1 = \{1, \dots, \lfloor m/2 \rfloor\}$ and $I_2 = \{\lfloor m/2 \rfloor + 1, \dots, m\}$, and write $\tau_I = \sum_{i \in I} \xi_i$. On $\{\tau_m = k\}$ one of τ_{I_1}, τ_{I_2} is at least $k/2$, so by independence

$$\mathbb{P}(\tau_m = k, \max_i \xi_i < y) \leq \sum_{i=1,2} \mathbb{P}(\tau_{I_i} \geq k/2, \max_{j \in I_i} \xi_j < y) \sup_\ell \mathbb{P}(\tau_{I_{3-i}} = \ell).$$

Each half has at least $\lfloor m/2 \rfloor \geq m/3$ indices, so by (ii) the supremum is at most $\bar{u}/a_{\lfloor m/2 \rfloor} \leq 3^{1/\alpha} \bar{u}/a_m$. For the first factor we use (17) for the sum over I_i , with $x_0 = k/2$ and at most m terms. Now $\mathbb{E}[\xi; \xi < y] \leq \sum_{\ell < y} \bar{F}(\ell) \leq C y^{1-\alpha}$ and $\mathbb{E}[\xi^2; \xi < y] \leq C y^{2-\alpha}$, and $m y^{-\alpha} = r^\alpha/t$. So the exponent is at most $-sr/2 + C(s + s^2 e^s)/t$, and $s = \frac{1}{2} \log t$ makes the second term bounded. Hence the first factor is at most $C t^{-r/4} = C t^{-1-1/\alpha}$, and $t^{-1-1/\alpha}/a_m = m k^{-\alpha-1}$.

(iv) Fix $y > 1$ and let $k_m = \lceil y a_m \rceil$. By (iii) and $m a_m^{-\alpha} = 1$, $a_m u_m(k_m) \leq C_B m a_m (y a_m)^{-\alpha-1} = C_B y^{-1-\alpha}$. By (ii) and the continuity of g , the left side tends to $g(y)$. The integral converges because g is bounded. $\square$

Idea of the proof of Theorem 4.8. With a fresh start, Proposition 4.2 reads $C_N = \sum_{m \geq 1} u_m(h) [q_m(h) + q_{m+1}(h)]$. Put $t_m = h/a_m$. Step 1 finds $q_m(h)$ on the scale where t_m is of order 1, namely $m q_m(h) \approx t_m g(t_m)/\alpha$. Step 2 sums over m . The terms with t_m very large or very small are negligible, and the others form a Riemann sum in the variable $\log m$. Step 3 identifies the limit $4 \int_0^\infty t g(t)^2 dt$ as the density at $\frac{1}{2}$ of $G_1/(G_1 + G_2)$ and computes it.

Proof of Theorem 4.8. Fix $M > 1$.

Step 1. Let $g_1(t) = \int_0^t w^{-\alpha} g(t-w) dw$. We claim that $m q_m(h) = g_1(t_m) + o(1)$ as $h \rightarrow \infty$, uniformly in m with $t_m \in [1/(2M), 2M]$. Such m tend to infinity. Since $|k/a_{m-1} - k/a_m| \leq 2M(a_m/a_{m-1} - 1) \rightarrow 0$ for $0 \leq k \leq h$, Lemma D.4(ii) for u_{m-1} and the uniform continuity of g give $a_m u_{m-1}(k) = g(k/a_m) + o(1)$ uniformly in $0 \leq k \leq h$. As $m j^{-\alpha} = (j/a_m)^{-\alpha}$ and $u_{m-1}(0) = 0$, the sum for q_m in (16) becomes

$$m q_m(h) = \frac{1}{a_m} \sum_{j=1}^{h-1} \left(\frac{j}{a_m} \right)^{-\alpha} \left(g(t_m - \frac{j}{a_m}) + o(1) \right).$$

For $x > 0$ we have $\frac{1}{a_m} \sum_{j \leq x a_m} (j/a_m)^{-\alpha} \leq \int_0^x w^{-\alpha} dw = x^{1-\alpha}/(1-\alpha)$. With $x = t_m \leq 2M$ this shows that the $o(1)$ terms add up to $o(1)$. With $x = \eta$ it shows that the terms with $j \leq \eta a_m$ are

$O(\eta^{1-\alpha})$, and so is $\int_0^\eta w^{-\alpha} g(t_m - w) dw$. The other terms form a Riemann sum with mesh $1/a_m$ of the function $w^{-\alpha} g(t - w)$, which is uniformly continuous on $\{\eta \leq w \leq t \leq 2M\}$. Letting $\eta \rightarrow 0$ proves the claim.

Next, the Laplace transform of g_1 is the product of those of $w^{-\alpha}$ and g , namely $\Gamma(1 - \alpha)s^{\alpha-1}e^{-\Gamma(1-\alpha)s^\alpha}$. This is $\frac{1}{\alpha}$ times $-\frac{d}{ds}e^{-\Gamma(1-\alpha)s^\alpha}$, which is the transform of $tg(t)$. Both g_1 and $tg(t)$ are continuous on $(0, \infty)$, so $g_1(t) = tg(t)/\alpha$.

Step 2. If $t_m > M$, then $h > Ma_m$ and Lemma D.4(iii) gives $u_m(h) \leq C_B m h^{-\alpha-1} \leq C_B M^{-\alpha}/h$. Since $\sum_m (q_m(h) + q_{m+1}(h)) \leq 2$, these m contribute at most $2C_B M^{-\alpha}/h$ to C_N . If $t_m < 1/M$, then $u_m(h) \leq \bar{u}/a_m < \bar{u}/(Mh)$, and these m contribute at most $2\bar{u}/(Mh)$. For the remaining m we have $t_{m+1} = t_m(1 + o(1)) \in [1/(2M), 2M]$ and $(m+1)/m \rightarrow 1$, and g_1 is uniformly continuous on $[1/(2M), 2M]$. So Step 1, $1/a_m = t_m/h$ and $\delta_m \rightarrow 0$ give, uniformly,

$$u_m(h)[q_m(h) + q_{m+1}(h)] = \frac{t_m}{h}(g(t_m) + o(1)) \frac{2}{m} \left(\frac{t_m g(t_m)}{\alpha} + o(1) \right) = \frac{2}{\alpha h m} (t_m^2 g(t_m)^2 + o(1)).$$

As m runs over these values, $\log m = \alpha \log(h/t_m)$ runs over an interval of length $2\alpha \log M$ with steps $\log(1 + 1/m) = (1 + o(1))/m$. So the sum of $1/m$ is $O(\log M)$, the $o(1)$ terms contribute $o(1/h)$, and the rest is a Riemann sum in the variable $\log m$. With $dm/m = -\alpha dt/t$ it is $\frac{2}{\alpha h} \left(\int_{1/M}^M t^2 g(t)^2 \frac{\alpha dt}{t} + o(1) \right)$. Hence

$$hC_N = 2 \int_{1/M}^M tg(t)^2 dt + O(M^{-\alpha} + M^{-1}) + o(1),$$

where the O term is uniform in h . By Lemma D.4(iv) we may let $M \rightarrow \infty$, and we get $NC_N = 2hC_N \rightarrow 4 \int_0^\infty tg(t)^2 dt$.

Step 3. Let G_1, G_2 be independent copies of G and $Y = G_1/(G_1 + G_2)$. The map $(G_1, G_2) \mapsto (Y, G_1 + G_2)$ has inverse $(y, r) \mapsto (yr, (1 - y)r)$ with Jacobian r . So Y has the density $f(y) = \int_0^\infty r g(yr) g((1 - y)r) dr$ on $(0, 1)$, and $f(\frac{1}{2}) = \int_0^\infty r g(r/2)^2 dr = 4 \int_0^\infty tg(t)^2 dt$. Hence $NC_N \rightarrow f(\frac{1}{2})$. The function f is continuous on $[\frac{1}{4}, \frac{3}{4}]$. Indeed, for such y both yr and $(1 - y)r$ are at least $r/4$, so by Lemma D.4(iv) the integrand is at most $r \min(\|g\|_\infty^2, Cr^{-2-2\alpha})$. This is integrable, and dominated convergence applies.

We now identify the law of Y . It does not change if G is multiplied by a constant, so we may take $\mathbb{E}e^{-sG} = e^{-s^\alpha}$. For $x > 0$ and $0 < p < 1$,

$$x^p = \frac{p}{\Gamma(1 - p)} \int_0^\infty s^{-p-1} (1 - e^{-sx}) ds, \quad x^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty s^{p-1} e^{-sx} ds.$$

Taking expectations and substituting $w = s^\alpha$ gives $\mathbb{E}G^p = \Gamma(1 - p/\alpha)/\Gamma(1 - p)$ for $|p| < \alpha$ (see also [23, Theorem 2.6.3]). Let $W = G_1/G_2$, so that $Y = W/(1 + W)$. By independence and $\Gamma(1 - z)\Gamma(1 + z) = \pi z / \sin(\pi z)$,

$$\mathbb{E}W^p = \frac{\Gamma(1 - \frac{p}{\alpha})\Gamma(1 + \frac{p}{\alpha})}{\Gamma(1 - p)\Gamma(1 + p)} = \frac{\sin \pi p}{\alpha \sin(\pi p/\alpha)}, \quad 0 < |p| < \alpha.$$

Let $\rho_\alpha(w) = \frac{\sin \pi \alpha}{\pi} \frac{w^{\alpha-1}}{w^{2\alpha} + 2w^\alpha \cos \pi \alpha + 1}$ for $w > 0$. We use the classical integral

$$\int_0^\infty \frac{x^\beta dx}{x^2 + 2x \cos \theta + 1} = \frac{\pi \sin \beta \theta}{\sin \theta \sin \pi \beta}, \quad 0 < |\beta| < 1, \quad 0 < \theta < \pi,$$

whose value at $\beta = 0$ is the limit $\theta/\sin\theta$. For $-1 < \beta < 0$ it follows from the partial fractions $\frac{1}{x^2+2x\cos\theta+1} = \frac{1}{2i\sin\theta}\left(\frac{1}{x+e^{-i\theta}} - \frac{1}{x+e^{i\theta}}\right)$ and $\int_0^\infty x^\beta(x+e^{\pm i\theta})^{-1}dx = -\pi e^{\pm i\beta\theta}/\sin(\pi\beta)$. The substitution $x \mapsto 1/x$ shows that the left side is even in β , and so is the right side. With $x = w^\alpha$, $\beta = p/\alpha$ and $\theta = \pi\alpha$ it gives $\int_0^\infty w^p \rho_\alpha(w) dw = \mathbb{E}W^p$ for $|p| < \alpha$. At $p = 0$ this shows that ρ_α is a probability density. So $\log W$ and the logarithm of a variable with density ρ_α have the same moment generating function near 0, and W has density ρ_α . Then Y has density $\rho_\alpha(\frac{y}{1-y})(1-y)^{-2} = \ell_\alpha(y)$. Both f and ℓ_α are densities of Y that are continuous near $\frac{1}{2}$, so

$$f(\tfrac{1}{2}) = \ell_\alpha(\tfrac{1}{2}) = \frac{2\sin\pi\alpha}{\pi(1+\cos\pi\alpha)} = \frac{2}{\pi} \tan \frac{\pi\alpha}{2}.$$

Dividing $E_N = \frac{1}{2}N^{-\alpha}$ by C_N gives the formula for R_N . $\square$

E Proof of Proposition 5.7

We use the setting of Section 5 with two states and a fair first step. So $0 < c < 2$, X_1 is a fair sign, and the switch indicators $\chi_k = \mathbf{1}\{X_k \neq X_{k-1}\}$, $k \geq 2$, are independent Bernoulli(c/k) variables, independent of X_1 . Put

$$\rho(a, b) = \prod_{k=a+1}^b \left(1 - \frac{c}{k}\right) = \frac{\Gamma(b+1-c)\Gamma(a+1)}{\Gamma(b+1)\Gamma(a+1-c)}, \quad \lambda_{t,N} = \frac{c}{t} \rho(t, N)$$

for integers $1 \leq a \leq b$ and $2 \leq t \leq N$. So $\lambda_{t,N}$ is the probability of a switch at time t and none in $(t, N]$, and $E_N = \frac{1}{2}\rho(1, N)$. Let T_s^+ be the time of the s -th step equal to $+1$. Let B be a Beta(c, c) variable, $d_c = 4^{1-c}/B(c, c)$ its density at $\frac{1}{2}$, and $\Theta(c) = \log(2\Gamma(2-c)d_c)$. For even N we write $h = N/2$ and $C_N = \mathbb{P}(A_N = h)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. We call this flip symmetry.

Idea of the proof. A walk that ends in the center with last step -1 takes its h -th step $+1$ at some time r , switches at time $r+1$, and never switches again. So C_N is an average of the explicit numbers $\lambda_{r+1,N}$ over the law of T_h^+ (Lemma E.1). For $c \leq 1$ these numbers are monotone in r , which already traps c_N between two explicit roots and gives $1 - c_N = O(1/\log N)$ (Proposition E.2). For the constant we need the law of T_h^+ . We have $T_h^+ \leq r$ exactly when $A_r \geq h$, and A_r/r is close to a Beta(c, c) variable with an error that is a power of r (Lemma E.4). This is proved by coupling the walk with a sign process in continuous time that switches at the points of a Poisson process of intensity $c dt/t$, whose occupation fraction is exactly Beta(c, c) at every time (Proposition E.3). Summation by parts then gives $NC_N = d_c + O(N^{-0.4})$ (Proposition E.5).

E.1 A gamma ratio

Gautschi's inequality [29, 5.6.4] says that $x^{1-s} < \Gamma(x+1)/\Gamma(x+s) < (x+1)^{1-s}$ for $x > 0$ and $0 < s < 1$. We claim that for $0 < c < 2$ and real $y \geq 2$,

$$\frac{\Gamma(y+1-c)}{\Gamma(y+1)} = y^{-c}(1+\vartheta) \quad \text{with} \quad -\frac{1}{y} \leq \vartheta \leq \frac{2}{y}. \quad (18)$$

If $c = 1$ the left side is $1/y$ and $\vartheta = 0$. If $c < 1$ we take $x = y$ and $s = 1 - c$. Then the left side lies between $(y+1)^{-c}$ and y^{-c} , and $(1+1/y)^{-c} \geq 1 - c/y \geq 1 - 1/y$. If $c > 1$ we take $x = y$ and $s = 2 - c$

and use $\Gamma(y+1-c) = \Gamma(y+2-c)/(y+1-c)$. Then the left side lies between $(y+1)^{1-c}/(y+1-c)$ and $y^{1-c}/(y+1-c)$. Here $y/(y+1-c) = 1 + (c-1)/(y+1-c) \leq 1 + 1/(y-1) \leq 1 + 2/y$, and $(1 + 1/y)^{1-c} y/(y+1-c) \geq (1 + 1/y)^{-1} \geq 1 - 1/y$.

E.2 Averaging at the center and a bracket

Lemma E.1 (averaging at the center). *Let $N \geq 2$ be even and $T = T_h^+$. Then*

$$C_N = 2 \sum_{r=h}^{N-1} \mathbb{P}(T=r) \lambda_{r+1,N} \quad \text{and} \quad \sum_{r=h}^{N-1} \mathbb{P}(T=r) = \frac{1}{2}. \quad (19)$$

If $0 < c \leq 1$, then $c/N \leq C_N \leq \lambda_{h+1,N}$.

Proof. On the event $\{A_N = h, X_N = -1\}$ the last step equal to $+1$ is the h -th one, at a time $T = r \leq N-1$. It is followed by a switch at $r+1$ and no switch in $(r+1, N]$. Conversely, these conditions give $A_N = h$ and $X_N = -1$. The event $\{T = r\}$ depends only on $X_1, \chi_2, \dots, \chi_r$. So $\mathbb{P}(A_N = h, X_N = -1) = \sum_{r=h}^{N-1} \mathbb{P}(T=r) \lambda_{r+1,N}$, and flip symmetry gives the same value for $X_N = +1$. Next, $T \leq N-1$ exactly when $A_{N-1} \geq h$. This has probability $\frac{1}{2}$, because $N-1$ is odd and A_{N-1} is symmetric about $(N-1)/2$. Finally $\lambda_{t+1,N}/\lambda_{t,N} = t/(t+1-c)$, which is at most 1 for $c \leq 1$. So $\lambda_{N,N} = c/N \leq \lambda_{t,N} \leq \lambda_{h+1,N}$ for $h+1 \leq t \leq N$, and by (19) C_N is a weighted average of these values. $\square$

For an integer $M \geq 2$ put $\eta_M(c) = M\rho(1, M)/(2c)$. Since $\rho(1, M)$ decreases in c and equals $1/M$ at $c = 1$, the function η_M decreases strictly on $(0, 1]$, from ∞ at 0 to $\eta_M(1) = \frac{1}{2}$. Let $c^{(M)} \in (0, 1)$ be the root of $\eta_M = 1$.

Proposition E.2 (bracket). *The numbers $c^{(M)}$ increase with M , and $1 - c^{(M)} = \frac{\log 2}{\log M + 1 + \gamma} + O((\log M)^{-3})$. For every even $N \geq 2$ we have $c^{(h+1)} \leq c_N \leq c^{(N)}$. So $(1 - c_N) \log N \rightarrow \log 2$, and $c_N \geq c^{(3)} = (9 - \sqrt{57})/2 > 0.725$ for $N \geq 4$.*

Proof. For $c < 1$ we have $\eta_{M+1}/\eta_M = (M+1-c)/M > 1$. So $\eta_{M+1}(c^{(M)}) > 1$ and $c^{(M+1)} > c^{(M)}$. By Proposition 5.6(b), $C_N < E_N$ for $c < c_N$ and $C_N > E_N$ for $c > c_N$, and $c_N < 1$. So it is enough to compare C_N with E_N for $0 < c < 1$. If $c^{(N)} < c < 1$, then $\eta_N(c) < 1$, that is $E_N = \frac{1}{2}\rho(1, N) < c/N \leq C_N$ by Lemma E.1. So $c_N \leq c^{(N)}$. If $c < c^{(h+1)}$, then $\eta_{h+1}(c) > 1$. With $\rho(1, N) = \rho(1, h+1)\rho(h+1, N)$ this gives $E_N > \frac{c}{h+1}\rho(h+1, N) = \lambda_{h+1,N} \geq C_N$. So $c_N \geq c^{(h+1)}$.

The number $c^{(3)}$ solves $3\rho(1, 3) = (2-c)(3-c)/2 = 2c$, that is $c^2 - 9c + 6 = 0$. By (18) with $y = M$, the equation $\eta_M(c) = 1$ reads $(1-c) \log M = \Theta_0(c) + O(1/M)$ with $\Theta_0(c) = \log(2c\Gamma(2-c))$, uniformly for $0 < c \leq 1$. For $M \geq 3$ we have $c^{(M)} \in [c^{(3)}, 1)$, where Θ_0 is bounded. So $e = 1 - c^{(M)}$ is $O(1/\log M)$, and we may expand at $c = 1$. We have $\Theta_0(1) = \log 2$ and $\Theta_0'(1) = 1 - \psi(1) = 1 + \gamma$, where $\psi = \Gamma'/\Gamma$. Taylor's theorem gives $e(\log M + 1 + \gamma) = \log 2 + O(e^2) + O(1/M)$, which is the expansion. The limit of $(1 - c_N) \log N$ follows since $\log(h+1) \sim \log N$. $\square$

E.3 The continuum occupation

Proposition E.3 (continuum occupation). *Let Π be a Poisson process on $(0, \infty)$ with intensity $c dt/t$, and let $Y(1)$ be a fair sign independent of Π . Put $Y(t) = Y(1)(-1)^{\Pi((1,t])}$ for $t \geq 1$ and $Y(t) = Y(1)(-1)^{\Pi((t,1])}$ for $0 < t < 1$. Then for every $r > 0$ the fraction $Z_r = \frac{1}{r} \int_0^r \mathbf{1}\{Y(t) = +1\} dt$ has exactly the Beta(c, c) law.*

This agrees with [34, Remark 6.2], which obtains the $\text{Beta}(c, c)$ one-dimensional marginals of the zigzag process of [34, Definition 6.1] from a scaling limit and mentions a direct proof by E. Crane that is not written there. We give a direct proof.

Proof. The points $\log t$, $t \in \Pi$, form a Poisson process $\tilde{\Pi}$ of rate c on $\mathbb{R}$, and $W(u) = Y(e^u)$ satisfies $W(v) = W(u)(-1)^{\tilde{\Pi}((u,v])}$ for $u < v$. For each real a , $W(a)$ is $W(0)$ times a function of $\tilde{\Pi}$, so it is a fair sign independent of $\tilde{\Pi}$. As $\tilde{\Pi} - a$ has the law of $\tilde{\Pi}$, the pair $(W(a), \tilde{\Pi} - a)$ has the law of $(W(0), \tilde{\Pi})$, and so $W(a + \cdot)$ has the law of W . Hence $Z_r = \int_0^1 \mathbf{1}\{Y(rt) = +1\} dt$ has the law of $Z_1 = \int_0^\infty e^{-s} \mathbf{1}\{V_s = +1\} ds$, where $V_s = W(-s)$ starts at a fair sign and switches at rate c . Let $\mu_\pm$ be the law of Z_1 given $V_0 = \pm 1$, let τ be the first switch of V , and let $U = e^{-\tau}$. Then $\mathbb{P}(U \leq u) = u^c$ for $0 \leq u \leq 1$, so U is $\text{Beta}(c, 1)$. After τ the process starts afresh from $-V_0$, independently of τ . Splitting the integral at τ shows that (μ_+, μ_-) is a fixed point of the map Ψ that sends a pair (ν_+, ν_-) of laws on $[0, 1]$ to the laws of $1 - U + UZ_-$ and UZ_+ , where $Z_\pm \sim \nu_\pm$ are independent of U . Coupling $Z_\pm$ optimally with $Z'_\pm \sim \nu'_\pm$, independently of U , gives $\mathbb{E}|UZ_\pm - UZ'_\pm| = \mathbb{E}U \mathbb{E}|Z_\pm - Z'_\pm|$. So Ψ contracts the larger of the two Wasserstein distances by the factor $\mathbb{E}U = c/(c+1)$, and it has only one fixed point.

Let G_a, G_b, G_d be independent Gamma variables with shapes a, b, d and unit scale. A change of variables shows that $G_a/(G_a + G_b)$ is $\text{Beta}(a, b)$ and independent of $G_a + G_b$. Hence it is independent of the pair $(G_a + G_b, G_d)$ and of $(G_a + G_b)/(G_a + G_b + G_d)$, which is $\text{Beta}(a + b, d)$. Their product $G_a/(G_a + G_b + G_d)$ is $\text{Beta}(a, b + d)$. With $(a, b, d) = (c, 1, c)$ this says that UZ is $\text{Beta}(c, c + 1)$ whenever Z is $\text{Beta}(c + 1, c)$ and independent of U . Applying it to $Z = Z_+$ and to $Z = 1 - Z_-$, and using $1 - U + UZ_- = 1 - U(1 - Z_-)$, we see that Ψ maps the pair $\text{Beta}(c + 1, c)$, $\text{Beta}(c, c + 1)$ to itself. So Z_1 is the equal mixture of these two laws. As $\text{B}(c + 1, c) = \text{B}(c, c + 1) = \frac{1}{2}\text{B}(c, c)$, the mixture has density $x^{c-1}(1-x)^{c-1}(x+1-x)/\text{B}(c, c)$, which is the $\text{Beta}(c, c)$ density. $\square$

Lemma E.4 (coupling). *Let $J \subset (0, 2)$ be compact, $a_J = \min(\min J, 1)$ and $\kappa_J = a_J/(1 + a_J)$. There is C_J such that for $c \in J$ and every integer $r \geq 2$,*

$$\sup_{y \in \mathbb{R}} |\mathbb{P}(A_r \geq y) - \mathbb{P}(rB \geq y)| \leq C_J (\log r) r^{-\kappa_J}.$$

Proof. Take Π and Y as in Proposition E.3 and an integer $1 \leq K \leq r$. The variables $\chi'_k = \mathbf{1}\{\Pi((k-1, k]) \text{ is odd}\}$, $k > K$, are independent of each other and of $Y(K)$, and $\mathbb{P}(\chi'_k = 1) = p'_k = \frac{1}{2}(1 - (1 - 1/k)^{2c})$. By Taylor's theorem $|p'_k - c/k| = \frac{1}{2}c|2c - 1|(1 - \vartheta)^{2c-2}k^{-2}$ for some $0 < \vartheta < 1/k$. This is at most $3k^{-2}$. Indeed, if $c \geq 1$ then $\frac{1}{2}c|2c - 1| < 3$ and the power of $1 - \vartheta$ is at most 1. If $c < 1$ then $\frac{1}{2}c|2c - 1| \leq \frac{1}{2}$ and the power is at most $(1 - \frac{1}{2})^{-2} = 4$, as $k \geq 2$. For $K < k \leq r$ we couple $\chi_k \sim \text{Bernoulli}(c/k)$ with χ'_k so that $\mathbb{P}(\chi_k \neq \chi'_k) = |p'_k - c/k|$, using independent extra randomness. We put $X_K = Y(K)$ and $X_k = X_{k-1}(-1)^{\chi_k}$ for $K < k \leq r$, and we draw $X_1, \dots, X_{K-1}$ from the law of the walk given X_K , again with independent randomness. In the walk X_K is a fair sign and $(\chi_k)_{k>K}$ is independent of $(X_1, \dots, X_K)$. So $(X_1, \dots, X_r)$ has the law of the walk.

Let $\mathcal{A}$ be the event that $\chi_k = \chi'_k$ for $K < k \leq r$. On $\mathcal{A}$ we have $X_k = Y(k)$ for $K \leq k \leq r$, and $\mathbb{P}(\mathcal{A}^c) \leq \sum_{k>K} 3k^{-2} \leq 3/K$. If also $\Pi((k-1, k]) = 0$, then $Y = Y(k)$ on $(k-1, k]$, so step k adds the same amount to A_r as the interval $(k-1, k]$ adds to rZ_r . The other steps $k \leq r$ number at most $K + \Pi((K, r])$, and each changes the difference by at most 1. Hence $|A_r - rZ_r| \leq K + \Pi((K, r])$ on $\mathcal{A}$. The variable $\Pi((K, r])$ is Poisson with mean $c \log(r/K) \leq 2 \log r$, so by Markov's inequality it exceeds K with probability at most $2 \log r/K$. With $b_K = (3 + 2 \log r)/K$ we get

$$\mathbb{P}(rZ_r \geq y + 2K) - b_K \leq \mathbb{P}(A_r \geq y) \leq \mathbb{P}(rZ_r \geq y - 2K) + b_K,$$

and rZ_r has the law of rB . So the left side of the lemma is at most $b_K + \sup_x \mathbb{P}(x \leq B \leq x + 2K/r)$.

For $c \leq 1$ the density of B is at most $2^{1-c}(x^{c-1} + (1-x)^{c-1})/B(c, c)$, and x^{c-1} has integral at most δ^c/c over an interval of length δ . For $c \geq 1$ the density is at most d_c . The mass of an interval is also at most 1. So there is C'_J such that an interval of length δ has mass at most $C'_J \delta^{a_J}$ for all $\delta > 0$ and $c \in J$. Take $K = \lceil r^{\kappa_J} \rceil$. Then $b_K \leq (3 + 2 \log r) r^{-\kappa_J}$ and $2K/r \leq 4r^{\kappa_J-1}$, and $(r^{\kappa_J-1})^{a_J} = r^{-\kappa_J}$ because $(1 - \kappa_J)a_J = \kappa_J$. $\square$

E.4 A local limit at the center

Proposition E.5 (local limit at the center). *For each compact $J \subset (0, 2)$ there is C_J such that $|NC_N - d_c| \leq C_J(\log N) N^{-\kappa_J}$ for $c \in J$ and even $N \geq 4$.*

Proof. Let $T = T_h^+$, $F(r) = \mathbb{P}(T \leq r) = \mathbb{P}(A_r \geq h)$, $\Phi(r) = \mathbb{P}(rB \geq h)$ and $\Lambda_r = N\lambda_{r+1, N}$. Then $F(h-1) = \Phi(h-1) = \Phi(h) = 0$. For $h \leq r \leq N-1$ we have $2 \leq r$ and $N/2 \leq r < N$, so Lemma E.4 gives $|F(r) - \Phi(r)| \leq \delta_N := C_J(\log N) N^{-\kappa_J}$ after enlarging C_J . By Lemma E.1, $NC_N = 2 \sum_{r=h}^{N-1} (F(r) - F(r-1)) \Lambda_r$. The ratio $\lambda_{t+1, N}/\lambda_{t, N} = t/(t+1-c)$ shows that Λ_r is monotone in r . Also $\Lambda_{N-1} = c$ and $0 < \Lambda_h < 2c$. For $c \leq 1$ this holds because $\rho \leq 1$ and $N < 2(h+1)$, and for $c > 1$ because Λ_r increases. So summation by parts shows that replacing F by Φ changes NC_N by at most $2\delta_N(\Lambda_{N-1} + |\Lambda_{N-1} - \Lambda_h|) \leq 4c\delta_N$. For $h < r \leq N-1$ we have $\Phi(r) - \Phi(r-1) = \mathbb{P}(h/r \leq B < h/(r-1))$, and on this event $\lceil h/B \rceil = r$. So the new main term is $2\mathbb{E}[\Lambda_{\lceil h/B \rceil}; B \geq h/(N-1)]$.

By (18) with $y = N$ and $y = r+1 \geq 3$, $\Lambda_r = c((r+1)/N)^{c-1}(1 + O(1/N))$ for $h \leq r \leq N-1$. On the event above with $r = \lceil h/B \rceil$, the number $(r+1)/N$ lies in $[\frac{1}{2}, 2]$ and within $2/N$ of $1/(2B)$, which also lies in $[\frac{1}{2}, 2]$. The function x^{c-1} is 4-Lipschitz on $[\frac{1}{2}, 2]$. So $\Lambda_{\lceil h/B \rceil} = c(2B)^{1-c} + O(1/N)$. Also $\mathbb{P}(\frac{1}{2} \leq B < h/(N-1)) = O_J(1/N)$, since the density of B is bounded near $\frac{1}{2}$ uniformly on J . So NC_N is within $4c\delta_N$ of

$$2c\mathbb{E}[(2B)^{1-c}; B \geq \frac{1}{2}] + O_J(1/N) = \frac{2c2^{1-c}}{B(c, c)} \int_{1/2}^1 (1-b)^{c-1} db + O_J(1/N) = d_c + O_J(1/N). \quad \square$$

E.5 The expansion of the crossing

Proof of Proposition 5.7. Let $J = [c^{(3)}, 1]$. Then $\kappa_J = c^{(3)}/(1 + c^{(3)}) > 0.42$, and d_c is bounded below on J by a positive constant. By (18) with $y = N$, $\log E_N = \log(\frac{1}{2}\rho(1, N)) = -\log 2 - \log \Gamma(2-c) - c \log N + O(1/N)$. By Proposition E.5, $\log C_N = \log d_c - \log N + O(N^{-0.4})$ for large N . Both hold uniformly for $c \in J$, so

$$\log C_N - \log E_N = (c-1) \log N + \Theta(c) + O(N^{-0.4}) \quad (c \in J). \quad (20)$$

Here $\Theta(1) = \log 2$ because $d_1 = 1$. Next, $\Theta'(c) = -\psi(2-c) - \log 4 - 2\psi(c) + 2\psi(2c)$, and with $\psi(1) = -\gamma$ and $\psi(2) = 1 - \gamma$ this gives $\Theta'(1) = \gamma + 2 - 2\log 2$. Put $M = \log N + \Theta'(1)$. By Proposition E.2, $c_N \in J$ for $N \geq 4$ and $e = 1 - c_N$ is $O(1/\log N)$. The left side of (20) vanishes at $c = c_N$. As Θ is smooth on J , Taylor's theorem gives $eM = \log 2 + O(e^2) + O(N^{-0.4}) = \log 2 + O(M^{-2})$. So $e = \log 2/M + O(M^{-3})$. $\square$

With one more term in Taylor's theorem and $e^2 = (\log 2)^2/M^2 + O(M^{-4})$, the same argument adds the term $-\Theta''(1)(\log 2)^2/(2M^3)$ to the expansion of c_N , with error $O(M^{-4})$. Here $\Theta''(1) = \pi^2/2 - 4$, since $\Theta''(c) = \psi'(2-c) - 2\psi'(c) + 4\psi'(2c)$, $\psi'(1) = \zeta(2)$ and $\psi'(2) = \zeta(2) - 1$. At $N = 200$ the bracket of Proposition E.2 is $[0.88749, 0.89878]$, Proposition 5.7 gives 0.89319, and with the M^{-3} term we get 0.89236. The high-precision value is $c_{200} = 0.89253$. The constants C_J and the constants in the O -terms of this appendix are not computed.

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